

## Chapter 10

# Elements of optimal control theory

### 10.1 Introduction

The feedback control techniques examined so far are based on the idea of allocating the eigenvalues of the feedback system. While it would be desirable to choose their position according to time or frequency domain properties one wants to attribute to the closed-loop system, this can be achieved only empirically and, often, in an imprecise way.

Control optimal methodologies, the focus of the present chapter, allow to express directly in the time domain (at least in principle) specifics on a dynamical system, even a non-linear one, by characterizing the input to be applied to the system as the one that minimises, or maximises, a suitable cost (or quality) index. Once the problem is formulated in this terms, the optimal control theory provides analytical tools to compute, either directly or through numerical procedures, the optimal input, and in many case to realise such an input as a feedback from the state. The three essential elements of an optimal control problem are, therefore:

- i) the system model,
- ii) the constraints on state and input variables,
- iii) the index to be minimised (or maximised) by choosing a suitable input function.

Most available control techniques employ state models, both for linear and non-linear dynamics, and both for continuous and discrete time.

Constraints can be of very different kind, and they can be expressed, in the most common cases, by bounding the amplitude of input and internal variables along the evolution of the system, or by specifying the final values of the state variables. These constraints characterise, for each problem, the set of inputs that are considered feasible, in the sense that the optimal solution must be an element of such a set.

The performance of a system under any admissible input is evaluated by means of an index, usually represented by a functional that associates a real number to any admissible input. The structure of the index itself depends on the objective to be reached by solving the control problem, and must consider the aspects of the system dynamics that characterise the evolution of the involved variables, as well as their relative weight.

Typical examples of indices derive from energy-related analysis, and correspond to determining the input that steers the system to a preassigned objective by using in an optimal way the available resource, e.g., the fuel.

In other cases the indices provide, for any input to the system, an evaluation of the corresponding economical costs or benefits and/or other advantages (for instance, of biological, ecological, or societal nature) that cannot be strictly quantified in economical terms. For example, in natural resource management (such as, fishing areas, woodlands, etc.) the state equations express the dynamics of the exploited species, while the index to maximise involves both the economical profit due to exploitation and the survival of the involved species.

In other scenarios the index corresponds to the duration of the time interval, or time horizon, needed to reach an objective, and the optimal input is the one that minimises such a duration.

**Example 10.1.1** [MAXIMISATION OF THE ENERGY EXTRACTED FROM A NETWORK] In the electrical network  $\Sigma$ , at time  $t = 0$ , the current in the inductor is  $i_L(0) = 1\text{A}$ . The task is to determine which voltage  $v(\cdot)$  has to be applied to the external terminals in such a way that the maximum possible energy is extracted during the interval  $[0, T]$ . It is possible to formulate the problem as an optimal problem in which:

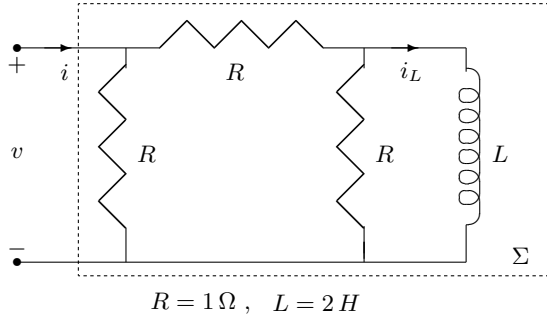


Figure 10.1.1

- i) the state model is represented by the equation

$$\frac{di_L(t)}{dt} = -\frac{1}{4}i_L(t) + \frac{1}{4}v(t); \quad (10.1)$$

- ii) no constraints are imposed on the input function;
- iii) the index to maximise is the energy supplied by the two terminal component (bipole) in  $[0, T]$ , that is,

$$J = - \int_0^T v(t)i(t)dt. \quad (10.2)$$

Bearing in mind that the current  $i(t)$  entering the bipole can be expressed as

$$i(t) = \frac{1}{2}i_L(t) + \frac{3}{2}v(t), \quad (10.3)$$

the index to maximise is expressed, as a function of  $v(\cdot)$ ,

$$J = - \int_0^T \left\{ v(t) \frac{3}{2}v(t) + v(t) \frac{1}{2}i_L(t) \right\} dt = - \int_0^T \begin{bmatrix} v(t) & i_L(t) \end{bmatrix} \begin{bmatrix} \frac{3}{2} & \frac{1}{4} \\ \frac{1}{4} & 0 \end{bmatrix} \begin{bmatrix} v(t) \\ i_L(t) \end{bmatrix} dt. \quad (10.4)$$

**Example 10.1.2** [MINIMISATION OF THE CONTROL EFFORT] In order to transfer a system from the initial state  $\mathbf{x}(0) = \mathbf{x}_i$  to the final state  $\mathbf{x}_f$  minimising the control effort, it is customary to adopt indices taking the form

$$(\mathbf{x}(T) - \mathbf{x}_f)^T S (\mathbf{x}(T) - \mathbf{x}_f) + \int_0^T L(\mathbf{u}(t))dt. \quad (10.5)$$

The first term, where the matrix  $S$  is positive definite or positive semidefinite, represents the cost associated to failing to perfectly reach the objective. The second term represents the control cost. The structure of  $L(\mathbf{u})$  depends on the quantity adopted to measure the control effort. As an example,

let the control  $u$  be the force exerted by an engine and the control cost be the fuel cost, and assume that the magnitude of the force is proportional to the fuel consumption per unit time. Then

$$\int_0^T L(u(t))dt = \alpha \int_0^T |u(t)|dt,$$

where  $\alpha > 0$  is a suitable multiplicative constant.

**Example 10.1.3** [ENERGY DISSIPATION MINIMISATION] A different structure of the function

$L(i)$  emerges when the goal is to determine what current  $i(\cdot)$  the generator has to supply in order to discharge the capacitor, whose initial charge is  $v_C(0)$ , during the interval  $[0, T]$ , minimising the energy  $\int_0^T Ri^2(t)dt$  dissipated in the resistor. If failure to reach the final state  $v_C(T) = 0$  is penalised via a quadratic term, such as  $a^2v_C^2(T)$ , the index structure becomes

$$J = a^2v_C^2(T) + \int_0^T Ri^2(t)dt. \quad (10.6)$$

Notice that an analogous optimal control problem has been solved in Exercise 5.4.2 (under the assumption that the final state needs to be exactly reached).

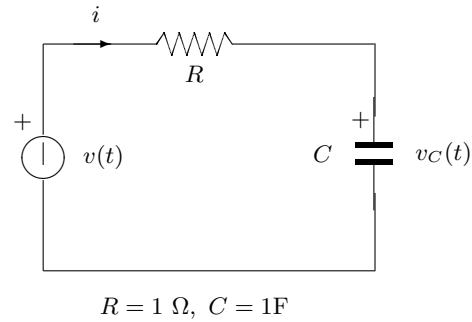


Figura 10.1.2

**Example 10.1.4** [MANAGEMENT OF A FISH RESERVE] We consider in this example the problem of the optimal management and planning of a small natural fish reserve<sup>1</sup> over the horizon  $[0, T]$ . Denote by  $x(t)$  and  $h(t)$  the quantity of fish in the fishery and the quantity of fish caught per unit time, respectively, at time  $t$ . The dynamics of the fish reserve is represented by the equation (1.110) provided in Example 1.8.1 of Chapter 1:

$$\dot{x}(t) = rx(t)\left(1 - \frac{x(t)}{K}\right) - h(t), \quad x(0) = x_0.$$

The rate at which fish is caught depends on both the fish quantity and the employed effort in terms of manpower and equipment  $u$ :

$$h = f(x, u). \quad (10.7)$$

If we denote by  $p$  the unit selling price and we assume that the costs are proportional to the effort by means of the constant  $c$ , the net gain over the interval  $[t, t + \Delta t]$  turns out to be

$$[ph(t) - cu(t)]\Delta t. \quad (10.8)$$

A typical problem is to determine  $u(\cdot)$  in such a way to maximise the profit, namely

$$\max_{u(\cdot)} \int_0^T [ph(t) - cu(t)]dt. \quad (10.9)$$

**Example 10.1.5** [DISCRETE TIME MODEL: FOREST PLANNING] Management policies of a forest designated to timber production can be classified into two categories: “uniform age” management and “disuniform age” management. The former management strategy consists of the simultaneous cut of all the trees of the forest, followed by seeding or planting young saplings, all of the same age. In such a way, the whole forest is made of plants of the same age at any moment of time. The latter management strategy consists instead of cutting down a certain number of trees belonging to given size classes in rotation, at the same time setting the conditions for natural regeneration of the cut area by seeding from the plants that have not been cut down. In general, such a procedure implies that the wood always contains trees belonging to different age clusters.

<sup>1</sup>That is, whose dimensions are such that they cannot influence the prices of the fish market.

Growth classes model When building a model, it is worth noticing that it is difficult, in general, to estimate a certain plant age before its cut, while other quantities correlated with age, such as the trunk diameter or the biomass of the tree, can be more easily measured, and also are more relevant for the management problems that we are considering. Therefore, rather than to age classes, as introduced in Leslie's model (Example 1.5.1), we prefer to consider here different "growth classes". The first class includes plants with the smallest possible size, the second class those with the second smallest possible size, and so on. As time passes by, plants move to classes corresponding to indices of increasing number (we discard the possibility for a plant to reduce its size). In the model, we discretise the time in such a way that during each time step either a plant remains in the same growth class, or it moves to the very next one, while a direct jump to higher classes is not possible.

Considering the time interval  $[t, t + 1)$ ,  $t \in \mathbb{Z}$ , we denote by

- $x_i(t)$ ,  $i = 1, 2, \dots, n$ , the number of trees that are in the  $i$ -th class at the beginning of the interval;
- $\beta_i$ ,  $i = 1, 2, \dots, n$ , the survival rates within each class, namely, the fraction of individuals that belong to the  $i$ -th class at the beginning of the interval and that also exist at the beginning of the consecutive interval, having survived the whole period;
- $p_i$ ,  $0 < p_i \leq 1$ ,  $i = 1, 2, \dots, n - 1$ , the fraction of survived individuals that at the beginning of the interval belong to the  $i$ -th class, and that at the beginning of the next interval belong to the successive class. Therefore,  $1 - p_i$  represents the fraction of survivors of the  $i$ -th class that during the period has not moved to the next class.

The update equation – in absence of cutting or reforestation – is

$$x_{i+1}(t + 1) = p_i \beta_i x_i(t) + (1 - p_{i+1}) \beta_{i+1} x_{i+1}(t), \quad i = 1, 2, \dots, n - 1. \quad (10.10)$$

The first growth class at the beginning of the interval  $[t + 1, t + 2)$  includes

- the survivors of the first class in  $[t, t + 1)$  that did not move to the second class;
- the newborns, which result from individuals of all growth classes according to the fertility rates  $\alpha_1, \alpha_2, \dots, \alpha_n$ . Of the individuals born in the interval  $[t, t + 1)$ , only a fraction, denoted by  $\beta_0$ , survives until the beginning of the interval  $[t + 1, t + 2)$ .

We thus obtain the equation

$$x_1(t + 1) = (1 - p_1) \beta_1 x_1(t) + \beta_0 \sum_{i=1}^n \alpha_i x_i(t). \quad (10.11)$$

The whole model is linear, with structure

$$\begin{aligned} \mathbf{x}(t + 1) &= \begin{bmatrix} (1 - p_1) \beta_1 + \alpha_1 \beta_0 & \alpha_2 \beta_0 & \alpha_3 \beta_0 & \dots & \dots & \alpha_{n-1} \beta_0 & \alpha_n \beta_0 \\ p_1 \beta_1 & (1 - p_2) \beta_2 & & & & & \\ & p_2 \beta_2 & (1 - p_3) \beta_3 & & & & \\ & & & \ddots & \ddots & & \\ & & & & (1 - p_{n-1}) \beta_{n-1} & & \\ & & & & p_{n-1} \beta_{n-1} & \beta_n & \end{bmatrix} \mathbf{x}(t) \\ &= F \mathbf{x}(t). \end{aligned} \quad (10.12)$$

Differently from Leslie's model, all the diagonal entries of the matrix  $F$  can be different from zero, to account for individuals that do not change class, while the rates  $\beta_i$  and  $\alpha_i$  refer to growth classes, instead of age classes.

If we separate in  $\mathbf{x}(t + 1)$  the contribution to the first class that is due to the plants already existent at the beginning of the interval  $[t, t + 1)$ , from the one due to newborns, and we set  $x_0 := \sum_{i=1}^n \alpha_i x_i$ , the model can be written in the form

$$\mathbf{x}(t+1) = A\mathbf{x}(t) + \begin{bmatrix} \beta_0 x_0(t) \\ 0 \\ \vdots \\ 0 \end{bmatrix} = A\mathbf{x}(t) + \phi(t) \quad (10.13)$$

In such a form, the growth and survival matrix  $A$  factorises as

$$A = \begin{bmatrix} (1-p_1) & & & & \\ p_1 & (1-p_2) & & & \\ & p_2 & \ddots & \ddots & \\ & & \ddots & \ddots & (1-p_{n-1}) \\ & & & p_{n-1} & 1 \end{bmatrix} \begin{bmatrix} \beta_1 & & & & \\ & \beta_2 & & & \\ & & \beta_3 & & \\ & & & \ddots & \\ & & & & \beta_{n-1} \\ & & & & & \beta_n \end{bmatrix},$$

and the *regeneration* vector  $\phi(t)$  can be interpreted as a “forcing” term, which is a linear function of the state variables.

The growth classes model – as well as Leslie’s linear model – has been generalised to the non-linear setting. We will only briefly consider the assumption that the entries of the matrix  $A$  are not constant, but rather depend on the state  $\mathbf{x}$  of the woodland, and that the first component of the regeneration vector  $\phi$  is a non-linear function of the state, of the type  $\beta_0 x_0 \psi(x_0)$ . Then, the model (10.13) becomes

$$\mathbf{x}(t+1) = A(\mathbf{x}(t))\mathbf{x}(t) + \begin{bmatrix} \beta_0 x_0(t)\psi(x_0(t)) \\ 0 \\ \vdots \\ 0 \end{bmatrix} = A(\mathbf{x}(t))\mathbf{x}(t) + \phi(\mathbf{x}(t)) \quad (10.14)$$

#### Harvesting, sowing and reforestation

Suppose we want to harvest, during  $[t, t+1)$ ,  $u_i(t)$  plants belonging to different growth classes  $1, 2, \dots, n$ . Assuming that the cut happens at the end of the interval, therefore after reproduction and death events that might happen during the interval itself, and in the absence of sowing and replanting interventions, the linear model (10.12) becomes

$$\mathbf{x}(t+1) = F\mathbf{x}(t) - \mathbf{u}(t), \quad (10.15)$$

where the components  $u_i(t)$  of  $\mathbf{u}(t)$  must satisfy the consistency constraint

$$0 \leq u_i(t) \leq (F\mathbf{x}(t))_i, \quad (10.16)$$

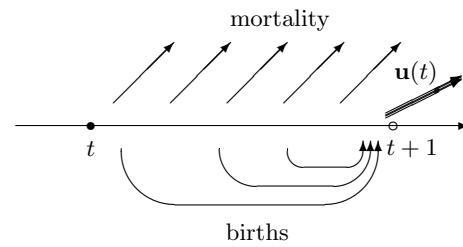
in order to guarantee that all the entries in the state vector in  $t+1$  are non-negative.

If, in addition to the harvest, one wants to also control the forest dynamics, one can directly intervene on the first class by sowing or planting new saplings, or even on successive classes, by planting already grown trees. By denoting by  $v_i(t)$  the number of plants belonging to the  $i$ -th growth class that are added to the woodland during the  $t$ -th time interval and by  $\mathbf{v}(t)$  the vector  $[v_1(t) \dots v_n(t)]^T$ , and also considering possible non-linearities described in the model (10.14), we get the update equation

$$\mathbf{x}(t+1) = A(\mathbf{x}(t))\mathbf{x}(t) + \phi(\mathbf{x}(t)) - \mathbf{u}(t) + \mathbf{v}(t). \quad (10.17)$$

#### Performance index maximisation

(i) Assume, as in the case of the linear model (10.15), that no sowing or replanting interventions are performed. Woodland management has the goal of maximising the net gain obtained during the interval  $[0, T]$ , therefore it needs to maximise the index



$$J_T(\mathbf{u}, \mathbf{x}_0) = \sum_{t=0}^T \mathbf{w}^T(t) \mathbf{u}(t), \quad (10.18)$$

where the  $i$ -th component of  $\mathbf{w}(t)$  represents the value (monetary, or in biomass unit) of a plant belonging to the  $i$ -th growth class that is cut (and sold) at time  $t$ . The initial distribution  $\mathbf{x}_0$  of the growth classes is a datum of the problem, and cut planning, that is, designing the function  $\mathbf{u}(\cdot)$ , can be done freely as long as it satisfies, at any time, the consistency constraint. In such a case a very heavy exploitation of the woodland is unavoidable, unless we impose additional constraints on the evolution of  $\mathbf{x}(\cdot)$ .

(ii) A possible additional objective is that the plants population remain at the same level in time, namely, that  $\mathbf{x}(\cdot)$  is a constant function of time. In this case, the initial state, which is also the desired steady state, is considered as a free variable, and its components are among the quantities to be optimised. If the vector  $\mathbf{w}(\cdot)$  is constant, among all the pairs  $(\mathbf{x}, \mathbf{u})$  satisfying the stationary condition

$$\mathbf{x} = F\mathbf{x} - \mathbf{u}, \quad \mathbf{x} \in \mathbb{R}_+^n, \quad \mathbf{u} \in \mathbb{R}_+^n, \quad (10.19)$$

one looks for the one that maximises  $\mathbf{w}^T \mathbf{x}$ . In order to satisfy the consistency condition (10.16), it necessarily holds true that  $\mathbf{u} = D F \mathbf{x}$ , where  $D$  is a diagonal matrix with diagonal entries both non-negative and not greater than 1. The optimal problem thus becomes

$$\max_{\mathbf{x} \in \mathbb{R}_+^n} \mathbf{w}^T D F \mathbf{x}, \quad (10.20)$$

where the search of the maximum has to satisfy the constraint (10.19), namely,  $\mathbf{x}$  must satisfy the equation

$$\mathbf{x} = (I - D) F \mathbf{x}. \quad (10.21)$$

If formulated in this way, the problem is undetermined: if  $\mathbf{x} \neq \mathbf{0}$  satisfies (10.21), then  $h\mathbf{x}$  satisfies the same constraint and if the solution  $\mathbf{u}$  corresponds to  $\mathbf{x}$ , then the solution  $h\mathbf{u}$  corresponds to  $h\mathbf{x}$ : therefore the index (10.20) does not have an upper bound. Uniqueness of the solution can be obtained by introducing additional constraints, such as setting the number of newborn corresponding to  $\mathbf{x}$ , by specifying  $\bar{x}_0 = \sum_{i=1}^n \alpha_i x_i$ .

(iii) We finally consider a rather general case, in which the linear model (10.13) is replaced by a non-linear one, and the maximisation time horizon is infinite. We assume that

- due to non-linearity, the elements of the matrix  $A$  and the coefficients  $\alpha_i$  that appear in the first component of the vector  $\phi$  depend on the  $n$  components of the vector  $\mathbf{x}$ . The update equation then becomes

$$\mathbf{x}(t+1) = A(\mathbf{x}(t))\mathbf{x}(t) + \phi(\mathbf{x}(t)) - \mathbf{u}(t), \quad (10.22)$$

with the constraint that the components of the harvesting vector  $\mathbf{u}(t)$  do not exceed the corresponding components of  $A(\mathbf{x}(t))\mathbf{x}(t) + \phi(\mathbf{x}(t))$ ;

- the index is given by

$$J(\mathbf{x}_0, \mathbf{u}) = \sum_{t=0}^{+\infty} \delta^t \mathcal{R}(\mathbf{x}(t), \mathbf{u}(t)) \quad (10.23)$$

where

- $\mathcal{R}(\mathbf{x}(t), \mathbf{u}(t))$ , the net profit in the  $t$ -th period, depends on the extent of harvesting and on the level  $\mathbf{x}(t)$  of the resource in the year in which harvesting is performed, assuming that the value of the resource increases when it runs low;
- $\delta \in (0, 1)$  is the discount rate, that allows to refer to  $t = 0$  the value of the revenues obtained during the years.

Alternatively, if one plans to exploit the resource in the interval  $[0, T]$ , the index to be considered is

$$J_{T-1}(\mathbf{x}_0, \mathbf{u}) = \sum_{t=0}^{T-1} \delta^t \mathcal{R}(\mathbf{x}(t), \mathbf{u}(t)) + \mathcal{L}(\mathbf{x}(T), T) \quad (10.24)$$

where the term  $\mathcal{L}(\mathbf{x}(T), T)$  accounts for the value that is assigned to the residual resource at the end of the interval.

In this chapter we present elements of the optimal control theory, by limiting ourselves to a rather narrow class of problems, that is, linear discrete time systems with indices represented by a quadratic functional. We will then extend the analysis to continuous time systems, for which we will have to resort to more advanced conceptual tools.

## 10.2 Finite-horizon optimal control of linear discrete time systems

One of the most studied branches of optimal control theory deals with problems in which

- the update equation of state variables is linear and time is discretised, and
- indices (representing cost or quality) are quadratic functions.

Such an interest is first and foremost due to technical reasons, since for this class of problems it is possible to explicitly provide the solution as a control law in terms of linear feedback from the state. In addition, the results for the linear case can be applied to non-linear systems, with the goal of determining how to modify a pre-determined optimal control law when the system dynamics is perturbed. More explicitly, let us assume that the optimal control law  $\mathbf{u}_N(\cdot)$  for a non-linear system with initial state  $\mathbf{x}_N(0)$  has already been computed. Assume that the system is initialised with a perturbed state  $\mathbf{x}_P(0)$  that is sufficiently close to  $\mathbf{x}_N(0)$ , for which the optimal control law is  $\mathbf{u}_P(\cdot)$ . Then, on first approximation, the variation  $\Delta \mathbf{u}(\cdot) = \mathbf{u}_P(\cdot) - \mathbf{u}_N(\cdot)$  can be computed by solving an optimal control problem for the linearised system.

The first paragraphs will be devoted to the discrete time case, while the continuous time case is postponed to the successive ones.

### 10.2.1 Problem formulation

We consider a discrete time linear system with state equation

$$\mathbf{x}(t+1) = F\mathbf{x}(t) + G\mathbf{u}(t), \quad (10.25)$$

with assigned initial state

$$\mathbf{x}(0) = \mathbf{x}_0, \quad (10.26)$$

and we assume that no constraint is imposed on the input functions. We want to address the problem of finding an input  $\mathbf{u}(\cdot)$ , acting on the interval  $[0, T-1]$ , that minimises the *index (or objective function)*

$$J_{T-1}(\mathbf{u}, \mathbf{x}_0) = \sum_{t=0}^{T-1} [\mathbf{x}^T(t) Q \mathbf{x}(t) + \mathbf{u}^T(t) R \mathbf{u}(t)] + \mathbf{x}^T(T) S \mathbf{x}(T), \quad (10.27)$$

where  $Q$  and  $S$  are  $n \times n$  symmetric and positive semidefinite matrices, and  $R$  is an  $m \times m$  symmetric and positive definite matrix.

Once  $\mathbf{x}_0$  is fixed, the functional  $J_{T-1}(\cdot, \mathbf{x}_0)$  maps in a unique way every input sequence  $\mathbf{u}(\cdot)$  into a non-negative number. In fact, for every initial state and every input sequence, we obtain via (10.25) a well defined state evolution  $\mathbf{x}(\cdot)$ , and by evaluating (10.27) at  $\mathbf{x}(\cdot)$  and  $\mathbf{u}(\cdot)$ , we obtain the value of  $J_{T-1}$ . By interpreting  $J_{T-1}$  as a functional defined on the set of input sequences  $\mathbf{u}(\cdot)$  and on the set of state evolutions  $\mathbf{x}(\cdot)$ , the optimal problem can be seen as the problem of minimising  $J_{T-1}$  once  $\mathbf{x}(\cdot)$  and  $\mathbf{u}(\cdot)$  are constrained by (10.25) and (10.26).

The structure of  $J_{T-1}$  in (10.27) highlights the role of the two features of the system performance over  $[0, T]$ . In particular, the term  $\mathbf{x}^T(T)S\mathbf{x}(T)$  represents the cost of a gap between the zero state and the final state of the system, while  $\sum_{t=0}^{T-1} \mathbf{x}^T(t)Q\mathbf{x}(t) + \mathbf{u}^T(t)R\mathbf{u}(t)$  represents the cost of a gap between zero and the intermediate states, as well as the energy spent for controlling the system.

The different structural constraint imposed on  $R$  with respect to those on  $Q$  and  $S$ , namely, the former being required to be positive definite, while the latter being only required to be positive semidefinite, might seem questionable and requires justification. Indeed, it is obvious that for the cost  $J_{T-1}$  to assume only non-negative values it is sufficient to require  $R$  to be positive semidefinite. In such a case, the problem formulation would exhibit a higher degree of uniformity and the problem itself would be formulated in a more general setting, since it would allow to give zero weight to certain inputs, or to certain linear combinations of the inputs. However, letting  $R$  be singular, namely, assuming only positive semidefiniteness, would considerably complicate the analysis.

On the other side, it is not convenient to strengthen the constraints on  $Q$  and  $S$  by imposing them to be positive definite. Indeed, the index to minimise is often written in terms of inputs and outputs of the system, and takes the form

$$J_{T-1}(\mathbf{u}, \mathbf{x}_0) = \sum_{t=0}^{T-1} [\mathbf{y}^T(t)V\mathbf{y}(t) + \mathbf{u}^T(t)R\mathbf{u}(t)] + \mathbf{y}^T(T)W\mathbf{y}(T). \quad (10.28)$$

If the system is strictly proper, by replacing  $\mathbf{y}(t)$  with  $H\mathbf{x}(t)$  we obtain an index that has the structure (10.27) and for which  $Q = H^T V H$  and  $S = H^T W H$ , which can be positive semidefinite even if  $V$  and  $W$  are positive definite.

We conclude with two remarks on the index. First of all, notice that one of the main reasons for choosing a quadratic index, instead of indices with more general structures, is that it yields an optimal control law that can be expressed as a linear feedback from the state. Secondly, the weighting matrices  $Q, R$  and  $S$  cannot always be determined in a unique way on the basis of the nature of the problem and the specifics on the system dynamics. Instead, their design might be<sup>2</sup> the result of an iterative procedure, based on an initial choice of the matrices and successive refining and corrections after evaluation of the obtained optimal solutions.

---

<sup>2</sup>For example, when the index refers to quantities that are difficult to define or to measure.



### 10.2.2 Index minimisation

The first step to take to solve the optimal control problem is to modify the form of the index  $J_{T-1}$  in (10.27) to obtain an expression that simplifies the subsequent computations.

**Lemma 10.2.1** [INDEX REWRITING] *Let  $M(0), M(1), \dots, M(T) \in \mathbb{R}^{n \times n}$  be an arbitrary sequence of matrices. Then the index can be expressed as follows*

$$J_{T-1}(\mathbf{u}, \mathbf{x}_0) = \mathbf{x}^T(0)M(0)\mathbf{x}(0) + \mathbf{x}^T(T)[S - M(T)]\mathbf{x}(T) + \sum_{t=0}^{T-1} \begin{bmatrix} \mathbf{u}^T(t) & \mathbf{x}^T(t) \end{bmatrix} \begin{bmatrix} R + G^T M(t+1)G & G^T M(t+1)F \\ F^T M(t+1)G & Q + F^T M(t+1)F - M(t) \end{bmatrix} \begin{bmatrix} \mathbf{u}(t) \\ \mathbf{x}(t) \end{bmatrix}. \quad (10.29)$$

PROOF First notice that the index can be rewritten as

$$J_{T-1}(\mathbf{u}, \mathbf{x}_0) = \mathbf{x}^T(T)S\mathbf{x}(T) + \sum_{t=0}^{T-1} \begin{bmatrix} \mathbf{u}^T(t) & \mathbf{x}^T(t) \end{bmatrix} \begin{bmatrix} R & 0 \\ 0 & Q \end{bmatrix} \begin{bmatrix} \mathbf{u}(t) \\ \mathbf{x}(t) \end{bmatrix}. \quad (10.30)$$

Consider now the identity

$$0 = \mathbf{x}^T(0)M(0)\mathbf{x}(0) - \mathbf{x}^T(T)M(T)\mathbf{x}(T) + \sum_{t=0}^{T-1} [\mathbf{x}^T(t+1)M(t+1)\mathbf{x}(t+1) - \mathbf{x}^T(t)M(t)\mathbf{x}(t)] \quad (10.31)$$

and replace  $\mathbf{x}(t+1)$  with the expression yielded by the state update equation (10.25). One obtains in this way the new identity

$$0 = \mathbf{x}^T(0)M(0)\mathbf{x}(0) - \mathbf{x}^T(T)M(T)\mathbf{x}(T) + \sum_{t=0}^{T-1} \begin{bmatrix} \mathbf{u}^T(t) & \mathbf{x}^T(t) \end{bmatrix} \begin{bmatrix} G^T M(t+1)G & G^T M(t+1)F \\ F^T M(t+1)G & F^T M(t+1)F - M(t) \end{bmatrix} \begin{bmatrix} \mathbf{u}(t) \\ \mathbf{x}(t) \end{bmatrix} \quad (10.32)$$

and summing left and right hand sides of such an identity and of (10.30), one then obtains (10.29). ■

The matrices  $M(0), M(1), \dots, M(T)$  in the previous lemma can be arbitrarily chosen. This provides a degree of freedom that we exploit in order to determine for (10.29) a particular sequence  $M(0), M(1), \dots, M(T)$  that puts the index in a form that is suitable to solve the minimisation problem. The sequence determination is based on the following lemma.

**Lemma 10.2.2** [RDE: RICCATI DIFFERENCE EQUATION] *Let  $Q \in \mathbb{R}^{n \times n}$  and  $R \in \mathbb{R}^{m \times m}$  be symmetric matrices, positive semidefinite and positive definite, respectively. Given an arbitrary “final” condition  $M(T) \in \mathbb{R}^{n \times n}$  symmetric and positive semidefinite, for any  $t \leq T$  there exist matrices  $\dots, M(T-2), M(T-1), M(T)$  that solve the backward recursive equation*

$$M(t) = Q + F^T M(t+1)F - F^T M(t+1)G[R + G^T M(t+1)G]^{-1}G^T M(t+1)F. \quad (10.33)$$

All matrices  $M(t), t \leq T$ , are symmetric and positive semidefinite.

PROOF The “final” matrix  $M(T)$  is symmetric and positive semidefinite (p.s.d.) by assumption.

We proceed now by induction, namely, we assume that  $M(t+1)$  is symmetric and p.s.d., and we verify that  $M(t)$  exists and is also symmetric and p.s.d.:

- Existence follows from the fact that  $R + G^T M(t+1)G$  is positive definite, and, therefore, invertible.
- Symmetry is obvious.
- It remains to show that  $M(t)$  is p.s.d. To this aim, it suffices to verify that the matrix

$$M(t+1) - M(t+1)G[R + G^T M(t+1)G]^{-1}G^T M(t+1) \quad (10.34)$$

is positive semidefinite. By employing the matrix identity  $(I + XY)^{-1}X = X(I + YX)^{-1}$ , and by adopting for sake of simplicity the notation  $M := M(t+1)$ , one can rewrite (10.34) in the form

$$\begin{aligned} M - MGR^{-1}[I_m + G^T MGR^{-1}]^{-1}G^T M &= M - MGR^{-1}G^T M[I_m + GR^{-1}G^T M]^{-1} \\ &= \left( M[I_m + GR^{-1}G^T M] - MGR^{-1}G^T M \right) [I_m + GR^{-1}G^T M]^{-1} \\ &= M[I_m + GR^{-1}G^T M]^{-1} := Z. \end{aligned} \quad (10.35)$$

For any  $\mathbf{x} \in \mathbb{R}^n$ , it now holds true

$$\begin{aligned} \mathbf{x}^T \{Z + Z^T\} \mathbf{x} &= \mathbf{x}^T \left\{ M[I_m + GR^{-1}G^T M]^{-1} + [I_m + MGR^{-1}G^T]^{-1} M \right\} \mathbf{x} \\ &= \mathbf{x}^T [I_m + MGR^{-1}G^T]^{-1} \left\{ [I_m + MGR^{-1}G^T] M + M[I_m + GR^{-1}G^T M] \right\} [I_m + GR^{-1}G^T M]^{-1} \mathbf{x} \\ &= 2\mathbf{v}^T \left\{ M + MGR^{-1}G^T M \right\} \mathbf{v} = 2\mathbf{v}^T M \mathbf{v} + 2\mathbf{v}^T MGR^{-1}G^T M \mathbf{v} \geq 0, \end{aligned} \quad (10.36)$$

where we put  $\mathbf{v} := [I_m + GR^{-1}G^T M(t+1)]^{-1} \mathbf{x}$ . Therefore, (10.34) is p.s.d. ■

• EXERCISE 10.2.1 Show that

- (i) if  $I + XY$  is invertible, then the same holds true for  $I + YX$ ;
- (ii) the matrix identity  $(I + XY)^{-1}X = X(I + YX)^{-1}$  holds true.

‡ Hint: recall (see Section 6.4) that  $\det(I + YX) = \det(I + XY)$ , and apply the identity  $X(I + YX) = (I + XY)X$ .

Equation (10.33) is called Riccati Difference Equation (RDE). By substituting in (10.29) the matrix sequence defined by the (10.33), we obtain for the index  $J_{T-1}$  an expression which consists of

- two quadratic forms, which depend on the initial and final states of the system
- a sum of definite positive quadratic terms,

as shown in the following lemma.

**Lemma 10.2.3** [SQUARES COMPLETION LEMMA] Let  $M(0), M(1), \dots, M(T) \in \mathbb{R}^{n \times n}$  be the solution to the RDE associated with an arbitrary final symmetric and positive semidefinite condition  $M(T)$ . The index (10.27) can then be expressed in the form

$$J_{T-1}(\mathbf{u}, \mathbf{x}_0) = \mathbf{x}^T(0)M(0)\mathbf{x}(0) + \mathbf{x}^T(T)[S - M(T)]\mathbf{x}(T)$$

$$+ \sum_{t=0}^{T-1} \left( [R + G^T M(t+1)G] \mathbf{u}(t) + G^T M(t+1)F \mathbf{x}(t) \right)^T [R + G^T M(t+1)G]^{-1} \times \\ \times \left( [R + G^T M(t+1)G] \mathbf{u}(t) + G^T M(t+1)F \mathbf{x}(t) \right). \quad (10.37)$$

PROOF In (10.29), and keeping in mind (10.33), let us replace the matrix  $Q + F^T M(t+1)F - M(t)$  with  $F^T M(t+1)G[I_m + G^T M(t+1)G]^{-1}G^T M(t+1)F$ .

The index can be rewritten as

$$J_{T-1}(\mathbf{u}, \mathbf{x}_0) = \mathbf{x}^T(0)M(0)\mathbf{x}(0) + \mathbf{x}^T(T)[S - M(T)]\mathbf{x}(T) + \\ + \sum_{t=0}^{T-1} [\mathbf{u}^T(t) \quad \mathbf{x}^T(t)] \begin{bmatrix} R + G^T M(t+1)G & G^T M(t+1)F \\ F^T M(t+1)G & F^T M(t+1)G[R + G^T M(t+1)G]^{-1}G^T M(t+1)F \end{bmatrix} \begin{bmatrix} \mathbf{u}(t) \\ \mathbf{x}(t) \end{bmatrix} \quad (10.38)$$

Each of the  $(m+n) \times (m+n)$  matrices that appear in the sum in (10.38) can be factorised as follows

$$\begin{bmatrix} R + G^T M(t+1)G \\ F^T M(t+1)G \end{bmatrix} [R + G^T M(t+1)G]^{-1} \begin{bmatrix} R + G^T M(t+1)G & G^T M(t+1)F \end{bmatrix}. \quad (10.39)$$

By substituting (10.39) in (10.38) we obtain the expression (10.37) for  $J_{T-1}$ . ■

If we solve the RDE by choosing  $M(T) = S$  and we recall that the matrices  $R + G^T M(t+1)G$  in (10.37) are positive definite, we can minimise  $J_{T-1}$ , as shown in the next result.

**Proposition 10.2.4** [INDEX MINIMISATION] *Let  $M(0), M(1), \dots, M(T) \in \mathbb{R}^{n \times n}$  be the solution of the RDE associated with the final condition  $M(T) = S$ . Then*

- *the minimum value of the index, when the system has initial condition  $\mathbf{x}(0) = \mathbf{x}_0$ , is*

$$\min_{\mathbf{u}} J_{T-1}(\mathbf{u}, \mathbf{x}_0) = \mathbf{x}_0^T M(0) \mathbf{x}_0; \quad (10.40)$$

- *the optimal input that yields the minimum value of the index is unique, and is given by the state feedback control law*

$$\mathbf{u}_{\text{opt}}(t) = -[R + G^T M(t+1)G]^{-1}G^T M(t+1)F \mathbf{x}(t). \quad (10.41)$$

PROOF By setting  $M(T) = S$  in (10.37), the index becomes the sum of

- the term  $\mathbf{x}_0^T M(0) \mathbf{x}_0$ , which does not depend on the control,
- $T$  non-negative addends, corresponding to the quadratic positive definite forms in the sum, that depend on the control  $\mathbf{u}(\cdot)$  applied to the system.

All these addends vanish if and only if

$$[R + G^T M(t+1)G] \mathbf{u}(t) + G^T M(t+1)F \mathbf{x}(t) = \mathbf{0},$$

holds true, for  $t = 0, 1, \dots, T-1$ , namely if and only if the input has the structure given in (10.41). This choice also immediately yields the minimum value of  $J_{T-1}$ , which is the quadratic form  $\mathbf{x}_0^T M(0) \mathbf{x}_0$ . ■

It is important noticing that (10.41) provides the optimal control law in the form

$$\mathbf{u}_{\text{opt}}(t) = K(t)\mathbf{x}(t), \quad (10.42)$$

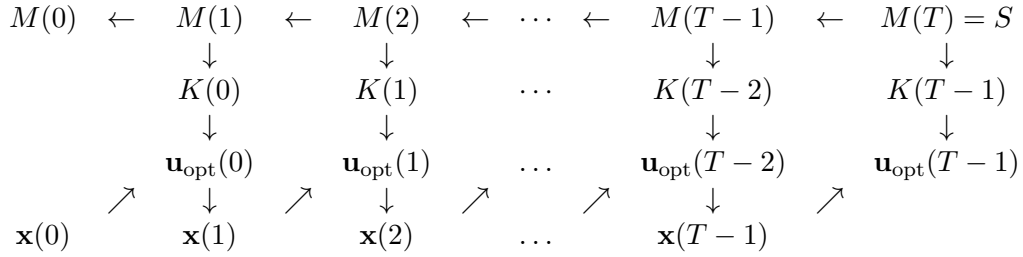
with

$$K(t) = -[R + G^T M(t+1)G]^{-1}G^T M(t+1)F, \quad (10.43)$$

namely as a time-varying state feedback. Once one has computed, setting  $M(T) = S$ ,

- the sequence of matrices  $M(T-1), \dots, M(1), M(0)$ ,
- the sequence of feedback matrices  $K(T-1), K(T-2), \dots, K(0)$ ,

the equation (10.42) and the state update equation with initial state  $\mathbf{x}(0) = \mathbf{x}_0$ , explicitly provide the whole input sequence  $\mathbf{u}_{\text{opt}}(0), \mathbf{u}_{\text{opt}}(1), \dots, \mathbf{u}_{\text{opt}}(T-1)$  that minimises  $J_{T-1}$ .



**Remark** The matrices that solve the RDE only depend on the value chosen as “final” condition and on the number of backwards steps with respect to the time at which the final condition holds, which, in the previous formulas, has been chosen to be the instant  $t = T$ .

If in the RDE we place the “final” condition at time  $t = 0$ , therefore obtaining the sequence  $M(-T), M(1-T), \dots, M(0)$ , then

(i) in Lemma 10.2.3 for any choice of input  $\mathbf{u}(\cdot)$  one has to adopt the notation

$$\begin{aligned} J_{T-1}(\mathbf{u}, \mathbf{x}_0) &= \mathbf{x}^T(0)M(-T)\mathbf{x}(0) + \mathbf{x}^T(T)[S - M(0)]\mathbf{x}(T) \\ &+ \sum_{t=0}^{T-1} \left( [R + G^T M(t+1-T)G]\mathbf{u}(t) + G^T M(t+1-T)F\mathbf{x}(t) \right)^T [R + G^T M(t+1-T)G]^{-1} \\ &\quad \left( [R + G^T M(t+1-T)G]\mathbf{u}(t) + G^T M(t+1-T)F\mathbf{x}(t) \right). \end{aligned} \quad (10.44)$$

(ii) In Proposition 10.2.4 one assumes  $M(0) = S$ , and the minimum value of the index with initial condition  $\mathbf{x}(0) = \mathbf{x}_0$  is expressed by

$$\min_{\mathbf{u}} J_{T-1}(\mathbf{u}, \mathbf{x}_0) = \mathbf{x}_0^T M(-T)\mathbf{x}_0; \quad (10.45)$$

(iii) the control law that minimises the index is given by

$$\mathbf{u}_{\text{opt}}(t) = -[R + G^T M(t+1-T)G]^{-1}G^T M(t+1-T)F\mathbf{x}(t). \quad (10.46)$$

While solving backwards the RDE assuming as final instant  $t = 0$  might seem just a change of notation, it often turns out to be more convenient than setting as final instant  $t = T$ , for reasons that will be evident in the rest of the chapter.

- EXERCISE 10.2.2 Without employing the minimisation procedure described above, verify that the minimum value of the index (10.27) associated to the system (10.25) exists, namely that, once the initial state  $\mathbf{x}_0$  is assigned, the infimum of the index value over the set of input functions is indeed attained.

‡ *Hint.* By employing an orthogonal matrix  $N$ , rewrite the matrix  $R$  in the form  $R = N^T \text{diag}\{\lambda_1, \dots, \lambda_m\}N$ , and let  $\lambda_\mu > 0$  be the smallest among the eigenvalues of  $R$ . Verify that

- for any  $\mathbf{v} \in \mathbb{R}^m$  with unitary norm,  $\mathbf{v}^T R \mathbf{v} = \mathbf{v}^T N^T \text{diag}\{\lambda_1, \lambda_1, \dots, \lambda_m\}N \mathbf{v} \geq \lambda_\mu$ ;
- if the norm of at least one among the input samples  $\mathbf{u}(0), \mathbf{u}(1), \dots, \mathbf{u}(T-1)$  is not smaller than 1, then the cost  $\sum_{t=0}^{T-1} \mathbf{u}^T(t) R \mathbf{u}(t)$  is not smaller than  $\lambda_\mu$ ;
- if  $\xi = J_{T-1}(\mathbf{0}, \mathbf{x}_0)$  is the cost of the unforced evolution of the system,  $J_{T-1}(\mathbf{u}(\cdot), \mathbf{x}_0)$  is not greater than  $\xi$  only if the input  $\mathbf{u}(\cdot)$  belongs to the compact set  $\mathcal{C} := \{\mathbf{u}(\cdot) \in \mathbb{R}^{mT} : \|\mathbf{u}(t)\| \leq \xi/\lambda_\mu, t = 0, 1, \dots, T-1\}$ , over which the continuous function  $J_{T-1} : \mathcal{C} \rightarrow \mathbb{R} : \mathbf{u}(\cdot) \mapsto J_{T-1}(\mathbf{u}(\cdot), \mathbf{x}_0)$  attains a minimum;
- the minimum cost when  $\mathbf{u}(\cdot)$  varies in  $\mathcal{C}$  coincides with the minimum cost when  $\mathbf{u}(\cdot)$  varies in  $\mathbb{R}^{mT}$ .

### 10.2.3 Complementary topics

The results that we obtained on the structure of the optimal control can be clarified and deepened by employing Bellman's optimality principle, which we now state and prove. Notice that the formulation remains valid under very broad conditions, such as, non-linear systems and non-quadratic indices.

**Proposition 10.2.5** [BELLMAN OPTIMALITY PRINCIPLE] *Let  $\mathbf{u}_{\text{opt}}(\cdot)$  be the optimal control corresponding to an arbitrary initial state  $\mathbf{x}(0) = \mathbf{x}_0$  and an arbitrary (and possibly infinite) time horizon  $[0, T]$ . If  $\tau$  is an arbitrary time between 0 and  $T$ , and  $\mathbf{x}(\tau) = \mathbf{x}_\tau$  is the state reached by the system by means of such a control at time  $t = \tau$ , then  $\mathbf{u}_{\text{opt}}(\cdot)$  also corresponds to the optimal control for the initial state  $\mathbf{x}(\tau) = \mathbf{x}_\tau$  over the interval  $[\tau, T]$ .*

PROOF The justification of the principle is rather straightforward. Recalling (10.25) and (10.27), the value of the index  $J_{T-1}(\mathbf{u}(\cdot), \mathbf{x}_0)$  over  $[0, T]$  is the sum of

$$\sum_{t=0}^{\tau-1} [\mathbf{x}^T(t) Q \mathbf{x}(t) + \mathbf{u}^T(t) R \mathbf{u}(t)]$$

and of the index value

$$J_{T-\tau-1}(\mathbf{u}(\cdot), \mathbf{x}_\tau) = \sum_{t=\tau}^{T-1} [\mathbf{x}^T(t) Q \mathbf{x}(t) + \mathbf{u}^T(t) R \mathbf{u}(t)] + \mathbf{x}^T(T) S \mathbf{x}(T).$$

By contradiction, if there were another input  $\bar{\mathbf{u}}(\cdot)$  that, over the interval  $[\tau, T]$ , yields a value of the objective function smaller than the one obtained using  $\mathbf{u}_{\text{opt}}(\cdot)$ , it would suffice to consider the control

$$\mathbf{u}(t) = \begin{cases} \mathbf{u}_{\text{opt}}(t) & \text{for } t = 0, 1, \dots, \tau-1 \\ \bar{\mathbf{u}}(t) & \text{for } t \geq \tau \end{cases} \quad (10.47)$$

in order to obtain, over the interval  $[0, T]$ , and index value that is smaller than the one obtained using  $\mathbf{u}_{\text{opt}}(\cdot)$ , which contradicts the definition of optimal control. ■

It follows from Bellman's principle that we can reduce the index minimisation over the whole interval  $[0, T]$  to a sequence of single step minimisations:

$$\min_{\mathbf{u}(\cdot)} J_{T-1}(\mathbf{u}(\cdot), \mathbf{x}_0) = \min_{\mathbf{u}(0)} \left\{ \mathbf{u}^T(0) R \mathbf{u}(0) + \mathbf{x}_0^T Q \mathbf{x}_0 + \min_{\mathbf{u}(\cdot)} J_{T-2}(\mathbf{u}(\cdot), \mathbf{x}(1)) \right\}. \quad (10.48)$$

If we assume that the minimum value of the index over an arbitrary interval  $[0, T]$  and for an arbitrary initial state  $\mathbf{x}_0$  has the following quadratic structure

$$\min_{\mathbf{u}(\cdot)} J_{T-1}(\mathbf{u}(\cdot), \mathbf{x}_0) = \mathbf{x}_0^T M(-T) \mathbf{x}_0,$$

with  $M(-T)$  symmetric and positive definite, then

$$\begin{aligned} \min_{\mathbf{u}(\cdot)} J_{T-1}(\mathbf{u}(\cdot), \mathbf{x}_0) &= \mathbf{x}_0^T M(-T) \mathbf{x}_0 \\ &= \min_{\mathbf{u}(0)} \left\{ \mathbf{u}^T(0) R \mathbf{u}(0) + \mathbf{x}_0^T Q \mathbf{x}_0 + \mathbf{x}^T(1) M(-T+1) \mathbf{x}(1) \right\} \\ &= \min_{\mathbf{u}(0)} \left\{ \mathbf{u}^T(0) R \mathbf{u}(0) + \mathbf{x}_0^T Q \mathbf{x}_0 + [F \mathbf{x}_0 + G \mathbf{u}(0)]^T M(-T+1) [F \mathbf{x}_0 + G \mathbf{u}(0)] \right\} \\ &= \min_{\mathbf{u}(0)} \left[ \mathbf{u}^T(0) \quad \mathbf{x}_0^T \right] \left\{ \begin{bmatrix} R & 0 \\ 0 & Q \end{bmatrix} + \begin{bmatrix} G^T \\ F^T \end{bmatrix} M(-T+1) \begin{bmatrix} G & F \end{bmatrix} \right\} \begin{bmatrix} \mathbf{u}(0) \\ \mathbf{x}_0 \end{bmatrix} \end{aligned} \quad (10.49)$$

namely, upon setting

$$A := R + G^T M(-T+1)G, \quad B := G^T M(-T+1)F, \quad C := Q + F^T M(-T+1)F, \quad (10.50)$$

then

$$\mathbf{x}_0^T M(-T) \mathbf{x}_0 = \min_{\mathbf{u}(0)} [\mathbf{u}(0)^T A \mathbf{u}(0) + \mathbf{x}_0^T B^T \mathbf{u}(0) + \mathbf{u}(0)^T B \mathbf{x}_0 + \mathbf{x}_0^T C \mathbf{x}_0] \quad (10.51)$$

We now look for a vector  $\mathbf{u}^* \in \mathbb{R}^m$  that allows us to rewrite the expression in the square brackets in (10.51) in the form

$$[\mathbf{u}(0) - \mathbf{u}^*]^T A [\mathbf{u}(0) - \mathbf{u}^*] - (\mathbf{u}^*)^T A \mathbf{u}^* + \mathbf{x}_0^T C \mathbf{x}_0. \quad (10.52)$$

Once we determine  $\mathbf{u}^*$ , the problem is solved, as (10.52) can be minimised with respect to  $\mathbf{u}(0)$  simply choosing  $\mathbf{u}(0) = \mathbf{u}^*$ , since  $A$  is positive definite. To compute such a  $\mathbf{u}^*$ , we equate (10.51) and (10.52), thus obtaining  $\mathbf{u}(0)^T [A \mathbf{u}^* + B \mathbf{x}_0] = 0$ . This equation holds for any value of  $\mathbf{u}(0)$  if

$$\mathbf{0} = A \mathbf{u}^* + B \mathbf{x}_0 = [R + G^T M(-T+1)G] \mathbf{u}^* + G^T M(-T+1)F \mathbf{x}_0,$$

which yields  $\mathbf{u}^*$

$$\mathbf{u}^* = -[R + G^T M(-T+1)G]^{-1} G^T M(-T+1)F \mathbf{x}_0. \quad (10.53)$$

Substituting now (10.53) in (10.52) and then choosing  $\mathbf{u}(0) = \mathbf{u}^*$ , we have

$$\min_{\mathbf{u}(\cdot)} J_{T-1}(\mathbf{u}(\cdot), \mathbf{x}_0) = \mathbf{x}_0^T M(-T) \mathbf{x}_0$$

$$\begin{aligned}
&= -(\mathbf{u}^*)^T A \mathbf{u}^* + \mathbf{x}_0^T C \mathbf{x}_0 \\
&= -\mathbf{x}_0 F^T M(-T+1) G [R + G^T M(-T+1) G]^{-1} G^T M(-T+1) F \mathbf{x}_0 \\
&+ \mathbf{x}_0^T [Q + F^T M(-T+1) F] \mathbf{x}_0
\end{aligned} \tag{10.54}$$

which establishes that the matrices  $M(\cdot)$  solve the RDE.

By considering (11.125), in particular, for  $T = 1$ , it is clear that the minimum value of  $J_0(\mathbf{u}(\cdot), \mathbf{x}_0)$  is

$$\begin{aligned}
\mathbf{x}_0^T M(-1) \mathbf{x}_0 &= \min_{\mathbf{u}(0)} \{ \mathbf{u}^T(0) R \mathbf{u}(0) + \mathbf{x}_0^T Q \mathbf{x}_0 + \mathbf{x}^T(1) M(0) \mathbf{x}(1) \} \\
&= \min_{\mathbf{u}(\cdot)} J_0(\mathbf{u}(\cdot), \mathbf{x}_0) \\
&= \min_{\mathbf{u}(0)} \{ \mathbf{u}^T(0) R \mathbf{u}(0) + \mathbf{x}_0^T Q \mathbf{x}_0 + \mathbf{x}^T(1) S \mathbf{x}(1) \},
\end{aligned} \tag{10.55}$$

and by comparing now (11.125) and (10.55), we conclude that it must hold  $M(0) = S$  in the RDE.

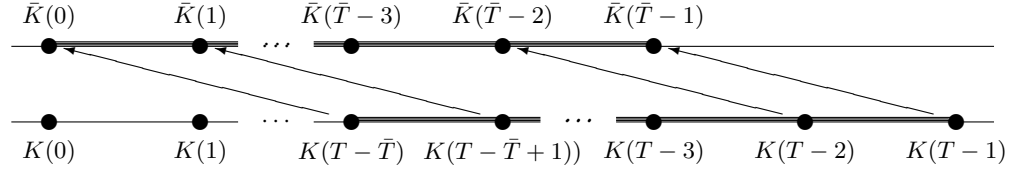
**Remark** [OPTIMAL CONTROL OVER A TIME SUBINTERVAL] We have seen that the feedback matrices  $K(t)$ , which provide the optimal control law  $\mathbf{u}_{\text{opt}}(t) = K(t)\mathbf{x}(t)$  at time  $t$ , only depend on the difference  $T - t$  between the final and current times, once we are given a) the matrices  $R$ ,  $Q$ , and b) the cost associated with the gap between the final state and  $\mathbf{0}$ , expressed via the weighting matrix  $S$ .

Consequently, the solution to the optimal problem over the horizon  $[0, T]$  can be used to solve the same problem over any subinterval  $[0, \bar{T}]$ , with  $\bar{T} < T$ . In order to compare the two cases, it is convenient to solve the Riccati Difference Equation setting the final condition  $M(0) = S$  at time 0. To minimise the index

$$J_{\bar{T}-1}(\mathbf{u}, \mathbf{x}_0) = \sum_{t=0}^{\bar{T}-1} [\mathbf{x}^T(t) Q \mathbf{x}(t) + \mathbf{u}^T(t) R \mathbf{u}(t)] + \mathbf{x}^T(\bar{T}) S \mathbf{x}(\bar{T}), \tag{10.56}$$

one employs the matrices of the sequence  $\dots, M(-2), M(-1), M(0)$ , until the term  $M(-\bar{T})$ , while for the minimisation problem over the whole  $[0, T]$ , one carries on until the term  $M(-T)$ . The corresponding feedback matrices, which provide the optimal control law over the time horizon  $[0, \bar{T}]$ , are

$$\begin{aligned}
\bar{K}(\bar{T}-1) &= -[R + G^T M(0) G]^{-1} G^T M(0) F &= K(T-1) \\
\bar{K}(\bar{T}-2) &= -[R + G^T M(-1) G]^{-1} G^T M(-1) F &= K(T-2) \\
&\dots &\dots \\
\bar{K}(1) &= -[R + G^T M(-\bar{T}+2) G]^{-1} G^T M(-\bar{T}+2) F &= K(T-\bar{T}+1) \\
\bar{K}(0) &= -[R + G^T M(-\bar{T}+1) G]^{-1} G^T M(-\bar{T}+1) F &= K(T-\bar{T})
\end{aligned}$$



**Remark 2** [OPEN LOOP SOLUTION] An alternative and conceptually very simple method to solve the optimal control problem consists in writing the states  $\mathbf{x}(1), \mathbf{x}(2), \dots, \mathbf{x}(T)$  as functions of the initial state  $\mathbf{x}(0) = \mathbf{x}_0$ , which we assume to be given, and of the unknown input values  $\mathbf{u}(0), \mathbf{u}(1), \dots, \mathbf{u}(T-1)$

$$\begin{aligned} \mathbf{x}(1) &= F\mathbf{x}(0) + G\mathbf{u}(0) \\ \mathbf{x}(2) &= F\mathbf{x}(1) + G\mathbf{u}(1) = F^2\mathbf{x}(0) + FG\mathbf{u}(0) + G\mathbf{u}(1) \\ &\dots \quad \dots \end{aligned} \tag{10.57}$$

By replacing the (10.57) in the index expression, the latter becomes a function of the  $mT$  scalar variables that constitute the input sequence. By equating to zero the partial derivatives of  $J_{T-1}$  with respect to such variables, one can then determine the sequence that minimises the index.

While conceptually very simple, such a procedure has several drawbacks. First of all, it provides an open loop control sequence uniquely based on the initial condition, which does not consider uncertainties and noises that could perturb the system state during the interval  $[0, T]$ , after which the control sequence would lose the optimality property. On the contrary, the feedback control law (10.41) still provides, after the time  $T'$  at which the perturbation is applied to the system, the optimal input for the new dynamics in  $[T', T]$ . Furthermore, determining the optimal input by setting to zero the derivatives of  $J_{T-1}$  is computationally heavy, except when  $mT$  is very small, and, in addition, has to be repeated for every choice of the initial state.

**Example 10.2.1** Consider the one dimensional system, with scalar input

$$x(t+1) = x(t) + u(t), \quad x(0) = x_0 = 1, \tag{10.58}$$

and the problem of determining the inputs  $u(0)$  e  $u(1)$  that minimise the index

$$J_1(u, x_0) = \sum_{t=0}^1 [u^2(t) + x^2(t)] + 2x(2)^2. \tag{10.59}$$

The RDE to be solved is

$$M(t) = 1 + M(t+1) - M(t+1)^2[1 + M(t+1)]^{-1}$$

with final condition  $M(2) = S = 2$ .

We obtain  $M(1) = 5/3$ ,  $M(0) = 13/8$ , therefore

$$\begin{aligned} K(0) &= -[1 + M(1)]^{-1}M(1) = -5/8 \\ K(1) &= -[1 + M(2)]^{-1}M(2) = -2/3 \end{aligned}$$

and the values of  $u(\cdot)$  that minimise the index are

$$u_{\text{opt}}(0) = K(0)x(0) = -5/8$$



$$u_{\text{opt}}(1) = K(1)x(1) = K(1)(x(0) + u_{\text{opt}}(0)) = -1/4. \quad (10.60)$$

The corresponding minimum value of the index is thus

$$J_1(u_{\text{opt}}, x_0) = M(0)x_0^2 = 13/8. \quad (10.61)$$

The same result can be obtained by substituting in the index the expressions for  $x(1)$  and  $x(2)$

$$\begin{aligned} J_1(u, x_0) &= x_0^2 + u(0)^2 + x(1)^2 + u(1)^2 + 2x(2)^2 \\ &= x_0^2 + u(0)^2 + [x_0 + u(0)]^2 + u(1)^2 + 2[x_0 + u(0) + u(1)]^2 \\ &= 1 + u(0)^2 + [1 + u(0)]^2 + u(1)^2 + 2[1 + u(0) + u(1)]^2 \end{aligned} \quad (10.62)$$

and computing the values of  $u(0)$  and  $u(1)$  that solve the system of equations

$$\frac{\partial J_1}{\partial u(0)} = 0, \quad \frac{\partial J_1}{\partial u(1)} = 0.$$

### 10.3 Infinite horizon optimal control of linear discrete time systems

In this chapter we consider the optimal control problem for the linear discrete time system (10.25) - (10.26) when  $T \rightarrow +\infty$ . In this case, and under certain assumptions, the optimal control law turns out to be in the form of a time-invariant linear state feedback law. This results in two advantages, namely, an easier computation process, as one has only to determine a feedback matrix with time-invariant entries, and a simplified implementation of the control law.

It is worth noticing that this type of time-invariant feedback control can also be useful when dealing with control problems over a finite horizon, if the control horizon is long enough so that the matrix feedback  $K(t)$ , solution of the finite horizon optimal control problem, converges to its steady state for most of the control horizon itself.

#### 10.3.1 Preliminary considerations

When we let  $T$  increase to infinity, the main problems we have to deal with, and that do not emerge in the case of finite horizon control, can be outlined as follows:

- i) we must guarantee that, for any initial state, there exists at least one input sequence for which the index  $J$  has a finite value, as otherwise the problem of minimising  $J$  is meaningless;
- ii) we shall show that, if an optimal control exists, it can be implemented as a linear and time-invariant state feedback. It is thus crucial to determine the conditions under which the resulting feedback system is internally stable;
- iii) the feedback matrix can be computed from the solution to a non-linear matrix equation (Algebraic Riccati Equation or ARE), which can be formally obtained by assuming in the RDE  $M(t+1) = M(t) := M$ . In general the Algebraic Riccati Equation admits more than one solution, and only one of them yields the optimal control law. The problem is thus to characterize the “optimal” solution, namely, the one that is associated with the optimal control law.

The following examples are meant to illustrate these remarks.

**Example 10.3.1** [INFINITE INDEX FOR ANY INPUT] Consider the system

$$\mathbf{x}(t+1) = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t), \quad \mathbf{x}(0) = \mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad (10.63)$$

and the index

$$J(u, \mathbf{x}_0) = \sum_{t=0}^{+\infty} \left( \mathbf{x}^T(t) Q \mathbf{x}(t) + u^2(t) \right), \quad Q = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}. \quad (10.64)$$

The state at time  $t$  corresponding to an input sequence  $u(\cdot)$  is  $\mathbf{x}(t) = \begin{bmatrix} 2^t \\ x_2(t) \end{bmatrix}$ , so for the index it holds true

$$J(u, \mathbf{x}_0) = \sum_{t=0}^{+\infty} \left( 2^{2t} + u^2(t) \right) = +\infty. \quad (10.65)$$

Notice that  $J$  diverges independent of the applied input sequence, which is the consequence of the fact that the following three simultaneous facts hold:

1. the initial state  $\mathbf{x}(0)$  is not controllable to zero. If it were so, there would exist at least one input sequence steering the system to zero in a finite time, therefore yielding a finite value for the index;
2. the unforced evolution from the initial state  $\mathbf{x}(0)$  contains the diverging mode  $\sum_{t=0}^{\infty} 2^t z^{-t}$  (equivalently,  $2^t, t \in \mathbb{Z}$ );
3. due to the structure of the matrix  $Q$ , such a diverging mode appears in the product  $\mathbf{x}^T(t) Q \mathbf{x}(t)$ , and thus contributes to the index value.

It will be shown that if in an optimal control problem at least one of such three conditions does not hold true, then there exist input sequences for which  $J$  takes a finite value.

**Example 10.3.2** [NON-STABILIZING OPTIMAL CONTROL] Consider the system

$$\mathbf{x}(t+1) = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u(t), \quad \mathbf{x}(0) = \mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad (10.66)$$

and the index

$$J(u, \mathbf{x}_0) = \sum_{t=0}^{+\infty} \left( [x_1(t) \ x_2(t)] \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + u^2(t) \right),$$

as in (10.64).

In this case the system is reachable, therefore for any initial state there exist inputs for which  $J$  takes a finite value. In particular, if the initial state is  $\mathbf{x}(0) = \mathbf{x}_0 = \mathbf{e}_1$ , the input  $u(0) = 4, u(1) = -12, u(2) = u(3) = \dots = 0$  steers (and holds) the system in the zero state in two steps.

We observe that minimising over the input sequence  $u(\cdot)$  the index

$$J(u, \mathbf{x}_0) = \sum_{t=0}^{+\infty} \left( x_1^2(t) + u^2(t) \right) \quad (10.67)$$

subject to

$$\begin{aligned} x_1(t+1) &= 2x_1(t) + u(t) & x_1(0) &= 1 \\ x_2(t+1) &= 3x_2(t) + u(t) & x_2(0) &= 0 \end{aligned} \quad (10.68)$$

is equivalent to minimising the index (10.67) only subject to the condition

$$x_1(t+1) = 2x_1(t) + u(t) \quad x_1(0) = 1, \quad (10.69)$$

since the component  $x_2(t)$  does not appear in the index (10.67) and does not influence the dynamics of  $x_1(t)$ . The problem thus reduces to an optimal control problem for the unidimensional system

(10.69) and for the index (10.67). To solve it, we employ a heuristic procedure, in particular assuming that an optimal input exists and has the structure of a time-invariant feedback from the state<sup>3</sup>

$$u(t) = kx_1(t). \quad (10.70)$$

In this case, being the model very simple, it is easy to determine the expression of  $J$  as a function of the feedback gain  $k$  and the optimal input  $u_{\text{opt}}(t) = \bar{k}x_1(t)$  by minimising  $J$  with respect to  $k$ . It is worth noticing, however, that the basic procedure that we employ is impractical for any slightly more complex model.

By making use of (10.70), the expression (10.67) can be rewritten as

$$J(u, \mathbf{x}_0) = \sum_{t=0}^{+\infty} (x_1^2(t) + k^2 x_1^2(t)) = (1+k^2) \sum_{t=0}^{+\infty} [(2+k)^t x_1(0)]^2 = (1+k^2) x_1^2(0) \sum_{t=0}^{+\infty} (2+k)^{2t}. \quad (10.71)$$

When  $|2+k| < 1$ , namely when  $k$  lies between  $-3$  and  $-1$ , the sequence in (10.71) converges and the value of the index is

$$J = \frac{k^2 + 1}{1 - (2+k)^2} x_1^2(0), \quad (10.72)$$

while for any other value of  $k$  the index does not converge. Therefore, the value  $\bar{k}$  of  $k$  for which the index is minimum must lie within  $-3$  e  $-1$ , and can be computed by solving the equation

$$0 = \frac{dJ}{dk} = -4x_1^2(0) \frac{k^2 + k - 1}{[1 - (2+k)^2]^2}$$

which has two solutions,  $\frac{-1 \pm \sqrt{5}}{2}$ . However, the only one that belongs to the admissible interval  $(-3, -1)$  is  $\bar{k} = \frac{-1 - \sqrt{5}}{2}$ , therefore this is the solution we are looking for. The optimal control feedback law for the second order system (10.66) must provide the same input  $u_{\text{opt}}(\cdot)$  that we have already computed, and in particular the second component of the state,  $x_2$ , cannot perturb the inputs. We have thus

$$K = [\bar{k} \quad 0] = \left[ \frac{-1 - \sqrt{5}}{2} \quad 0 \right] \quad (10.73)$$

and the closed loop state matrix becomes

$$F + \mathbf{g}K = \begin{bmatrix} \frac{3 - \sqrt{5}}{2} & 0 \\ \frac{-1 - \sqrt{5}}{2} & 3 \end{bmatrix}. \quad (10.74)$$

In this case we could compute an optimal control law, but the resulting closed loop system would be unstable.

### 10.3.2 Problem solution and Algebraic Riccati Equation (ARE)

Having set the foundations for our analysis, we aim now to solve the optimal problem over infinite horizon for the linear system (10.25) with arbitrary initial state

$$\mathbf{x}(t+1) = F\mathbf{x}(t) + G\mathbf{u}(t), \quad \mathbf{x}(0) = \mathbf{x}_0,$$

and quadratic index

$$J(\mathbf{u}, \mathbf{x}_0) = \sum_{t=0}^{+\infty} (\mathbf{u}^T(t) R \mathbf{u}(t) + \mathbf{x}^T(t) Q \mathbf{x}(t)), \quad (10.75)$$

---

<sup>3</sup>That an optimal control exists and has the structure (10.70) will be established by the theory presented later in this chapter.

where  $R$  is symmetric positive definite and  $Q$  is symmetric positive semidefinite.

As a first step, we verify that if the system (10.25) is stabilisable, then the optimal control problem is well posed, namely, for any  $\mathbf{x}_0$  there exist inputs that yield a finite value for the index. We then show that the optimal control exists, that its structure is a state feedback (with time-invariant matrix) and that the feedback matrix can be computed by solving a non-linear equation known as Algebraic Riccati Equation (ARE)<sup>4</sup>.

**Lemma 10.3.1** [STABILISABILITY AND FINITENESS OF THE INDEX] *If the system (10.25) is stabilisable and  $K$  is a stabilising feedback matrix, then, for any initial state  $\mathbf{x}(0) = \mathbf{x}_0$ ,*

- *the value of the index (10.75) which corresponds to the input  $\bar{\mathbf{u}}(\cdot) = K\mathbf{x}(\cdot)$  is finite;*
- *the Lyapunov equation*

$$(F + GK)^T X (F + GK) - X = -(K^T R K + Q) \quad (10.76)$$

*admits a unique solution  $P$ , symmetric and positive semidefinite, and it holds true that*

$$J(\bar{\mathbf{u}}, \mathbf{x}_0) = \mathbf{x}_0^T P \mathbf{x}_0. \quad (10.77)$$

PROOF By replacing the input expression with  $\bar{\mathbf{u}}(\cdot) = K\mathbf{x}(\cdot)$  in (10.75), we obtain

$$\begin{aligned} J(\bar{\mathbf{u}}, \mathbf{x}_0) &= \sum_{t=0}^{+\infty} \mathbf{x}^T(t) Q \mathbf{x}(t) + \bar{\mathbf{u}}^T(t) R \bar{\mathbf{u}}(t) \\ &= \sum_{t=0}^{+\infty} \mathbf{x}^T(t) Q \mathbf{x}(t) + \mathbf{x}^T(t) K^T R K \mathbf{x}(t) = \sum_{t=0}^{+\infty} \mathbf{x}^T(t) (Q + K^T R K) \mathbf{x}(t) \end{aligned}$$

namely, writing  $\mathbf{x}(t) = (F + GK)^t \mathbf{x}_0$ , we obtain

$$J(\bar{\mathbf{u}}, \mathbf{x}_0) = \mathbf{x}_0^T \sum_{t=0}^{+\infty} ((F + GK)^T)^t (Q + K^T R K) (F + GK)^t \mathbf{x}_0. \quad (10.78)$$

Since the eigenvalues of  $F + GK$  have modulus strictly smaller than 1, by reasoning along the same lines as in [Section 3.6](#) one proves that the sum in (10.78) converges to a symmetric and positive semidefinite matrix  $P$ , which is the unique solution to the equation (10.76). Therefore,  $J$  takes the finite value  $\mathbf{x}_0^T P \mathbf{x}_0$  when the input to the system is  $\bar{\mathbf{u}}(\cdot)$ . ■

EXERCISE 10.3.1 Show that

- (i) (10.76) admits a unique solution, which is symmetric and positive semidefinite;
- (ii) if  $\mathbf{x}_0^T P \mathbf{x}_0 = 0$ , then the feedback input  $\bar{\mathbf{u}}(\cdot)$  is identically zero.

<sup>4</sup>To distinguish between the version for discrete-time systems and the one for continuous-time systems, we will encounter later on, the previous equation is often referred to as DARE.

**Lemma 10.3.2** [LIMIT SOLUTION TO RDE AND ALGEBRAIC RICCATI EQUATION (ARE)]  
 Let (10.25) be a stabilisable system, for which  $K$  is a stabilising feedback matrix and  $P$  is the solution to the Lyapunov equation (10.76). Consider the matrix sequence

$$\dots, M(-2), M(-1), M(0) \quad (10.79)$$

which is obtained by solving the RDE (10.33) starting from the “final” condition  $M(0) = 0$ .

- i) For any  $t \geq 0$  the matrix  $M(-t-1) - M(-t)$  is positive semidefinite, namely, the sequence (10.79) is monotonically non decreasing when  $-t$  decreases.
- ii) For any  $t \geq 0$ , the matrix  $P - M(-t)$  is positive semidefinite, namely the sequence  $M(-t)$  is upper bounded by  $P$ .
- iii) The limit of the sequence (10.79)

$$\lim_{t \rightarrow \infty} M(-t) =: M_{\infty}. \quad (10.80)$$

exists, is finite, and is a symmetric and p.s.d. solution to the ARE

$$M = Q + F^T M F - F^T M G [R + G^T M G]^{-1} G^T M F. \quad (10.81)$$

PROOF (i) Consider the indices

$$\begin{aligned} J_{T-1}(\mathbf{u}, \mathbf{x}_0) &= \sum_{t=0}^{T-1} [\mathbf{u}^T(t) R \mathbf{u}(t) + \mathbf{x}^T(t) Q \mathbf{x}(t)], \\ J_T(\mathbf{u}, \mathbf{x}_0) &= \sum_{t=0}^T [\mathbf{u}^T(t) R \mathbf{u}(t) + \mathbf{x}^T(t) Q \mathbf{x}(t)]. \end{aligned}$$

We start by verifying that  $\min_{\mathbf{u}} J_T(\mathbf{u}, \mathbf{x}_0) \geq \min_{\mathbf{u}} J_{T-1}(\mathbf{u}, \mathbf{x}_0)$ . In fact, if  $\tilde{\mathbf{u}}(\cdot)$  is the sequence that minimises  $J_T$  for the system with initial state  $\mathbf{x}_0$ , it holds true

$$\begin{aligned} \min_{\mathbf{u}} J_T(\mathbf{u}, \mathbf{x}_0) &= \sum_{t=0}^{T-1} \left( \tilde{\mathbf{u}}^T(t) R \tilde{\mathbf{u}}(t) + \mathbf{x}^T(t) Q \mathbf{x}(t) \right) + \tilde{\mathbf{u}}^T(T) R \tilde{\mathbf{u}}(T) + \mathbf{x}^T(T) Q \mathbf{x}(T) \\ &\geq \sum_{t=0}^{T-1} \left( \tilde{\mathbf{u}}^T(t) R \tilde{\mathbf{u}}(t) + \mathbf{x}^T(t) Q \mathbf{x}(t) \right) \geq \min_{\mathbf{u}} J_{T-1}(\mathbf{u}, \mathbf{x}_0). \end{aligned} \quad (10.82)$$

Now, from the equalities

$$\min_{\mathbf{u}} J_T(\mathbf{u}, \mathbf{x}_0) = \mathbf{x}_0^T M(-T-1) \mathbf{x}_0, \quad \min_{\mathbf{u}} J_{T-1}(\mathbf{u}, \mathbf{x}_0) = \mathbf{x}_0^T M(-T) \mathbf{x}_0$$

and from (10.82) it follows that

$$\mathbf{x}_0^T \left( M(-T-1) - M(-T) \right) \mathbf{x}_0 \geq 0$$

and, since  $\mathbf{x}_0$  arbitrary, this establishes that  $M(-T-1) - M(-T)$  is p.s.d. for any  $T \geq 0$ .

(ii) Consider the stabilising matrix  $K$  and the control law  $\bar{\mathbf{u}}(t) = K\mathbf{x}(t)$ . By Lemma 10.3.1, the value of the index  $J(\bar{\mathbf{u}}, \mathbf{x}_0)$  relative to such an input is  $\mathbf{x}_0^T P \mathbf{x}_0$ , where  $P$  is the p.s.d. solution to the Lyapunov equation (10.76), and moreover

$$\begin{aligned} \mathbf{x}_0^T P \mathbf{x}_0 &= J(\bar{\mathbf{u}}, \mathbf{x}_0) = \sum_{t=0}^{+\infty} \left( \bar{\mathbf{u}}^T(t) R \bar{\mathbf{u}}(t) + \mathbf{x}^T(t) Q \mathbf{x}(t) \right) \geq \sum_{t=0}^{T-1} \left( \bar{\mathbf{u}}^T(t) R \bar{\mathbf{u}}(t) + \mathbf{x}^T(t) Q \mathbf{x}(t) \right) \\ &\geq \min_{\mathbf{u}} J_{T-1}(\mathbf{u}, \mathbf{x}_0) = \mathbf{x}_0^T M(-T) \mathbf{x}_0 \end{aligned} \quad (10.83)$$

The first inequality establishes that, for a given input function  $\bar{\mathbf{u}}(\cdot)$ , the value of the index computed over  $[0, +\infty]$  is not smaller than the value computed over  $[0, T]$ , while the second follows from the possible non-optimality of  $\bar{\mathbf{u}}(\cdot)$  over the time horizon  $[0, T]$ .

Since for any  $\mathbf{x}_0$  and for any  $T \geq 0$ , it holds true

$$\mathbf{x}_0^T (P - M(-T)) \mathbf{x}_0 \geq 0$$

then  $P - M(-T)$  is p.s.d. for any  $T \geq 0$ .

(iii) The criterion that guarantees the existence and finiteness of the limit of a non-decreasing and upper bounded scalar sequence can be generalised to the matrix case (see exercise 10.3.2). Consequently, the limit of the sequence of symmetric matrices  $M(-t)$  exists, and it is finite and symmetric, as  $t \rightarrow +\infty$ . Denote by  $M_\infty$  such a limit.

We claim that  $M_\infty$  satisfies the ARE (10.81). This can be proven simply by rewriting the RDE (10.33) as

$$M(-t-1) = Q + F^T M(-t) F - F^T M(-t) G [R + G^T M(-t) G]^{-1} G^T M(-t) F, \quad (10.84)$$

and letting  $t$  grow unbounded on both sides of it. Since  $\mathbf{x}_0^T M(-t) \mathbf{x}_0$  is non-negative for any  $t \geq 0$  and for any  $\mathbf{x}_0$ , by moving to the limit, one obtains

$$\mathbf{x}_0^T M_\infty \mathbf{x}_0 = \lim_{t \rightarrow +\infty} \mathbf{x}_0^T M(-t) \mathbf{x}_0 \geq 0$$

therefore  $M_\infty$  is positive semidefinite. ■

- EXERCISE 10.3.2 Prove in a thorough way point (iii) of Lemma 10.3.2.

‡ *Hint: consider, for  $i = 1, 2, \dots, n$ , the products  $\mathbf{e}_i^T M(-t) \mathbf{e}_i = m_{ii}(-t)$ ,  $t \geq 0$  and the product  $\mathbf{e}_i^T P \mathbf{e}_i = p_{ii}$ . Verify that  $m_{ii}(-t-1) - m_{ii}(-t) \geq 0$ ,  $\forall t \geq 0$  and  $p_{ii} \geq m_{ii}(-t)$ ,  $\forall t \geq 0$ , and infer that the limit  $\lim_{t \rightarrow \infty} m_{ii}(-t) := [M_\infty]_{ii}$  exists finite.*

*Then, for  $i \neq j$ , consider the products  $(\mathbf{e}_i + \mathbf{e}_j)^T M(-t) (\mathbf{e}_i + \mathbf{e}_j) = m_{ii}(-t) + m_{jj}(-t) + 2m_{ij}(-t)$ ,  $t \leq 0$  and the product  $(\mathbf{e}_i + \mathbf{e}_j)^T P (\mathbf{e}_i + \mathbf{e}_j) = p_{ii} + p_{jj} + 2p_{ij}$ . The sequence  $\{(\mathbf{e}_i + \mathbf{e}_j)^T M(-t) (\mathbf{e}_i + \mathbf{e}_j)\}_{t \leq 0}$  is also non decreasing when  $t \rightarrow \infty$ , and it is upper bounded by  $(\mathbf{e}_i + \mathbf{e}_j)^T P (\mathbf{e}_i + \mathbf{e}_j)$ . Therefore, it admits a finite limit, which we denote it by  $r_{ij}$ . Verify that  $2 \lim_{t \rightarrow +\infty} m_{ij}(-t) = r_{ij} - [M_\infty]_{ii} - [M_\infty]_{jj}$ .*

- EXERCISE 10.3.3 [CONSTANT SOLUTIONS OF RDE] If  $\bar{M}$  is a p.s.d. solution to the ARE (10.81) – in particular, if  $\bar{M} = M_\infty$  – and if in the Riccati Difference Equation (10.84) we set the final condition  $M(0) = \bar{M}$ , then the matrix sequence that solves the difference equation has, for any  $t \geq 0$ , the constant value  $M(-t) = \bar{M}$ .

We can now prove the following fundamental result.

**Proposition 10.3.3** [STRUCTURE OF OPTIMAL CONTROL OVER INFINITE HORIZON] *Let (10.25) be a stabilisable system and, by referring to the matrices of the index (10.75), let  $M_\infty$  be the symmetric positive semi-definite solution to the ARE obtained in Lemma 10.3.2. Then*

i) *under the control law*

$$\mathbf{u}_\infty(t) := K_\infty \mathbf{x}(t) \quad \text{with} \quad K_\infty = -(R + G^T M_\infty G)^{-1} G^T M_\infty F, \quad (10.85)$$

*the index takes the value*

$$J(\mathbf{u}_\infty, \mathbf{x}_0) = \mathbf{x}_0^T M_\infty \mathbf{x}_0; \quad (10.86)$$

ii) *the input  $\mathbf{u}_\infty(\cdot)$  defined above minimises the index (10.75) and thus provides the optimal control over the infinite horizon  $[0, +\infty]$ .*

PROOF By solving the RDE (10.84) corresponding to final condition  $M(0) = M_\infty$ , we get the constant solution  $M(-t) = M_\infty$  for every  $t \geq 0$ .

We write now the index at time  $T-1$  as  $J_{T-1}(\mathbf{u}_\infty, \mathbf{x}_0) = \sum_{t=0}^{T-1} (\mathbf{u}_\infty^T(t) R \mathbf{u}_\infty(t) + \mathbf{x}^T(t) Q \mathbf{x}(t))$  by employing the square completion formula (10.37), and then we plug in it the constant solution to the EDR  $M(\cdot) = M_\infty$ , thus obtaining:

$$\begin{aligned} J_{T-1}(\mathbf{u}_\infty, \mathbf{x}_0) &= \mathbf{x}_0^T M_\infty \mathbf{x}_0 - \mathbf{x}^T(T) M_\infty \mathbf{x}(T) \\ &+ \sum_{t=0}^{T-1} [(R + G^T M_\infty G) \mathbf{u}_\infty(t) + G^T M_\infty F \mathbf{x}(t)]^T [R + G^T M_\infty G]^{-1} [(R + G^T M_\infty G) \mathbf{u}_\infty(t) + G^T M_\infty F \mathbf{x}(t)] \end{aligned}$$

Since the sum on the right hand side is zero, the input  $\mathbf{u}_\infty(\cdot)$  yields, for any  $T > 0$ , the index value over the finite interval

$$J_{T-1}(\mathbf{u}_\infty, \mathbf{x}_0) = \mathbf{x}_0^T M_\infty \mathbf{x}_0 - \mathbf{x}^T(T) M_\infty \mathbf{x}(T). \quad (10.87)$$

On the other side, when  $\mathbf{u}(\cdot)$  varies, the minimum value of the index  $J_{T-1}(\mathbf{u}, \mathbf{x}_0)$  can be expressed by employing the solution to the RDE (10.84) with final condition  $M(0) = 0$ :

$$\min_{\mathbf{u}} J_{T-1}(\mathbf{u}, \mathbf{x}_0) = \mathbf{x}_0^T M(-T) \mathbf{x}_0. \quad (10.88)$$

Since  $M_\infty$  is positive semidefinite and  $\mathbf{u}_\infty(\cdot)$  is not necessarily the optimal input over  $[0, T]$  corresponding to the initial condition  $\mathbf{x}_0$ , the following chain of inequalities holds true

$$\mathbf{x}_0^T M(-T) \mathbf{x}_0 \leq \sum_{t=0}^{T-1} (\mathbf{u}_\infty^T(t) R \mathbf{u}_\infty(t) + \mathbf{x}^T(t) Q \mathbf{x}(t)) = \mathbf{x}_0^T M_\infty \mathbf{x}_0 - \mathbf{x}^T(T) M_\infty \mathbf{x}(T) \leq \mathbf{x}_0^T M_\infty \mathbf{x}_0 \quad (10.89)$$

By considering the limit as  $T \rightarrow \infty$ , it follows from (10.89) that

$$\mathbf{x}_0^T M_\infty \mathbf{x}_0 \leq \sum_{t=0}^{+\infty} (\mathbf{u}_\infty^T(t) R \mathbf{u}_\infty(t) + \mathbf{x}^T(t) Q \mathbf{x}(t)) \leq \mathbf{x}_0^T M_\infty \mathbf{x}_0 \quad (10.90)$$

which establishes (10.86).

Concerning (ii), we shall verify that for any input  $\mathbf{u}(\cdot) \neq \mathbf{u}_\infty(\cdot)$  and for any  $\mathbf{x}_0$  it holds true

$$J(\mathbf{u}, \mathbf{x}_0) \geq J(\mathbf{u}_\infty, \mathbf{x}_0).$$

By contradiction, assume that there exists an input sequence  $\bar{\mathbf{u}}(\cdot)$  for which  $J(\bar{\mathbf{u}}, \mathbf{x}_0) < J(\mathbf{u}_\infty, \mathbf{x}_0)$ , and denote by  $\bar{\mathbf{x}}(\cdot)$  the corresponding system evolution:

$$\begin{aligned} J(\bar{\mathbf{u}}, \mathbf{x}_0) &= \sum_{t=0}^{+\infty} \left( \bar{\mathbf{u}}^T(t) R \bar{\mathbf{u}}(t) + \bar{\mathbf{x}}^T(t) Q \bar{\mathbf{x}}(t) \right) \\ &< \sum_{t=0}^{+\infty} \left( \mathbf{u}_\infty^T(t) R \mathbf{u}_\infty(t) + \mathbf{x}^T(t) Q \mathbf{x}(t) \right) = \mathbf{x}_0^T M_\infty \mathbf{x}_0 \end{aligned} \quad (10.91)$$

namely

$$\lim_{T \rightarrow +\infty} \sum_{t=0}^{T-1} \left( \bar{\mathbf{u}}^T(t) R \bar{\mathbf{u}}(t) + \bar{\mathbf{x}}^T(t) Q \bar{\mathbf{x}}(t) \right) < \lim_{T \rightarrow +\infty} \mathbf{x}_0^T M(-T) \mathbf{x}_0. \quad (10.92)$$

By the “sign permanence theorem”, when  $T$  is large enough it holds true that

$$J_{T-1}(\bar{\mathbf{u}}, \mathbf{x}_0) = \sum_{t=0}^{T-1} \left( \bar{\mathbf{u}}^T(t) R \bar{\mathbf{u}}(t) + \bar{\mathbf{x}}^T(t) Q \bar{\mathbf{x}}(t) \right) < \mathbf{x}_0^T M(-T) \mathbf{x}_0, \quad (10.93)$$

which is a contradiction, since  $\mathbf{x}_0^T M(-T) \mathbf{x}_0$  represents the minimum value of the index  $J_{T-1}$  associated with initial state  $\mathbf{x}_0$ . ■

By the previous proposition, there exists a symmetric and positive semidefinite solution to the Algebraic Riccati Equation that determines the structure of the optimal control over a finite horizon. The following two questions naturally arise, concerning the existence and the properties of other possible symmetric and p.s.d. solutions of the ARE and other optimal control inputs:

1. If the solution  $M_\infty$  of the ARE that we have found is not the unique symmetric and p.s.d. solution to the equation, which characteristics distinguish it from the others?
2. Is the sequence  $\mathbf{u}_\infty(\cdot)$  given by (10.85) the unique input that minimises the index, as in the optimal control over finite interval  $[0, T]$ , or are there other inputs that achieve the same goal?

The next proposition provides an answer to both previous questions, by characterising the  $M_\infty$  as the “smallest” p.s.d. solution to the ARE and the corresponding input  $\mathbf{u}_\infty(\cdot)$  resulting from (10.85) as the unique input that minimises the index.

**Proposition 10.3.4** [MINIMALITY OF  $M_\infty$  AND UNIQUENESS OF THE OPTIMAL INPUT] (i) For any symmetric and positive semidefinite solution  $\bar{M}$  of the ARE, the matrix  $\bar{M} - M_\infty$  is positive semidefinite.

(ii) For a given initial state  $\mathbf{x}_0$ , the input that minimises the index (10.75) is unique, and therefore coincides with  $\mathbf{u}_\infty(\cdot)$  as given by (10.85).



PROOF In order to show (i), assume, by contradiction, that  $\bar{M} - M_\infty$  is not p.s.d. Then there exists an initial state  $\mathbf{x}_0$  such that

$$\mathbf{x}_0^T \bar{M} \mathbf{x}_0 < \mathbf{x}_0^T M_\infty \mathbf{x}_0. \quad (10.94)$$

We assume now to control the system by means of the feedback input

$$\bar{\mathbf{u}}(t) = -(R + G^T \bar{M} G)^{-1} G^T \bar{M} F \mathbf{x}(t). \quad (10.95)$$

and we evaluate the index  $J_{T-1}(\bar{\mathbf{u}}, \mathbf{x}_0)$  by employing the square completion Lemma 10.2.3, by plugging in it the solution to the RDE given by  $M(\cdot) = \bar{M}$  and the input (10.95). We obtain, for any  $T > 0$ ,

$$J_{T-1}(\bar{\mathbf{u}}, \mathbf{x}_0) = \sum_{t=0}^{T-1} \left( \bar{\mathbf{u}}(t) R \bar{\mathbf{u}}(t) + \mathbf{x}(t) Q \mathbf{x}(t) \right) = \mathbf{x}_0^T \bar{M} \mathbf{x}_0 - \mathbf{x}(T)^T \bar{M} \mathbf{x}(T) \leq \mathbf{x}_0^T \bar{M} \mathbf{x}_0 \quad (10.96)$$

and, moving to the limit as  $T \rightarrow +\infty$ ,

$$J(\bar{\mathbf{u}}, \mathbf{x}_0) \leq \mathbf{x}_0^T \bar{M} \mathbf{x}_0 < \mathbf{x}_0^T M_\infty \mathbf{x}_0. \quad (10.97)$$

However, this clearly contradicts the fact that  $\mathbf{x}_0^T M_\infty \mathbf{x}_0$  represents the minimum value of  $J(\cdot, \mathbf{x}_0)$  for the cost with initial state  $\mathbf{x}_0$ .

We prove now (ii). To this aim, assume that the input  $\bar{\mathbf{u}}(\cdot)$  achieves the minimum value of the index over  $[0, +\infty]$  when the initial state of the system is  $\mathbf{x}_0$ . By applying the square completion lemma to the index  $J_{T-1}(\bar{\mathbf{u}}, \mathbf{x}_0)$ , where  $T$  is an arbitrary positive integer, with  $M(\cdot) = M_\infty$ , solution to the RDE, we get

$$\begin{aligned} J_{T-1}(\bar{\mathbf{u}}, \mathbf{x}_0) + \bar{\mathbf{x}}(T)^T M_\infty \bar{\mathbf{x}}(T) &= \mathbf{x}_0^T M_\infty \mathbf{x}_0 \\ + \sum_{t=0}^{T-1} [(R + G^T M_\infty G) \bar{\mathbf{u}}(t) + G^T M_\infty F \bar{\mathbf{x}}(t)]^T [R + G^T M_\infty G]^{-1} &[(R + G^T M_\infty G) \bar{\mathbf{u}}(t) + G^T M_\infty F \bar{\mathbf{x}}(t)] \end{aligned} \quad (10.98)$$

The previous expression has the following interpretation: over the interval  $[T, \infty]$ , consider the evolution of the system initialised at  $\bar{\mathbf{x}}(T)$ , the state that the system reaches when the input  $\bar{\mathbf{u}}(\cdot)$  is applied over the interval  $[0, T-1]$ . The term  $\bar{\mathbf{x}}(T)^T M_\infty \bar{\mathbf{x}}(T)$  in (10.98) represents the minimum value of the index over the interval  $[T, +\infty]$ , if we can choose the input sequence  $\mathbf{u}(T), \mathbf{u}(T+1), \dots$  to minimise the cost starting from the state  $\bar{\mathbf{x}}(T)$  that the system reaches as a result of the input  $\bar{\mathbf{u}}(\cdot)$  applied in the time interval  $[0, T-1]$ . As a consequence, by employing a control law that coincides with the sequence  $\bar{\mathbf{u}}(0), \bar{\mathbf{u}}(1), \dots, \bar{\mathbf{u}}(T-1)$  in its initial part yields an index value over  $[0, +\infty]$  that is not smaller than

$$\begin{aligned} \mathbf{x}_0^T M_\infty \mathbf{x}_0 + \sum_{t=0}^{T-1} [(R + G^T M_\infty G) \bar{\mathbf{u}}(t) + G^T M_\infty F \bar{\mathbf{x}}(t)]^T [R + G^T M_\infty G]^{-1} \\ [(R + G^T M_\infty G) \bar{\mathbf{u}}(t) + G^T M_\infty F \bar{\mathbf{x}}(t)]. \end{aligned} \quad (10.99)$$

Since  $\mathbf{x}_0^T M_\infty \mathbf{x}_0$  is the minimum cost over  $[0, +\infty]$ , if the sequence  $\bar{\mathbf{u}}(\cdot)$  achieves such a minimum, then it must necessarily cancel off the non-negative term that is represented by the sum in (10.99). For this to be true,  $\bar{\mathbf{u}}(\cdot)$  must necessarily satisfy, for  $t = 0, 1, \dots, T-1$ , the condition

$$\bar{\mathbf{u}}(t) = -(R + G^T M_\infty G)^{-1} G^T M_\infty F \bar{\mathbf{x}}(t) = \mathbf{u}_\infty(t). \quad (10.100)$$

By the arbitrariness of  $T$ , this implies that  $\bar{\mathbf{u}}(t) = \mathbf{u}_\infty(t)$ ,  $\forall t \geq 0$ , therefore establishing the uniqueness of the optimal input.  $\blacksquare$

- EXERCISE 10.3.4 Let the system (10.25) be stabilisable and with index (10.75), and assume that the solution to the RDE (10.84), initialised at  $M(0) = 0$ , satisfies, for some  $h \geq 0$ ,  $M(-h) = M(-h-1)$ . Verify that
  - (i)  $M(-h) = M(-h-t)$ ,  $\forall t \geq 0$ ;
  - (ii)  $M_\infty = M(-h)$ ;
  - (iii) the optimal control over the finite interval  $[0, h+T]$  in feedback form has as feedback matrices  $K(t) = K_\infty$  for  $t = 0, 1, \dots, T-1$ , and subsequently  $K(t) = -(R + G^T M(t+1-h-T)G)^{-1} G^T M(t+1-h-T)F$ , for  $t = T, \dots, T+h-1$ .

### 10.3.3 Positive definiteness of the matrix $M_\infty$ and observability

As we have shown in the previous paragraph, the solution  $M_\infty$  to the ARE is a positive semidefinite matrix. When it is not positive definite, there exist non-zero initial states  $\mathbf{x}_0$  corresponding to which the minimum value of the index is zero, namely  $\min_{\mathbf{u}} J(\mathbf{u}, \mathbf{x}_0) = \mathbf{x}_0^T M_\infty \mathbf{x}_0 = 0$ .

Our aim is to investigate the properties of such initial states and their influence on the optimal control. First of all, it is easy to see that the optimal control must be the zero control. In fact, due to the positive definiteness of the matrix  $R$  any non-zero input would yield a positive contribution to the index value, thus contradicting  $\min_{\mathbf{u}} J(\mathbf{u}, \mathbf{x}_0) = 0$ . Additionally, for any factorisation<sup>5</sup> of the matrix  $Q$  of the form

$$Q = C^T C \quad (10.101)$$

we obtain

$$0 = \min_{\mathbf{u}} J(\mathbf{u}, \mathbf{x}_0) = J(\mathbf{0}, \mathbf{x}_0) = \sum_{t=0}^{+\infty} \mathbf{x}^T(t) Q \mathbf{x}(t) = \sum_{t=0}^{+\infty} \mathbf{x}_0^T (F^T)^t C^T C F^t \mathbf{x}_0. \quad (10.102)$$

Since all the terms of the sum are non-negative, by (10.102) it follows that  $(F^T)^t C^T C F^t \mathbf{x}_0 = 0$ ,  $t = 0, 1, 2, \dots$ , so that

$$\begin{bmatrix} C \\ CF \\ \vdots \\ CF^{n-1} \end{bmatrix} \mathbf{x}_0 = \mathbf{0}, \quad (10.103)$$

establishing the fact that  $\mathbf{x}_0$  belongs to the non observable subspace of the pair  $(F, C)$ .

<sup>5</sup>If  $Q$  is symmetric p.s.d. with rank  $\nu$ , it is shown in Section A.13 that for any integer  $\mu$  with  $\mu \geq \nu$  there exist matrices  $C \in \mathbb{R}^{\mu \times n}$  such that  $Q = C^T C$ .

The inverse result also holds true. In fact, if  $Q = C^T C$  and the pair  $(F, C)$  is not observable, then any non observable state  $\mathbf{x}_0$  satisfies

$$\mathbf{0} = CF^t \mathbf{x}_0, \quad \forall t \geq 0,$$

therefore  $(F^T)^t C^T CF^t \mathbf{x}_0 = 0$ ,  $t = 0, 1, 2, \dots$ , and in unforced evolution the index

$$J(\mathbf{0}, \mathbf{x}_0) = \sum_{t=0}^{+\infty} \mathbf{x}_0^T (F^T)^t C^T CF^t \mathbf{x}_0$$

takes the zero value. Then it is also zero the minimum value  $\mathbf{x}_0^T M_\infty \mathbf{x}_0$  of the index corresponding to the initial condition  $\mathbf{x}(0) = \mathbf{x}_0$ , and such a minimum follows by applying the identically zero input.

This establishes the following result.

**Proposition 10.3.5** [POSITIVE DEFINITENESS OF  $M_\infty$  AND OBSERVABILITY] *Let the system (10.25) be stabilisable and let  $M_\infty$  be the minimum symmetric p.s.d. solution to the ARE, which provides the control law that minimises the index (10.75). Let moreover be  $C$  a matrix such that  $Q = C^T C$ .*

*Then  $M_\infty$  is positive definite if and only if the pair  $(F, C)$  is observable.* ■

EXERCISE 10.3.5 Consider two different factorisations of the p.s.d. matrix  $Q = C_1^T C_1 = C_2^T C_2$ . Verify that the non observable subspaces of the pairs  $(F, C_1)$  and  $(F, C_2)$  coincide.

**Corollary 10.3.6** *Let the system*

$$\begin{aligned} \mathbf{x}(t+1) &= F\mathbf{x}(t) + G\mathbf{u}(t) \\ \mathbf{y}(t) &= H\mathbf{x}(t) \end{aligned} \quad (10.104)$$

*be stabilisable and consider the quadratic index*

$$J(\mathbf{u}, \mathbf{x}_0) = \sum_{t=0}^{\infty} [\mathbf{y}^T(t) V \mathbf{y}(t) + \mathbf{u}^T(t) R \mathbf{u}(t)], \quad (10.105)$$

*with  $V$  and  $R$  positive definite. If  $M_\infty$  is the minimum symmetric p.s.d. solution to the ARE*

$$M = H^T V H + F^T M F - F^T M G [R + G^T M G]^{-1} G^T M F, \quad (10.106)$$

*to which corresponds the control law that minimises the index (10.105), then  $M_\infty$  is positive definite if and only if the pair  $(F, H)$  is observable.* ■

- EXERCISE 10.3.6 Let  $(F, G)$  be stabilisable.

- (i) The matrix  $F + GK_\infty$  factorises into the form  $F + GK_\infty = UF$  where  $U$  is an invertible matrix;
- (ii) extend the result to matrices  $F + G\bar{K}$ , where  $\bar{K} = -(R + G^T \bar{M} G)^{-1} G^T \bar{M} F$  and where  $\bar{M}$  is any positive semidefinite solution to the ARE;
- (iii) If  $F$  is non singular, then the same holds true for  $F + GK_\infty$ .

‡ Hint:  $F + G\bar{K} = F - G(R + G^T \bar{M} G)^{-1} G^T \bar{M} F = [I - G(R + G^T \bar{M} G)^{-1} G^T \bar{M}] F = UF$ . To verify that  $U$  is invertible, recall Exercise 10.2.1:

$$I - G(R + G^T \bar{M} G)^{-1} G^T \bar{M} = I - GR^{-1}(I + G^T \bar{M} GR^{-1})^{-1} G^T \bar{M} = I - GR^{-1} G^T \bar{M} (I + GR^{-1} G^T \bar{M})^{-1}.$$

It is straightforward that  $I + GR^{-1} G^T \bar{M}$  is the inverse of  $I - GR^{-1} G^T \bar{M} (I + GR^{-1} G^T \bar{M})^{-1}$ . Thus  $F + G\bar{K} = [I + GR^{-1} G^T \bar{M}]^{-1} F$ .

- EXERCISE 10.3.7 Let  $(F, G)$  be stabilisable
  - (i) If  $F$  is non singular, then the same holds true for  $F + GK_\infty$  and no non-zero initial state can be steered to zero in finite time by the optimal control.
  - (ii) If  $F$  is singular and  $\mathbf{x}_0 \in \ker F$ , then the optimal input is  $\mathbf{u}_\infty(t) = \mathbf{0}$ ,  $\forall t$ , and  $J(\mathbf{u}_\infty, \mathbf{x}_0) = \min_{\mathbf{u}} J(\mathbf{u}, \mathbf{x}_0) = \mathbf{x}_0^T Q \mathbf{x}_0$ . Moreover, notice that  $\mathbf{x}(1) = \mathbf{0}$ .

### 10.3.4 Additional examples and new questions

Proposition 10.3.3 shows that the optimal control law has the form of a linear state feedback, and thus justifies, *a posteriori*, our assumption in Example 10.3.2, where we chose and optimised an input with such a structure. Instead of proceeding as in the example, we can compute the feedback matrix by solving the ARE (10.81) associated with the system (10.25) and with the index (10.75). Such a methodology is always valid, and almost always represents a forced step in the solution of an optimal control problem for a linear system. It is worth noticing that the solution to the ARE can be obtained in closed form only in very simple cases, while in general it is necessary to resort to numerical methods.

In the following examples we introduce a number of questions concerning the set of solutions of the ARE. They show that, in some cases, there is no solution  $\bar{M}$  of the ARE that is “stabilising”, namely, such that the feedback matrix  $\bar{K} = -(R + G^T \bar{M} G)^{-1} G^T \bar{M} F$  stabilises the system. In particular, the “optimal” solution  $M_\infty$  might, or might not, be stabilising, depending on the particular case considered.

**Example 10.3.2 (cont.d)** [NON STABILISING  $M_\infty$  OPTIMAL SOLUTION, EXISTENCE OF STABILISING  $M_S$ ] The ARE for the considered system and index is

$$M = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} M \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} M \begin{bmatrix} 1 \\ 1 \end{bmatrix} \left( 1 + \begin{bmatrix} 1 & 1 \end{bmatrix} M \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 1 \end{bmatrix} M \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}. \quad (10.107)$$

Since the system is stabilisable, the ARE admits at least one symmetric and p.s.d. solution, which can be obtained as the limit as  $t \rightarrow -\infty$  of the solution to the RDE, initialised with  $M(0) = 0$ . By setting in (10.107)

$$M = \begin{bmatrix} m & p \\ p & q \end{bmatrix}, \quad (10.108)$$

we obtain the following system of three non-linear equations in three unknowns

$$\begin{aligned} (1 + 3m)(1 + m + 2p + q) &= 4(m + p)^2 \\ 5p(1 + m + 2p + q) &= 6(m + p)(p + q) \\ 8q(1 + m + 2p + q) &= 9(p + q)^2. \end{aligned} \quad (10.109)$$

Even in such an apparently simple case, determining all the solutions of the ARE is not straightforward. To start with, we notice that, since we are interested in the solution that provides the optimal control, we need to look for the solution to the ARE that is the limit of the sequence obtained by solving the RDE (10.84). In the present case, the matrices of such a sequence have all zero entries except the one in position (1,1), and the limit must have the same structure. We can thus set  $p = q = 0$  in (10.109), and notice that while the second and the third equations are satisfied identicamente, the first one becomes

$$(1 + 3m)(1 + m) = 4m^2 \quad (10.110)$$

whose solutions are  $m = 2 \pm \sqrt{5}$ . Among such solutions, we need to choose the positive one, since  $M_\infty$  must be p.s.d. One thus obtains

$$M_\infty = \begin{bmatrix} 2 + \sqrt{5} & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
K_\infty &= -(R + G^T M_\infty G)^{-1} G^T M_\infty F = \begin{bmatrix} -\frac{1+\sqrt{5}}{2} & 0 \end{bmatrix} \\
F + GK_\infty &= \begin{bmatrix} \frac{3-\sqrt{5}}{2} & 0 \\ -\frac{1-\sqrt{5}}{2} & 3 \end{bmatrix}
\end{aligned} \tag{10.111}$$

which agrees with our conclusions in the first part of Example 10.3.2.

Another symmetric and positive definite solution to (10.107), whose interpretation will be investigated below, can be obtained by means of numerical methods, and takes the value

$$M_S = \begin{bmatrix} 113,901 & -177,442 \\ -177,442 & 287,108 \end{bmatrix}. \tag{10.112}$$

Such a solution is associated with the feedback matrix

$$K_S = -(R + G^T M_S G)^{-1} G^T M_S F = [2,696 \quad -6,981] \tag{10.113}$$

and the corresponding closed loop system matrix, which is

$$F + GK_S = \begin{bmatrix} 4.696 & -6,981 \\ 2.696 & -3.981 \end{bmatrix}, \tag{10.114}$$

has eigenvalues 0.382 and 0.333, both in modulus smaller than 1. Therefore, the ARE has, in this example, a p.s.d. optimising solution,  $M_\infty$ , and a p.s.d. – in particular, positive definite – stabilising solution  $M_S$ , which are different.

**Example 10.3.3** [NON STABILISING  $M_\infty$  OPTIMAL SOLUTION, NONEXISTENCE OF STABILISING SOLUTION  $M_S$ ] Consider the stabilisable system

$$\mathbf{x}(t+1) = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u(t), \tag{10.115}$$

with the same index (10.64) as in Examples 10.3.1 and 10.3.2.

The ARE associated with (10.64) and (10.115) is

$$M = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} M \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} M \begin{bmatrix} 1 \\ 1 \end{bmatrix} \left( 1 + [1 \quad 1] M \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right)^{-1} [1 \quad 1] M \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \tag{10.116}$$

and corresponds to the system of equations

$$\begin{aligned}
(1+3m)(1+m+2p+q) &= 4(m+p)^2 \\
p(1+m+2p+q) &= 2(m+p)(p+q) \\
0 &= (p+q)^2.
\end{aligned} \tag{10.117}$$

In this case, we can easily compute all the solutions, which are given by

$$\begin{aligned}
m &= 2 \pm \sqrt{5} \\
p &= 0 \\
q &= 0
\end{aligned} \tag{10.118}$$

and the ARE (10.116) has a unique symmetric p.s.d. solution, which coincides with the optimal solution:

$$M_\infty = \begin{bmatrix} 2+\sqrt{5} & 0 \\ 0 & 0 \end{bmatrix}. \tag{10.119}$$

Such a matrix corresponds to the same feedback matrix that we obtained in (10.111):

$$K_\infty = -(R + G^T M_\infty G)^{-1} G^T M_\infty F = \begin{bmatrix} -\frac{1+\sqrt{5}}{2} & 0 \end{bmatrix} \tag{10.120}$$

which in turn yields the optimal control law  $u_\infty(t) = K_\infty \mathbf{x}(t)$ . In this case too, one eigenvalue of the closed loop system matrix

$$F + GK_\infty = \begin{bmatrix} 0,382 & 0 \\ -1,610 & 1 \end{bmatrix} \tag{10.121}$$

does not have modulus smaller than 1, and thus the closed loop system is not asymptotically stable. However in this example, differently from the previous one, the ARE does not admit any other symmetric and p.s.d. solution that is stabilising.

**Example 10.3.4** [STABILISING OPTIMAL SOLUTION  $M_\infty$ ] Consider the stabilisable system in Example 10.3.2 and the index

$$J(u, \mathbf{x}_0) = \sum_{t=0}^{+\infty} (\mathbf{x}^T(t) Q \mathbf{x}(t) + u^2(t)), \quad Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \quad (10.122)$$

The solution  $M_\infty$  of the ARE can be obtained through numerical methods, as limit solution to the RDE (10.33), and it is given by

$$M_\infty = \begin{bmatrix} 126, 245 & -196, 030 \\ -196, 030 & 316, 114 \end{bmatrix} \quad (10.123)$$

The feedback matrix

$$K_\infty = -(R + G^T M_\infty G)^{-1} G^T M_\infty F = [2, 720 \quad -7, 022] \quad (10.124)$$

provides the optimal control law and the corresponding feedback system matrix

$$F + GK_\infty = \begin{bmatrix} 4, 720 & -7, 022 \\ 2, 720 & -4, 022 \end{bmatrix}$$

has eigenvalues 0.279 and 0.418, with modulus smaller than 1. In this case the solution  $M_\infty$  not only provides the optimal control, but it also produces a closed loop system that is internally stable.

EXERCISE 10.3.8 Consider the system (10.115) in Example 10.3.3 and assume as index

$$J(u, \mathbf{x}_0) = \sum_{t=0}^{+\infty} u^2(t) + \mathbf{x}^T(t) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{x}(t) \quad (10.125)$$

Compute  $M_\infty$  and determine whether the spectrum of  $F + GK_\infty$  is contained in the disk  $\{z \in \mathbb{C} : |z| < 1\}$ .

‡ *Hint. By using the notation in (10.108), and putting  $1 + m + 2p + q \neq 0$ , from*

$$\begin{cases} (1 + 3m)(1 + m + 2p + q) &= 4(m + p)^2 \\ p(1 + m + 2p + q) &= 2(m + p)(p + q) \\ 1 + m + 2p + q &= (p + q)^2 \end{cases} \quad \text{one has} \quad \begin{cases} m = \frac{1}{3}(p^2 - 1) \\ q = -\frac{1}{3}p + 2 - \frac{2}{3p} \\ 1 + m + 2p + q = (\frac{2}{3}p + 2 - \frac{2}{3p})^2 \end{cases}$$

and thus  $0 = p^4 + 9p^3 + 4p^2 - 18p + 4 = (p - 1)(p + 2)(p + 4 - \sqrt{18})(p + 4 + \sqrt{18})$ . Only the root  $p = -4 - \sqrt{18}$  corresponds to a p.s.d. solution – in particular, positive definite – of the ARE. Both eigenvalues of the matrix  $F + GK_\infty$  have modulus smaller than 1.

To summarise, the three previous examples show that searching for the optimal control law by means of the solution to the ARE leads to three very different situations:

- in Example 10.3.2, the equation has both a positive semidefinite solution  $M_\infty$ , associated with an optimal control law that is not stabilising, and a symmetric and positive semidefinite  $M_s$ , which yields a stabilising, but not optimal, control law;
- in Example 10.3.3, the solution  $M_\infty$  is associated with a non stabilising optimal control law, but there are no other positive semidefinite solutions to the ARE that yield a stabilising control law;
- finally, in Example 10.3.4, the solution  $M_\infty$  is associated with a control law that is both optimal and stabilising.

The diversity of these results is related to the structural properties of the pair  $(F, Q)$ , and will be thoroughly justified in Section 10.4, where we analyse the relation between the stabilising solutions and the optimal solution to the ARE.

## 10.4 Optimal control and stabilisation of discrete time linear systems

In this section we study which properties the dynamical system (10.25) and the index (10.75) must possess for the optimal control law  $\mathbf{u}_\infty(t) = K_\infty \mathbf{x}(t)$  to induce an internally stable feedback system. When the optimal control law is not stabilising, we analyse instead the possibility of designing stabilising control laws by employing symmetric and p.s.d. solutions of the ARE that are different from  $M_\infty$  and we will establish that, whenever they exit, they exhibit suboptimal characteristics.

**Lemma 10.4.1** [LYAPUNOV EQUATION AND POSITIVE SEMIDEFINITE SOLUTIONS TO ARE] *Let  $(F, G)$  be stabilisable,  $\bar{M}$  a positive semidefinite solution to the ARE (10.81) and  $\bar{K} = -(R + G^T \bar{M} G)^{-1} G^T \bar{M} F$ . Then the following identity holds true:*

$$(F + G\bar{K})^T \bar{M} (F + G\bar{K}) - \bar{M} = -Q - \bar{K}^T R \bar{K}. \quad (10.126)$$

PROOF From the definition of  $\bar{K}$  and the fact that  $\bar{M}$  solves the ARE, it follows

$$\begin{aligned} G^T \bar{M} G \bar{K} &= -R \bar{K} - G^T \bar{M} F, \\ F^T \bar{M} F &= \bar{M} - Q + F^T \bar{M} G (R + G^T \bar{M} G)^{-1} G^T \bar{M} F = \bar{M} - Q - F^T \bar{M} G \bar{K}. \end{aligned}$$

Therefore,

$$\begin{aligned} & (F + G\bar{K})^T \bar{M} (F + G\bar{K}) \\ &= F^T \bar{M} F + \bar{K}^T G^T \bar{M} F + F^T \bar{M} G \bar{K} + \bar{K}^T G^T \bar{M} G \bar{K} \\ &= (\bar{M} - Q - F^T \bar{M} G \bar{K}) + \bar{K}^T G^T \bar{M} F + F^T \bar{M} G \bar{K} + \bar{K}^T (-R \bar{K} - G^T \bar{M} F) \\ &= \bar{M} - Q - \bar{K}^T R \bar{K} \end{aligned}$$

from which it immediately follows (10.126). ■

**Proposition 10.4.2** [OPTIMAL STABILISING CONTROL AND DETECTABILITY OF  $(F, C)$ ] *Let  $(F, G)$  be stabilisable, and  $Q = C^T C$ . Consider the solution  $M_\infty$  of the ARE, and the associated optimal control feedback matrix  $K_\infty = -(R + G^T M_\infty G)^{-1} G^T M_\infty F$ . Such a control law is stabilising, namely the eigenvalues of the matrix of the feedback system  $F + GK_\infty$  have modulus smaller than 1, if and only if*

$$\begin{bmatrix} zI - F \\ C \end{bmatrix} \quad (10.127)$$

is of full rank for any  $z \in \mathbb{C}$ , with  $|z| \geq 1$ , namely if and only if the pair  $(F, C)$  is detectable.

PROOF By the previous lemma,  $M_\infty$  satisfies the identity

$$(F + GK_\infty)^T M_\infty (F + GK_\infty) - M_\infty = -C^T C - K_\infty^T R K_\infty. \quad (10.128)$$

To begin with, assume that  $F + GK_\infty$  has an eigenvalue  $\lambda$  with  $|\lambda| \geq 1$ , and let  $\mathbf{v}$  be an eigenvector corresponding to  $\lambda$ . Pre-multiplying (10.128) by  $\check{\mathbf{v}}^T$  and post-multiplying by  $\mathbf{v}$ , one obtains

$$(|\lambda|^2 - 1) \check{\mathbf{v}}^T M_\infty \mathbf{v} = -\check{\mathbf{v}}^T C^T C \mathbf{v} - \check{\mathbf{v}}^T K_\infty^T R K_\infty \mathbf{v}. \quad (10.129)$$

The right hand side of (10.129) is non-positive, while the left hand side is non-negative, thus the only possibility is that they are both zero. Therefore, we have

$$K_\infty \mathbf{v} = \mathbf{0}, \quad C\mathbf{v} = \mathbf{0}.$$

This implies

$$\lambda \mathbf{v} = (F + GK_\infty)\mathbf{v} = F\mathbf{v}$$

and finally

$$\begin{bmatrix} F - \lambda I \\ C \end{bmatrix} \mathbf{v} = \mathbf{0}. \quad (10.130)$$

Therefore (10.127) is not of full rank for  $z = \lambda$ , with  $|\lambda| \geq 1$ .

Conversely, assume that the PBH observability matrix (10.127) is not of full rank for  $z = \lambda$ , with  $|\lambda| \geq 1$ . We claim that  $\lambda$  is an eigenvalue of  $F + GK_\infty$ .

To prove it, perform a change of basis in the state space by means of a matrix  $P$  that puts the pair  $(F, C)$  in standard observability form:

$$\begin{aligned} \hat{F} = P^{-1}FP &= \begin{bmatrix} \hat{F}_{11} & 0 \\ \hat{F}_{21} & \hat{F}_{22} \end{bmatrix}, & \hat{G} = P^{-1}G &= \begin{bmatrix} \hat{G}_1 \\ \hat{G}_2 \end{bmatrix}, \\ \hat{C} = CP &= [\hat{C}_1 \quad 0]. \end{aligned} \quad (10.131)$$

Since the set of the eigenvalues of  $\hat{F}_{22}$  coincides with the set of values  $z$  that make the matrix (10.127) of the PBH criterion not full rank, then  $\lambda \in \Lambda(\hat{F}_{22})$ .

Since the relation between the components of the state vector in the new basis and in the old ones is expressed by  $\hat{\mathbf{x}}(t) = P^{-1}\mathbf{x}(t)$ , the index  $J$  becomes

$$\begin{aligned} J(\mathbf{u}, \mathbf{x}_0) &= \sum_{t=0}^{\infty} (\mathbf{x}^T(t) C^T C \mathbf{x}(t) + \mathbf{u}^T(t) R \mathbf{u}(t)) = \sum_{t=0}^{\infty} (\hat{\mathbf{x}}^T(t) P^T C^T C P \hat{\mathbf{x}}(t) + \mathbf{u}^T(t) R \mathbf{u}(t)) \\ &= \sum_{t=0}^{\infty} \left( \hat{\mathbf{x}}^T(t) \begin{bmatrix} \hat{C}_1^T \hat{C}_1 & 0 \\ 0 & 0 \end{bmatrix} \hat{\mathbf{x}}(t) + \mathbf{u}^T(t) R \mathbf{u}(t) \right). \end{aligned}$$

In order to determine the structure of  $\hat{M}_\infty$  with respect to the new basis, one can compute the limit as  $t \rightarrow +\infty$  of the sequence

$$\hat{M}(-t) = \begin{bmatrix} \hat{M}_{11}(-t) & \hat{M}_{12}(-t) \\ \hat{M}_{21}(-t) & \hat{M}_{22}(-t) \end{bmatrix}$$

obtained solving the RDE

$$\begin{aligned} \hat{M}(-t-1) &= \begin{bmatrix} \hat{C}_1^T \hat{C}_1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} \hat{F}_{11}^T & \hat{F}_{21}^T \\ 0 & \hat{F}_{22}^T \end{bmatrix} \hat{M}(-t) \begin{bmatrix} \hat{F}_{11} & 0 \\ \hat{F}_{21} & \hat{F}_{22} \end{bmatrix} \\ &\quad - \begin{bmatrix} \hat{F}_{11}^T & \hat{F}_{21}^T \\ 0 & \hat{F}_{22}^T \end{bmatrix} \hat{M}(-t) \begin{bmatrix} \hat{G}_1 \\ \hat{G}_2 \end{bmatrix} \left( R + \hat{G}^T \hat{M}(-t) \hat{G} \right)^{-1} [\hat{G}_1^T \quad \hat{G}_2^T] \hat{M}(-t) \begin{bmatrix} \hat{F}_{11} & 0 \\ \hat{F}_{21} & \hat{F}_{22} \end{bmatrix} \end{aligned}$$



initialised with  $\hat{M}(0) = 0$ . One can verify by induction that, in all the matrices of the sequence, only the block in position  $(1, 1)$  is different from zero. Therefore, the limit matrix  $\hat{M}_\infty$  has the same structure:

$$\hat{M}_\infty = \begin{bmatrix} \star & 0 \\ 0 & 0 \end{bmatrix}.$$

By partitioning the feedback matrix

$$\hat{K}_\infty = -\left(R + \hat{G}^T \hat{M}_\infty \hat{G}\right)^{-1} \hat{G}^T \hat{M}_\infty \hat{F}$$

in a way that is analogous to the partitioning of  $\hat{M}_\infty$ , one obtains

$$\hat{K}_\infty = [\hat{K}_1 \quad 0].$$

As a consequence, the matrix of the closed loop system

$$\hat{F} + \hat{G}\hat{K}_\infty = \begin{bmatrix} \hat{F}_{11} & 0 \\ \hat{F}_{21} & \hat{F}_{22} \end{bmatrix} + \begin{bmatrix} \hat{G}_1 \\ \hat{G}_2 \end{bmatrix} [\hat{K}_1 \quad 0] = \begin{bmatrix} \hat{F}_{11} + \hat{G}_1 \hat{K}_1 & 0 \\ \hat{F}_{21} + \hat{G}_2 \hat{K}_1 & \hat{F}_{22} \end{bmatrix}$$

has among its eigenvalues all the eigenvalues of the block  $\hat{F}_{22}$ , and therefore in particular

$$\lambda \in \Lambda(\hat{F}_{22}) \subset \Lambda(\hat{F} + \hat{G}\hat{K}_\infty). \quad \blacksquare$$

By Proposition 10.4.2, the existence of a symmetric and p.s.d. solution to the ARE that is simultaneously optimal and stabilising can be checked by evaluating the rank of (10.127), the PHB observability matrix of the pair  $(F, C)$ .

The statement of the following proposition shows that even the more general question of existence of a symmetric p.s.d. solution to the ARE that is stabilising (but not necessary optimal) can be answered by verifying the rank of (10.127).

**Proposition 10.4.3** [POSITIVE SEMIDEFINITE SOLUTION TO THE ARE AND STABILISATION] *If the pair  $(F, G)$  is stabilisable and  $Q = C^T C$ , then*

- i) *there exists at most one symmetric and p.s.d solution  $M_S$  of the ARE (10.81) that corresponds to a stabilising feedback matrix  $K_S = -(R + G^T M_S G)^{-1} G^T M_S F$ ;*
- ii) *such a solution exists if and only if the matrix  $\begin{bmatrix} zI - F \\ C \end{bmatrix}$  is of full rank for any complex  $z$  of modulus 1;*
- iii) *if the stabilising solution  $M_S$  exists, then for any other symmetric and p.s.d. solution  $\bar{M}$  the matrix  $M_S - \bar{M}$  is symmetric and p.s.d.;*
- iv) *if the stabilising solution  $M_S$  exists, it can be obtained as the limit as  $t \rightarrow +\infty$  of the solution  $M(-t)$  of the RDE (10.84), initialised by assuming that  $M(0)$  is any symmetric positive definite matrix.*

PROOF We shall only prove (i) and (iii).

We start with (iii), assuming that  $M_S$  is a (for the time being, not necessarily unique) stabilising solution. Set

$$K_S = -(R + G^T M_S G)^{-1} G^T M_S F, \quad (10.132)$$

and consider the control law, which by assumption is stabilising,

$$\mathbf{u}_S(t) = K_S \mathbf{x}_S(t). \quad (10.133)$$

Apply the square completion Lemma 10.2.3 in order to evaluate the index associated with the input (10.133) over a finite interval  $[0, T]$ . By plugging into (10.37) the constant  $M(-t) = M_S$ ,  $\forall t$ , which is the solution to the RDE, we obtain

$$J_{T-1}(\mathbf{u}_S, \mathbf{x}_0) = \sum_{t=0}^{T-1} [\mathbf{x}_S^T(t) Q \mathbf{x}_S(t) + \mathbf{u}_S^T(t) Q \mathbf{u}_S(t)] = \mathbf{x}_0^T M_S \mathbf{x}_0 - \mathbf{x}_S^T(T) M_S \mathbf{x}_S(T). \quad (10.134)$$

If instead we plug in the constant solution  $M(-t) = \bar{M}$ ,  $\forall t$ , we get the different expression

$$\begin{aligned} J_{T-1}(\mathbf{u}_S, \mathbf{x}_0) &= \sum_{t=0}^{T-1} [\mathbf{x}_S^T(t) Q \mathbf{x}_S(t) + \mathbf{u}_S^T(t) Q \mathbf{u}_S(t)] \\ &= \mathbf{x}_0^T \bar{M} \mathbf{x}_0 - \mathbf{x}_S^T(T) \bar{M} \mathbf{x}_S(T) \\ &\quad + \sum_{t=0}^{T-1} [(R + G^T \bar{M} G) \mathbf{u}_S(t) + G^T \bar{M} F \mathbf{x}_S(t)]^T [R + G^T \bar{M} G]^{-1} [(R + G^T \bar{M} G) \mathbf{u}_S(t) + G^T \bar{M} F \mathbf{x}_S(t)] \\ &\geq \mathbf{x}_0^T \bar{M} \mathbf{x}_0 - \mathbf{x}_S^T(T) \bar{M} \mathbf{x}_S(T). \end{aligned} \quad (10.135)$$

and from (10.134) and (10.135) we get

$$\mathbf{x}_0^T M_S \mathbf{x}_0 - \mathbf{x}_S^T(T) M_S \mathbf{x}_S(T) \geq \mathbf{x}_0^T \bar{M} \mathbf{x}_0 - \mathbf{x}_S^T(T) \bar{M} \mathbf{x}_S(T). \quad (10.136)$$

By considering to the limit (10.136) as  $T \rightarrow +\infty$ , and recalling that  $\mathbf{x}_S(T) \rightarrow \mathbf{0}$ , we obtain

$$\mathbf{x}_0^T M_S \mathbf{x}_0 \geq \mathbf{x}_0^T \bar{M} \mathbf{x}_0. \quad (10.137)$$

and by arbitrariness of  $\mathbf{x}_0$  we conclude that  $M_S - \bar{M}$  is positive semidefinite, and this establishes (iii).

An immediate consequence of this result is that, if there exists one symmetric, p.s.d. and stabilising solution of the ARE, it is unique. In fact, assume that  $M_{S1}$  and  $M_{S2}$  are both symmetric, p.s.d. and stabilising solutions. Then both  $M_{S1} - M_{S2}$  and  $M_{S2} - M_{S1}$  are positive semidefinite, and this implies that  $M_{S1} - M_{S2} = 0$ , thus establishing (i). ■

**Corollario 10.4.4** *If the solution  $M_\infty$  of the ARE is stabilising, it is the only symmetric and p.s.d. solution to the equation.*

PROOF If  $M_\infty$  is stabilising, by (i) of Proposition 10.4.3 any symmetric and p.s.d. solution  $\bar{M}$  is such that  $M_\infty - \bar{M}$  is symmetric and p.s.d. On the other hand, by (i) in Proposition 10.3.5,  $\bar{M} - M_\infty$  is symmetric and p.s.d. Therefore  $\bar{M} = M_\infty$ . ■

If the stabilising solution  $M_S$  exists, but it does not coincide with  $M_\infty$ , its computation is of great interest. In fact, in such a case, it is impossible to achieve simultaneously, namely, by means of the same control law, the two objectives of stabilising the system and minimising the cost. In such a case, it is often of prevailing interest to find a control law that produces an internally stable closed loop system. On the other side, within the set of all the stabilising control laws, the state feedback (10.133), which corresponds to the solution  $M_S$  of the ARE, is in fact the law that yields the minimum value of the index. In other words, (10.133) provides the optimal input when the control law is further constrained to be stabilising.

Verifying such a property is rather simple. To this aim, we compare over a finite interval  $[0, T]$  the values of the index corresponding to the control law (10.133) and to any other stabilising control law

$$\bar{\mathbf{u}}(t) = \bar{K}\bar{\mathbf{x}}(t). \quad (10.138)$$

We start by applying the square completion lemma using as solution to the Riccati Difference Equation  $M(-t) = M_S$ ,  $\forall t \geq 0$ . In the first case we get

$$J_{T-1}(\mathbf{u}_S, \mathbf{x}_0) = \sum_{t=0}^{T-1} [\mathbf{x}_S^T(t) Q \mathbf{x}_S(t) + \mathbf{u}_S^T(t) R \mathbf{u}_S(t)] = \mathbf{x}_0^T M_S \mathbf{x}_0 - \mathbf{x}_S^T(T) M_S \mathbf{x}_S(T) \quad (10.139)$$

and in the second case

$$\begin{aligned} J_{T-1}(\bar{\mathbf{u}}, \mathbf{x}_0) &= \sum_{t=0}^{T-1} [\bar{\mathbf{x}}^T(t) Q \bar{\mathbf{x}}(t) + \bar{\mathbf{u}}^T(t) R \bar{\mathbf{u}}(t)] \\ &= \mathbf{x}_0^T M_S \mathbf{x}_0 - \bar{\mathbf{x}}(T)^T M_S \bar{\mathbf{x}}(T) + \\ &\quad + \sum_{t=0}^{T-1} [(R + G^T M_S G) \bar{\mathbf{u}}(t) + G^T M_S F \bar{\mathbf{x}}(t)]^T [R + G^T M_S G]^{-1} [(R + G^T M_S G) \bar{\mathbf{u}}(t) + G^T M_S F \bar{\mathbf{x}}(t)] \\ &\geq \mathbf{x}_0^T M_S \mathbf{x}_0 - \bar{\mathbf{x}}^T(T) M_S \bar{\mathbf{x}}(T). \end{aligned} \quad (10.140)$$

Since both control laws are stabilising, the state of the system tends to  $\mathbf{0}$  as  $T \rightarrow +\infty$  both applying the control law (10.133) and the control law (10.138). Therefore, by moving to the limit as  $T \rightarrow +\infty$  in (10.139) and (10.140), we obtain

$$J(\mathbf{u}_S, \mathbf{x}_0) = \lim_{T \rightarrow +\infty} J_{T-1}(\mathbf{u}_S, \mathbf{x}_0) = \mathbf{x}_0^T M_S \mathbf{x}_0 \leq \lim_{T \rightarrow +\infty} J_{T-1}(\bar{\mathbf{u}}, \mathbf{x}_0) = J(\bar{\mathbf{u}}, \mathbf{x}_0) \quad (10.141)$$

which establishes the optimality of the control law  $\mathbf{u}_S(\cdot)$ . Moreover, it is easy to see that  $\mathbf{u}_S(\cdot)$  is the unique stabilising control law that minimises the index, since in (10.140) the equality holds true only if  $\bar{\mathbf{u}}(t)$  and  $\bar{\mathbf{x}}(t)$  are related via

$$\bar{\mathbf{u}}(t) = -(R + G^T M_S G)^{-1} G^T M_S F \bar{\mathbf{x}}(t) = K_S \bar{\mathbf{x}}(t), \quad (10.142)$$

which is exactly (10.133).

**Example 10.4.1.** [CONTROL OF A MIXER]

(i) Kwakernaak and Sivan's model We consider a scenario in which two flows, with volumetric flow rates,  $q_1(t)$  e  $q_2(t)$  respectively, are pushed and mixed in a cylindrical mixer. The same substance is dissolved in both flows, with constant concentrations  $c_1$  and  $c_2$ , respectively. We assume that at any time  $t$  the fluid inside the mixer is characterised by a uniform concentration  $c(t)$  of the substance dissolved in it, has volume  $v(t)$ , and flows out of the mixer with volumetric flow rate  $q(t)$ .

The mass balance equations are as follows:

$$\frac{dv(t)}{dt} = q_1(t) + q_2(t) - q(t) \quad (10.143)$$

$$\frac{d[c(t)v(t)]}{dt} = c_1 q_1(t) + c_2 q_2(t) - c(t)q(t) \quad (10.144)$$

If we relate the outflow rate  $q(t)$  to the level  $h(t)$  of the fluid through the equation

$$q(t) = k\sqrt{h(t)}, \quad (10.145)$$

where  $k$  is a constant that can be evaluated by experiments, and if we denote by  $S$  the mixer section,

we can express  $q(t)$  by means of the volume  $v(t)$  of the fluid, and we can rewrite (10.144) as a non-linear model, with state variables  $v(t)$  and  $c(t)$ , and inputs  $q_1(t)$  and  $q_2(t)$ :

$$\begin{aligned} \frac{dv(t)}{dt} &= q_1(t) + q_2(t) - k\sqrt{\frac{v(t)}{S}} \\ \frac{d[c(t)v(t)]}{dt} &= c_1 q_1(t) + c_2 q_2(t) - kc(t)\sqrt{\frac{v(t)}{S}}. \end{aligned} \quad (10.146)$$

Assume that, corresponding to the constant flow rates  $q_{10}$  and  $q_{20}$ , the system exhibits an equilibrium state characterised, inside the mixer, by volume  $v_0$  of fluid and concentration  $c_0$  of dissolved substance. We have therefore

$$0 = q_{10} + q_{20} - q_0 \quad (10.147)$$

$$0 = c_1 q_{10} + c_2 q_{20} - c_0 q_0, \quad (10.148)$$

where we denoted by  $q_0 = k\sqrt{v_0/S}$  the equilibrium outflow.

Assume now that when the mixer is operational the input flow rates' actual values are small perturbations of the equilibrium values  $q_{10}$  and  $q_{20}$  only, and that all the other variables are also small variations of the corresponding equilibrium values. Then

$$\begin{aligned} u_1 &= q_1 - q_{10} & u_2 &= q_2 - q_{20} \\ x_1 &= v - v_0 & x_2 &= c - c_0 \end{aligned}$$

are small quantities, and by (10.146) we obtain the following linear equations

$$\begin{aligned} \frac{dx_1}{dt} &= u_1(t) + u_2(t) - \frac{k}{2v_0}\sqrt{\frac{v_0}{S}}x_1(t) \\ x_0\frac{dx_2}{dt} + c_0\frac{dx_1}{dt} &= c_1 u_1(t) + c_2 u_2(t) - \frac{c_0 k}{2v_0}\sqrt{\frac{v_0}{S}}x_1(t) - k\sqrt{\frac{v_0}{S}}x_2(t). \end{aligned}$$

If we denote by  $\tau$  the ratio  $v_0/q_0$ , and we eliminate  $\dot{x}_1$  in the second equation, we obtain the linearised system

$$\frac{d\mathbf{x}}{dt} = \begin{bmatrix} -\frac{1}{2\tau} & 0 \\ 0 & -\frac{1}{\tau} \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 1 & 1 \\ \frac{c_1 - c_0}{v_0} & \frac{c_2 - c_0}{v_0} \end{bmatrix} \mathbf{u}(t) = F\mathbf{x}(t) + G\mathbf{u}(t). \quad (10.149)$$

We further take as outputs of the linearised system the variations, with respect to the values at equilibrium, of flow and of concentration in the drained fluid

$$y_1 = q - q_0, \quad y_2 = c - c_0 = x_2. \quad (10.150)$$

Figura1041.pdf

Since  $x_1/v_0 = (v - v_0)/v_0$  is smaller than 1, by applying (10.145) we can approximate  $y_1$  as

$$y_1 = q - q_0 = k\sqrt{\frac{v_0 + (v - v_0)}{S}} - k\sqrt{\frac{v_0}{S}} \sim k\sqrt{\frac{v_0}{S}} \frac{v - v_0}{2v_0} = \frac{1}{2\tau}x_1.$$

In this way the output vector is linearly related to the state according to

$$\mathbf{y}(t) = \begin{bmatrix} \frac{1}{2\tau} & 0 \\ 0 & 1 \end{bmatrix} \mathbf{x}(t) = H\mathbf{x}(t) \quad (10.151)$$

(ii) Sampled system Assume that such a mixer is part of a computer-controlled process. Since time must be slotted, the two input flows are changed only at discrete times, which we assume to be separated by intervals of constant duration  $\Delta$ . Since the inputs are constant between the times  $k\Delta$  and  $(k+1)\Delta$ ,  $k \in \mathbb{Z}$ , the signals of the resulting systems are sampled and, recalling the notation and formulas in Section 9.3, we obtain the matrices

$$\begin{aligned} \Phi &= e^{F\Delta} = \begin{bmatrix} e^{-\Delta/2\tau} & 0 \\ 0 & e^{-\Delta/\tau} \end{bmatrix} \\ \Gamma &= \int_0^\Delta e^{F\sigma} G d\sigma = \begin{bmatrix} 2\tau(1 - e^{-\Delta/2\tau}) & 2\tau(1 - e^{-\frac{\Delta}{\tau}}) \\ \frac{\tau(c_1 - c_0)}{v_0}(1 - e^{-\Delta/\tau}) & \frac{\tau(c_2 - c_0)}{v_0}(1 - e^{-\Delta/\tau}) \end{bmatrix}. \end{aligned} \quad (10.152)$$

Assume the following numerical values for nominal working conditions:

$$\begin{aligned} v_0 &= 1 \text{ m}^3 \\ q_{10} &= 0,015 \text{ m}^3/\text{sec} & q_{20} &= 0,005 \text{ m}^3/\text{sec} & q_0 &= 0,02 \text{ m}^3/\text{sec} \\ c_1 &= 1 \text{ kmole/m}^3 & c_2 &= 2 \text{ kmole/m}^3 & c_0 &= 1,25 \text{ kmole/m}^3 \end{aligned}$$

which corresponds to  $\tau = 50$  sec.

Assuming a sampling period of  $\Delta = 5$  sec, the matrices of the sampled signals model become

$$\Phi = \begin{bmatrix} 0,951 & 0 \\ 0 & 0,905 \end{bmatrix}, \quad \Gamma = \begin{bmatrix} 4,877 & 4,877 \\ -1,189 & 3,569 \end{bmatrix}, \quad H = \begin{bmatrix} 0,01 & 0 \\ 0 & 1 \end{bmatrix}. \quad (10.153)$$

(iii) Optimal control We formulate now the optimal regulation problem for the mixer in a neighborhood of the nominal working point, by minimising an index of the type

$$J = \sum_{t=0}^{\infty} \left( \mathbf{y}^T(t) V \mathbf{y}(t) + \mathbf{u}^T(t) R \mathbf{u}(t) \right), \quad (10.154)$$

where  $V$  and  $R$  are symmetric and positive definite matrices. The design of the entries of  $V$  and  $R$  can be performed based on the following specifics:

- a) concentration and exit flow rate variations contribute independently to the index value. This implies that the matrix  $V$  is diagonal, with positive diagonal entries  $v_{11}$  and  $v_{22}$ ;
- b) identical relative (percentage) variations of concentration and exit flow yield identical contribution to the index. Since a variation of  $\alpha$  percent of the exit flow is equal to  $\alpha \cdot 2 \cdot 10^{-4} \text{ m}^3/\text{sec}$ , and the same percent variation in concentration is equal to  $\alpha \cdot 12 \cdot 10^{-3} \text{ kmole/m}^3$ , for the two relative contributions to be the same we require

$$(2 \cdot 10^{-4} \cdot \alpha)^2 v_{11} = (12 \cdot 10^{-3} \cdot \alpha)^2 v_{22}.$$

The ratio between the terms  $v_{11}$  and  $v_{22}$ , and the resulting matrix  $V$ , are thus

$$\frac{v_{11}}{v_{22}} = \frac{(12 \cdot 10^{-3} \cdot \alpha)^2}{(2 \cdot 10^{-4} \cdot \alpha)^2} = 3600, \quad V = v_{22} \begin{bmatrix} 3600 & 0 \\ 0 & 1 \end{bmatrix}.$$

Obviously, it might be sometimes desirable to weight in different ways variations of flow and of concentration that are the same in percent. If, for example, we want to penalise in the same way the effect of a 10% flow variation and the effect of a 1% concentration variation, it must hold true

$$(2 \cdot 10^{-3})^2 v_{11} = (12 \cdot 10^{-3})^2 v_{22}$$

and therefore

$$\frac{v_{11}}{v_{22}} = \frac{(12 \cdot 10^{-3})^2}{(2 \cdot 10^{-3})^2} = 36, \quad V = v_{22} \begin{bmatrix} 36 & 0 \\ 0 & 1 \end{bmatrix}.$$

c) Variations of the two input flows contribute independently to the index value. Then the matrix  $R$  is also diagonal, with positive diagonal entries  $r_{11}$  and  $r_{22}$ .

d) The costs associated to percent variations of the same magnitude on first and second input flows are equal. A variation of  $\alpha$  percent on the first flow corresponds to  $\alpha \cdot 0,15 \cdot 10^{-3} \text{ m}^3/\text{sec}$ , while the same variation on the second flow corresponds to  $\alpha \cdot 0,05 \cdot 10^{-3} \text{ m}^3/\text{sec}$ .

In order for the costs to be equal, we require

$$(0,15 \cdot 10^{-3} \cdot \alpha)^2 r_{11} = (0,05 \cdot 10^{-3} \cdot \alpha)^2 r_{22}$$

and therefore

$$\frac{r_{11}}{r_{22}} = 9, \quad R = r_{22} \begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix}.$$

If, instead, we wanted to weigh differently the two variations, we would proceed as in (b).

Replacing  $y$  with  $H\mathbf{x}$  and setting  $\rho = r_{22}/v_{22}$ , the index (10.154) becomes

$$\begin{aligned} J &= \sum_{t=0}^{\infty} \left( \mathbf{x}^T(t) H^T V H \mathbf{x}(t) + \mathbf{u}^T(t) R \mathbf{u}(t) \right) \\ &= v_{22} \sum_{t=0}^{\infty} \left( \mathbf{x}^T(t) \begin{bmatrix} 0,36 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{x}(t) + \mathbf{u}^T(t) \rho \begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{u}(t) \right). \end{aligned} \quad (10.155)$$

As it is clear,  $v_{22}$  does not play any role in the minimisation process, and therefore it will be set to 1. On the other side,  $\rho$  determines the relative weight of the cost associated with the (deviation from the nominal value of the) control with respect to the cost associated with the (deviation from the nominal values of the) outputs.

Since  $Q = H^T V H$  is square and invertible, for any factorisation  $Q = C^T C$  the pair  $(\Phi, C)$  is observable. Since  $(\Phi, \Gamma)$  is reachable for any value of the sampling step, the optimal solution is stabilising for any value of the weight  $\rho$ .

The following table reports the feedback matrix  $K_{\infty}$  and the eigenvalues of the matrix  $\Phi + \Gamma K_{\infty}$  for decreasing values of  $\rho$ , namely, by attributing decreasing weight to control variations with respect to output deviations

| $\rho$    | $K_{\infty}$                                                          | autovalori di $\Phi + \Gamma K_{\infty}$  |
|-----------|-----------------------------------------------------------------------|-------------------------------------------|
| $10^4$    | $\begin{bmatrix} -0,0020 & 0,0001 \\ -0,0015 & -0,0016 \end{bmatrix}$ | 0,9435      0,8982                        |
| 10        | $\begin{bmatrix} -0,0351 & 0,0421 \\ -0,0540 & -0,1152 \end{bmatrix}$ | 0,7151      0,2452                        |
| 1         | $\begin{bmatrix} -0,0892 & 0,1138 \\ -0,0579 & -0,1669 \end{bmatrix}$ | 0,3680      0,0398                        |
| 0,1       | $\begin{bmatrix} -0,1341 & 0,1739 \\ -0,0510 & -0,1860 \end{bmatrix}$ | 0,0786      0,0043                        |
| $10^{-3}$ | $\begin{bmatrix} -0,1461 & 0,1900 \\ -0,0488 & -0,1902 \end{bmatrix}$ | $0,92 \cdot 10^{-3}$ $0,43 \cdot 10^{-4}$ |

As  $\rho$  tends to zero, the eigenvalues of  $\Phi + \Gamma K_{\infty}$  both tend to zero, and the matrix  $K_{\infty}$  tends to coincide with the one-step dead-beat controller. This phenomenon can be explained based on the following observations:

- when  $\rho$  tends to zero, only the output contributes to the index, therefore the latter becomes

$$\mathbf{x}_0^T H^T V H \mathbf{x}_0 + \mathbf{x}^T(1) H^T V H \mathbf{x}(1) + \mathbf{x}^T(2) H^T V H \mathbf{x}(2) + \dots \quad (10.156)$$

- the first addend of (10.156) does not depend on the input, and constitutes a lower bound on the index value;
- the system admits a dead-beat controller  $K$  that steers the system to zero after the first step;
- such a first step contributes to the index by

$$\mathbf{x}_0^T H^T V H \mathbf{x}_0 + \rho \mathbf{x}_0^T K^T R K \mathbf{x}_0 \sim \mathbf{x}_0^T H^T V H \mathbf{x}_0.$$

In any other step state and inputs are instead identically zero, and thus any other contribution to the index is zero. Therefore, the optimal input is the one that corresponds to the dead-beat control law.

The table further suggests that, as  $\rho$  tends to  $+\infty$ , the eigenvalues of  $\Phi + \Gamma K_\infty$  tend to those of  $\Phi$ . This is due to the fact that the open loop system is already asymptotically stable, and thus, as  $\rho$  diverges, the index only depends on the input. Consequently, the contribution of the input is minimised by choosing the zero input, namely  $K_\infty = 0$ .

**EXERCISE 10.4.1** [OPTIMAL CONTROL OF AN ASYMPTOTICALLY STABLE SYSTEM] Consider an asymptotically stable discrete time system (10.25), and the index

$$J(\mathbf{u}, \mathbf{x}_0) = \sum_{t=0}^{\infty} \left( \mathbf{x}^T(t) Q \mathbf{x}(t) + \rho \mathbf{u}^T(t) R \mathbf{u}(t) \right), \quad \rho > 0.$$

Assuming that the initial state  $\mathbf{x}_0$  is known, verify that, as  $\rho$  diverges, the optimal input tends to the zero input. In particular, show that, for any  $\epsilon > 0$ , there exists  $\bar{\rho} > 0$  such that if  $\rho > \bar{\rho}$  then the optimal control satisfies  $\sup_k \|\mathbf{u}(k)\| < \epsilon$ .

*# Hint: The value of the index for the unforced evolution is  $J(\mathbf{0}, \mathbf{x}_0) = \mathbf{x}_0^T Q \mathbf{x}_0 + \sum_{t=1}^{\infty} \mathbf{x}_t^T(t) Q \mathbf{x}_t(t) := M < \infty$ .*

*If we apply a bounded and non-zero input  $\mathbf{u}(\cdot)$ , and we set  $\|\mathbf{u}\|_\infty := \sup_k \|\mathbf{u}(k)\|$ , we obtain  $0 < \|\mathbf{u}\|_\infty < \infty$ . Since  $R$  is positive definite, there exists  $\alpha > 0$  for which  $\sum_{t=0}^{\infty} \mathbf{u}^T(t) R \mathbf{u}(t) \geq \alpha \|\mathbf{u}\|_\infty$ . Since for any applied input it holds true that  $\sum_{t=0}^{\infty} \mathbf{x}^T(t) R \mathbf{x}(t) \geq \mathbf{x}_0^T Q \mathbf{x}_0$ , we have  $M \geq \min_{\mathbf{u}} J(\mathbf{u}, \mathbf{x}_0) \geq \mathbf{x}_0^T Q \mathbf{x}_0 + \rho \alpha \|\mathbf{u}\|_\infty$ . Since at optimality the index cannot exceed its value in unforced evolution, it must hold true*

$$\frac{M - \mathbf{x}_0^T Q \mathbf{x}_0}{\alpha \rho} \geq \|\mathbf{u}\|_\infty,$$

*and, as  $\rho$  diverges, the inequality is satisfied only if  $\|\mathbf{u}\|_\infty$  tends to 0.*

## 10.5 Finite-horizon optimal control of linear continuous-time systems

We turn now to studying finite horizon optimal control for an  $n$ -dimensional linear continuous-time system

$$\frac{d\mathbf{x}}{dt} = F\mathbf{x}(t) + G\mathbf{u}(t), \quad (10.157)$$

with initial state

$$\mathbf{x}(0) = \mathbf{x}_0 \quad (10.158)$$

assuming that the cost index has the following structure:

$$J_T(\mathbf{u}, \mathbf{x}_0) = \int_0^T \left( \mathbf{x}^T(t) Q \mathbf{x}(t) + \mathbf{u}^T(t) R \mathbf{u}(t) \right) dt + \mathbf{x}^T(T) S \mathbf{x}(T), \quad (10.159)$$

where  $Q$  and  $S$  are  $n \times n$  symmetric positive semidefinite matrices, and  $R$  is a  $m \times m$  symmetric and positive definite matrix. As it happens for other branches of System Theory, the fundamental results are conceptually analogous to those for discrete-time systems. Since, however, the mathematical properties of the update equations are different, the solution techniques that we employ have to be different, too.

### 10.5.1 Index minimisation

In order to determine the input  $\mathbf{u}(\cdot)$  that minimises the index (10.159), we shall follow a procedure that is reminiscent of the argument in section 10.2.

**Lemma 2.5.1** [INDEX REWRITING] *If  $\mathbf{x}(\cdot)$  and  $\mathbf{u}(\cdot)$  satisfy the differential equation in (10.157) and if  $M(\cdot)$  is an  $n \times n$  matrix-valued function, differentiable in the interval  $[0, T]$ , then the index (10.159) can be expressed in the form*

$$\begin{aligned} J_T(\mathbf{u}, \mathbf{x}_0) &= \mathbf{x}_0^T M(0) \mathbf{x}_0 + \mathbf{x}^T(T) [S - M(T)] \mathbf{x}(T) \\ &+ \int_0^T [\mathbf{u}^T(t) \quad \mathbf{x}^T(t)] \begin{bmatrix} R & G^T M(t) \\ M(t)G & Q + F^T M(t) + M(t)F + \dot{M}(t) \end{bmatrix} \begin{bmatrix} \mathbf{u}(t) \\ \mathbf{x}(t) \end{bmatrix} dt. \end{aligned} \quad (10.160)$$

PROOF First of all, consider the chain of equalities

$$\begin{aligned} \mathbf{x}^T(t) M(t) \mathbf{x}(t) \Big|_0^T &= \int_0^T \frac{d}{dt} [\mathbf{x}^T(t) M(t) \mathbf{x}(t)] dt \\ &= \int_0^T [\dot{\mathbf{x}}^T(t) M(t) \mathbf{x}(t) + \mathbf{x}^T(t) \dot{M}(t) \mathbf{x}(t) + \mathbf{x}^T(t) M(t) \dot{\mathbf{x}}(t)] dt \\ &= \int_0^T [(F\mathbf{x}(t) + G\mathbf{u}(t))^T M(t) \mathbf{x}(t) + \mathbf{x}^T(t) \dot{M}(t) \mathbf{x}(t) + \mathbf{x}^T(t) M(t) (F\mathbf{x}(t) + G\mathbf{u}(t))] dt \\ &= \int_0^T [\mathbf{u}^T(t) \quad \mathbf{x}^T(t)] \begin{bmatrix} 0 & G^T M(t) \\ M(t)G & F^T M(t) + M(t)F + \dot{M}(t) \end{bmatrix} \begin{bmatrix} \mathbf{u}(t) \\ \mathbf{x}(t) \end{bmatrix} dt. \end{aligned} \quad (10.161)$$

Then (10.160) can be obtained by summing to both members of (10.161) the equation that defines the index

$$J_T(\mathbf{u}, \mathbf{x}_0) = \mathbf{x}^T(T) S \mathbf{x}(T) + \int_0^T [\mathbf{u}^T(t) \quad \mathbf{x}^T(t)] \begin{bmatrix} R & 0 \\ 0 & Q \end{bmatrix} \begin{bmatrix} \mathbf{u}(t) \\ \mathbf{x}(t) \end{bmatrix} dt$$

■

We can now exploit the arbitrariness of  $M(\cdot)$  in (10.160) in order to further reshape the index structure. Analogously to the discrete-time case, we consider the matrix function  $M(\cdot)$  obtained by integrating backwards the (non-linear) Riccati Differential Equation

$$-\dot{M}(t) = Q + F^T M(t) + M(t)F - M(t)GR^{-1}G^T M(t) \quad (10.162)$$

over the time horizon  $[0, T]$ , and with final condition  $M(T) = S$ . Under our assumptions on  $R, Q$ , and  $S$ , one can show that the solution to (10.162) exists and is finite over the time horizon  $(-\infty, T]$ , and that  $M(t)$  is symmetric and positive semidefinite everywhere in such an interval. Such a result allows us to state and prove the following Lemma.

**Lemma 10.5.2** [SQUARE COMPLETION] *Let  $M(\cdot)$  be a solution to the Riccati Differential Equation (10.162) over the interval  $[0, T]$ , corresponding to an arbitrary symmetric and*



p.s.d. final condition  $M(T)$ . For any initial state  $\mathbf{x}(0) = \mathbf{x}_0$  of the system, and for any input  $\mathbf{u}(\cdot)$ , the index can be expressed as

$$\begin{aligned} J_T(\mathbf{u}, \mathbf{x}_0) &= \mathbf{x}^T(0)M(0)\mathbf{x}(0) + \mathbf{x}^T(T)[S - M(T)]\mathbf{x}(T) \\ &+ \int_0^T [R\mathbf{u}(t) + G^T M(t)\mathbf{x}(t)]^T R^{-1} [R\mathbf{u}(t) + G^T M(t)\mathbf{x}(t)] dt. \end{aligned} \quad (10.163)$$

PROOF Since  $M(\cdot)$  is a symmetric and p.s.d. solution to the Riccati Differential Equation (10.162), but substituting in (10.160) the block  $Q + F^T M(t) + M(t)F + \dot{M}(t)$  with  $M(t)GR^{-1}G^T M(t)$ , one gets

$$\begin{aligned} J_T(\mathbf{u}, \mathbf{x}_0) &= \mathbf{x}_0^T M(0)\mathbf{x}_0 + \mathbf{x}^T(T)[S - M(T)]\mathbf{x}(T) \\ &+ \int_0^T [\mathbf{u}^T(t) \quad \mathbf{x}^T(t)] \begin{bmatrix} R & G^T M(t) \\ M(t)G & M(t)GR^{-1}G^T M(t) \end{bmatrix} \begin{bmatrix} \mathbf{u}(t) \\ \mathbf{x}(t) \end{bmatrix} dt \\ &= \mathbf{x}_0^T M(0)\mathbf{x}_0 + \mathbf{x}^T(T)[S - M(T)]\mathbf{x}(T) \\ &+ \int_0^T [\mathbf{u}^T(t) \quad \mathbf{x}^T(t)] \begin{bmatrix} R \\ M(t)G \end{bmatrix} R^{-1} [R \quad G^T M(t)] \begin{bmatrix} \mathbf{u}(t) \\ \mathbf{x}(t) \end{bmatrix} dt \end{aligned}$$

which is (10.163). ■

The following result is now straightforward.

**Proposition 10.5.3** [INDEX MINIMISATION] *Let  $M(\cdot)$  be the solution to the Riccati Differential Equation (10.162) over  $[0, T]$ , corresponding to the final condition  $M(T) = S$ . Then*

- the minimum value of the index with initial state  $\mathbf{x}(0) = \mathbf{x}_0$  is

$$\min_{\mathbf{u}} J_T(\mathbf{u}, \mathbf{x}_0) = \mathbf{x}_0^T M(0)\mathbf{x}_0; \quad (10.164)$$

- the optimal input that determines the minimum value of the index is unique and corresponds to the state feedback control law

$$\mathbf{u}_{\text{opt}}(t) = -R^{-1}G^T M(t)\mathbf{x}(t). \quad (10.165)$$

PROOF We start by plugging in the square completion lemma the solution to the Riccati Differential Equation corresponding to  $M(T) = S$ :

$$J_T(\mathbf{u}, \mathbf{x}_0) = \mathbf{x}^T(0)M(0)\mathbf{x}(0) + \int_0^T [R\mathbf{u}(t) + G^T M(t)\mathbf{x}(t)]^T R^{-1} [R\mathbf{u}(t) + G^T M(t)\mathbf{x}(t)] dt.$$

Since  $R^{-1}$  is positive definite, the integral is non-negative for any input  $\mathbf{u}(\cdot)$ , and it is zero only if  $R\mathbf{u}(t) + G^T M(t)\mathbf{x}(t) \equiv \mathbf{0}$ . Therefore, the minimum value of  $J_T$  is only obtained when  $\mathbf{u}(\cdot)$  satisfies (10.165), and the minimum is  $\mathbf{x}^T(0)M(0)\mathbf{x}(0)$ . ■

### 10.5.2 On the Riccati Differential Equation

In order to actually solve the optimal problem with the proposed method, we need to devise a procedure to integrate the Riccati Differential Equation, which is a non-linear equation. In other words, in order to obtain the control law we can overlook the mathematical machinery which is required to establish the existence of the solution, but we cannot overlook the methods that allow to compute it, whenever its existence is guaranteed. In practice, except for very simple cases, determining  $M(t)$  is done numerically.

Instead of studying the numerical methods that have been proposed to this aim, we consider an integration method for (10.162) that recasts the solution to the non-linear equation (10.162) into the solution to a linear equation with doubled dimension. Such a procedure, which is mostly of theoretical interest, requires the definition of the following *Hamiltonian matrix*

$$\mathcal{H} = \begin{bmatrix} F & -GR^{-1}G^T \\ -Q & -F^T \end{bmatrix}. \quad (10.166)$$

which we shall also employ at the end of this chapter. The method consists of the following steps:

- i) Associate to the Riccati Equation (10.162) the linear differential equation

$$\frac{d}{dt} \begin{bmatrix} \Phi_1(t) \\ \Phi_2(t) \end{bmatrix} = \begin{bmatrix} F & -GR^{-1}G^T \\ -Q & -F^T \end{bmatrix} \begin{bmatrix} \Phi_1(t) \\ \Phi_2(t) \end{bmatrix}, \quad \begin{bmatrix} \Phi_1(T) \\ \Phi_2(T) \end{bmatrix} = \begin{bmatrix} I_n \\ S \end{bmatrix} \quad (10.167)$$

in the matrix unknowns  $\Phi_1(\cdot)$  and  $\Phi_2(\cdot)$ , whose dimensions are  $n \times n$ .

- ii) Compute, by means of known integration methods, the solution to (10.167)

$$\begin{bmatrix} \Phi_1(t) \\ \Phi_2(t) \end{bmatrix} = e^{\mathcal{H}(t-T)} \begin{bmatrix} I_n \\ S \end{bmatrix}. \quad (10.168)$$

- iii) The solution to (10.162) can be then obtained as

$$M(t) = \Phi_2(t)\Phi_1^{-1}(t). \quad (10.169)$$

To verify that the method is correct, we shall proceed in an indirect way, by exploiting the fact that, under the given assumptions, the Riccati Equation (10.162), with final condition  $M(T) = S$ , is solvable over  $[0, T]$ , and that the optimal feedback control law can be obtained by solving (10.162) and assuming  $\mathbf{u}_{\text{opt}}(t) = -R^{-1}G^T M(t)\mathbf{x}(t)$ .

If  $\mathbf{u}_{\text{opt}}(t)$  is input to the system, the state evolves according to the linear time-varying equation

$$\dot{\mathbf{x}}(t) = (F - GR^{-1}G^T M(t))\mathbf{x}(t). \quad (10.170)$$

Consider the two matrices  $\Phi_1(\cdot)$  and  $\Phi_2(\cdot)$ , of which the former is the solution to the equation

$$\dot{\Phi}_1(t) = (F - GR^{-1}G^T M(t))\Phi_1(t), \quad \Phi_1(T) = I_n, \quad (10.171)$$

and the latter is defined as

$$\Phi_2(t) = M(t)\Phi_1(t). \quad (10.172)$$

For linear time-varying systems, such as (10.170), it is possible to show <sup>6</sup> that the transition matrix exists and is invertible for every  $t$ . Therefore  $\Phi_1(\cdot)$ , which is the transition matrix of (10.170), is invertible.

In order to establish that the matrices  $\Phi_1(\cdot)$  and  $\Phi_2(\cdot)$  can be obtained by computing the exponential of the matrix (10.166), it is sufficient to verify that they satisfy (10.167). In fact, it follows from (10.171) and (10.172) that

$$\dot{\Phi}_1(t) = F\Phi_1(t) - GR^{-1}G^T\Phi_2(t)$$

$$\Phi_1(T) = I_n,$$

whereas, by differentiating (10.172) and recalling that  $M(\cdot)$  solves the Riccati Equation (10.162), one has

$$\begin{aligned}\dot{\Phi}_2 &= \dot{M}\Phi_1 + M\dot{\Phi}_1 \\ &= -Q\Phi_1 - F^T M\Phi_1 - MF\Phi_1 + MGR^{-1}G^T M\Phi_1 + MF\Phi_1 - MGR^{-1}G^T M\Phi_1 \\ &= -Q\Phi_1 - F^T M\Phi_1 = -Q\Phi_1 - F^T \Phi_2\end{aligned}$$

$$\Phi_2(T) = M(T)\Phi_1(T) = S$$

Having therefore shown that  $\Phi_1(\cdot)$  and  $\Phi_2(\cdot)$  are in fact solutions to (10.167), (10.169) follows immediately from (10.172).

**Example 10.5.1** [THE RENDEZ-VOUS PROBLEM] Assume that at  $t = 0$  a friction-free, unitary mass cart has position  $s(0)$  and speed  $\dot{s}(0)$ . We want to apply, over the time horizon  $[0, T]$ , a force  $u(\cdot)$  to minimise the index

$$J_T = \int_0^T u^2(t)dt + [s(T) \quad \dot{s}(T)] \begin{bmatrix} c_1 & 0 \\ 0 & c_2 \end{bmatrix} \begin{bmatrix} s(T) \\ \dot{s}(T) \end{bmatrix}, \quad (10.173)$$

where  $c_1$  and  $c_2$  are positive constants.

Minimising the index  $J_T$  corresponds to minimising final position and final speed (squared), at the same time penalising the norm of the control applied over  $[0, T]$ . The relative values of  $c_1$  and  $c_2$  in the expression  $c_1 s^2(T) + c_2 \dot{s}^2(T)$  account for the relative importance of the gaps with respect to the value 0 of the final position and the speed, respectively. It is worth noticing that the techniques seen in Section 5.4 allow one to determine a function  $u(\cdot)$  such that  $s(T) = \dot{s}(T) = 0$ . However, the norm of such an  $u(\cdot)$  is in general large, therefore, due to the penalty on the control energy, this input does not minimise  $J_T$ .

Setting

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} s \\ \dot{s} \end{bmatrix},$$

the state model is

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t), \quad (10.174)$$

while the matrices  $Q, R, S$  in the index are

$$Q = 0, \quad R = 1, \quad S = \begin{bmatrix} c_1 & 0 \\ 0 & c_2 \end{bmatrix}.$$

The Riccati Equation associated to the minimisation problem is

$$-\dot{M}(t) = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} M(t) + M(t) \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} - M(t) \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \quad (10.175)$$

<sup>6</sup>By extending the methods discussed in Chapter 2.

with final condition

$$M(T) = \begin{bmatrix} c_1 & 0 \\ 0 & c_2 \end{bmatrix}. \quad (10.176)$$

Due to the particularly simple structure of the matrices that appear in (10.175), we can analytically integrate the equation, by computing the exponential of the Hamiltonian matrix

$$\mathcal{H} = \begin{bmatrix} F & -GR^{-1}G^T \\ -Q & -F^T \end{bmatrix} = \begin{bmatrix} 0 & 1 & | & 0 & 0 \\ 0 & 0 & | & 0 & -1 \\ - & - & - & - & - \\ 0 & 0 & | & 0 & 0 \\ 0 & 0 & | & -1 & 0 \end{bmatrix}.$$

Since  $\mathcal{H}$  is nilpotent, with  $\mathcal{H}^4 = 0$ , computing its exponential only requires the first terms of its series expansion. By denoting for the sake of simplicity  $\tau = t - T$ , one thus obtains:

$$\begin{bmatrix} \Phi_1(t) \\ \Phi_2(t) \end{bmatrix} = e^{\mathcal{H}\tau} \begin{bmatrix} I_2 \\ S \end{bmatrix} = \begin{bmatrix} 1 & \tau & | & \tau^3/6 & -\tau^2/2 \\ 0 & 1 & | & \tau^2/2 & -\tau \\ - & - & - & - & - \\ 0 & 0 & | & 1 & 0 \\ 0 & 0 & | & -\tau & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ - & - \\ c_1 & 0 \\ 0 & c_2 \end{bmatrix},$$

from which

$$\begin{aligned} M(t) = \Phi_2(t)\Phi_1^{-1}(t) &= \begin{bmatrix} c_1 & 0 \\ -c_1\tau & c_2 \end{bmatrix} \begin{bmatrix} 1 + c_1\tau^3/6 & \tau(1 - c_2\tau/2) \\ c_1\tau^2/2 & 1 - c_2\tau \end{bmatrix}^{-1} \\ &= \begin{bmatrix} c_1 & 0 \\ -c_1\tau & c_2 \end{bmatrix} \begin{bmatrix} 1 + c_2\tau & -\tau(1 - c_2\tau/2) \\ -c_1\tau^2/2 & 1 + c_1\tau^3/6 \end{bmatrix} \frac{1}{c_1c_2D(\tau)}, \end{aligned} \quad (10.177)$$

with

$$D(\tau) = \left(\frac{1}{c_2} - \tau\right) \left(\frac{1}{c_1} + \frac{\tau^3}{6}\right) - \frac{\tau^3}{2} \left(\frac{1}{c_2} - \frac{\tau}{2}\right)$$

so that the optimal control law  $u_{\text{opt}}(t) = [0 \quad -1] M(t)\mathbf{x}(t)$  takes the form

$$u_{\text{opt}}(t) = \frac{1}{D(t-T)} \begin{bmatrix} -\frac{(t-T)^2}{2} + \frac{t-T}{c_2} & \frac{(t-T)^3}{3} - \frac{(t-T)^2}{c_2} - \frac{1}{c_1} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}. \quad (10.178)$$

We examine now the behaviour of the optimal solution (10.178) for extreme values of the parameters  $c_1$  and  $c_2$  in the index  $J_T$ :

a) If  $c_1 \rightarrow 0$ , equation (10.177) tends to

$$M(t) = \begin{bmatrix} 0 & 0 \\ 0 & c_2 \end{bmatrix} \begin{bmatrix} 1 + c_2\tau & -\tau(1 - c_2\tau/2) \\ 0 & 1 \end{bmatrix} \frac{1}{1 - \tau c_2} = \begin{bmatrix} 0 & 0 \\ 0 & \frac{c_2}{1 - \tau c_2} \end{bmatrix}$$

and the optimal input tends to

$$u_{\text{opt}}(t) = \left(t - T - c_2^{-1}\right)^{-1} x_2(t).$$

In this case, since the index  $J_T$  does not depend on the final position  $x_1(T)$ , the problem is to reduce as much as possible the final speed  $x_2(T)$  of the cart while at the same time limiting the employed control energy. Notice that, if  $x_2(0) = 0$ , the optimal control is, obviously,  $u_{\text{opt}}(t) = 0$ ,  $\forall t \in [0, T]$ .

b) If  $c_2 \rightarrow 0$ , equation (10.177) tends to

$$M(t) = \begin{bmatrix} c_1 & 0 \\ -c_1\tau & 0 \end{bmatrix} \begin{bmatrix} 1 & -\tau \\ -c_1\tau^2/2 & 1 + c_1\tau^3/6 \end{bmatrix} \frac{1}{1 - c_1\tau^3/3}$$

and the optimal input tends to

$$u_{\text{opt}}(t) = [0 \quad -1] M(t) \mathbf{x}(t) = \frac{1}{c_1^{-1} - (t-T)^3/3} [t-T \quad -(t-T)^2] \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}.$$

In this case, since the  $J_T$  does not include the final speed  $x_2(T)$ , the goal is to steer the cart close to the position 0 at the final time, disregarding the actual speed of the cart at such an instant.

c) Letting both  $c_1$  and  $c_2$  tend to  $+\infty$  corresponds to requiring that, at time  $T$ , the system exactly reaches the zero state. Equation (10.178) then becomes

$$u_{\text{opt}}(t) = \frac{12}{(t-T)^4} \begin{bmatrix} -\frac{(t-T)^2}{2} & \frac{(t-T)^3}{3} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} -\frac{6}{(t-T)^2} & \frac{4}{t-T} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} \quad (10.179)$$

The control law coincides with that determined at the end of Section 5.4 to steer the system from the initial state  $\mathbf{x}(0) = \mathbf{x}_0$  to the final state  $\mathbf{x}(T) = \mathbf{0}$ , minimising the input norm  $(\int_0^T u^2(\sigma) d\sigma)^{1/2}$ . In fact, the feedback matrix  $K(t)$ , computed as in (5.59):

$$\begin{aligned} K(t) &= -G^T e^{F^T(T-t)} \mathcal{W}_{t,T}^{-1} e^{F(t-T)} = -G^T e^{F^T(T-t)} \left( \int_t^T e^{F(T-\xi)} G G^T e^{F^T(T-\xi)} d\xi \right)^{-1} e^{F(t-T)} \\ &= -[0 \quad 1] \begin{bmatrix} 1 & 0 \\ T-t & 1 \end{bmatrix} \begin{bmatrix} 12(T-t)^{-3} & -6(T-t)^{-2} \\ -6(T-t)^{-2} & 4(T-t)^{-1} \end{bmatrix} \begin{bmatrix} 1 & T-t \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -\frac{6}{(t-T)^2} & \frac{4}{t-T} \end{bmatrix} \end{aligned}$$

is the same obtained in (10.179). Observe that the elements of  $K(t)$  diverge as  $t \rightarrow T$ . However, this does not imply that the optimal input also diverges as  $t \rightarrow T$ , since it is obtained by multiplying the matrix  $K(t)$  by the state  $\mathbf{x}(t)$ , whose components tend to zero as  $t \rightarrow T$ .

## 10.6 Infinite horizon optimal control of continuous-time linear systems

We conclude the study of optimal control by addressing infinite horizon optimal control problems for continuous-time linear systems. As we shall see, the theory is based on solving an algebraic Riccati equation. Since the proofs are analogous to the discrete-time case, they are omitted and left as an exercise to the reader.

**Proposition 10.6.1** [ALGEBRAIC RICCATI EQUATION (CARE)] *If the linear continuous-time system (10.157) is stabilisable and the symmetric matrices  $R$  and  $Q$  are positive definite and positive semidefinite, respectively, then*

i) *the Riccati Differential Equation*

$$-\dot{M}(t) = Q + F^T M(t) + M(t)F - M(t)GR^{-1}G^T M(t)$$

*with final condition  $M(0) = 0$  can be solved over  $(-\infty, 0]$  and the solution converges, as  $t \rightarrow -\infty$ , to a symmetric p.s.d. matrix  $M_\infty$  satisfying the Algebraic Riccati Equation (CARE), i.e.,*

$$0 = Q + F^T M + MF - MGR^{-1}G^T M. \quad (10.180)$$

ii) *For any other symmetric and p.s.d. solution  $\bar{M}$  of (10.180), the difference  $\bar{M} - M_\infty$  is positive semidefinite.*

- iii) If  $C$  is any matrix such that  $Q = C^T C$ , then  $M_\infty$  is positive definite if and only if the pair  $(F, C)$  is observable.

**Proposition 10.6.2** [OPTIMAL CONTROL] *Let the system (10.157) be stabilisable and let  $M_\infty$  be the solution to the Algebraic Riccati Equation (10.180) defined in the previous Proposition 10.6.1. Then the control law*

$$\mathbf{u}_\infty(t) = K_\infty \mathbf{x}(t), \quad \text{with } K_\infty = -R^{-1}G^T M_\infty, \quad (10.181)$$

*minimises the quadratic index*

$$J(\mathbf{u}, \mathbf{x}_0) = \int_0^{+\infty} \left( \mathbf{x}^T(t) Q \mathbf{x}(t) + \mathbf{u}^T(t) R \mathbf{u}(t) \right) dt. \quad (10.182)$$

*The real part of the eigenvalues of the matrix  $F + GK_\infty$  is negative, namely the optimal control (10.181) is stabilisable, if and only if the matrix*

$$\begin{bmatrix} sI_n - F \\ C \end{bmatrix} \quad (10.183)$$

*is of full rank for any  $s \in \mathbb{C}$  with  $\operatorname{Re}(s) \geq 0$ , namely if the pair  $(F, C)$  is detectable.*

**Proposition 10.6.3** [STABILISING SOLUTIONS OF CARE] *Let  $(F, G)$  stabilisable and  $Q = C^T C$ .*

- i) *Equation (10.180) has at most one symmetric and p.s.d. solution  $M_s$  associated with a stabilising control law*

$$\mathbf{u}_s(t) = -R^{-1}G^T M_s \mathbf{x}(t) = K_s \mathbf{x}(t);$$

- ii) *such a solution exists if and only if the matrix (10.183) is of full rank for any imaginary value of the variable  $s$ ;*  
 iii) *if the stabilising solution  $M_s$  exists, and if  $\bar{M}$  is any symmetric and p.s.d. solution to (10.180), then the matrix  $M_s - \bar{M}$  is p.s.d.;*  
 iv) *if the stabilising solution  $M_s$  exists, it can be obtained by taking the limit as  $t \rightarrow -\infty$  of the solution  $M(t)$  of the Riccati Differential Equation, corresponding to an arbitrary positive definite final condition  $M(0)$ .*

**Example 10.6.1** Consider the first order system

$$\dot{x}(t) = ax(t) + u(t).$$

We want to design a control law  $u(\cdot)$  that minimises the index

$$J(u, x_0) = \int_0^{+\infty} (qx^2(t) + ru^2(t))dt,$$

where  $q$  and  $r$  are positive constants. The Algebraic Riccati Equation associated with the problem is

$$0 = -q - 2am + m^2/r$$

whose solutions are

$$m = ar \pm \sqrt{a^2 r^2 + qr}.$$

For the optimal problem the only admissible solution is the positive one, and the corresponding control law (10.181) is thus

$$u_\infty(t) = -r^{-1}(ar + \sqrt{a^2 r^2 + qr})x(t).$$

**Example 10.6.2** [POSITION CONTROL] The arm of an antenna with moment of inertia  $\mathcal{J}$  and fluid friction coefficient  $\beta$  is subject of a rotational movement by a direct current motor. The DC motor yields a torque  $c(\cdot)$  proportional to the current applied to the motor, and it is supplied by a voltage controlled linear amplifier.

By denoting by  $\theta$  the angular position of the arm and by  $v$  the voltage applied to the amplifier, we obtain the equations

$$\begin{aligned} \mathcal{J}\ddot{\theta}(t) + \beta\dot{\theta}(t) &= c(t), \\ c(t) &= \kappa v(t), \end{aligned}$$

from which, putting  $\xi_1 = \theta$  and  $\xi_2 = \dot{\theta}$ , and taking as output the angular position of the arm, we obtain the state model equation

$$\begin{aligned} \begin{bmatrix} \dot{\xi}_1(t) \\ \dot{\xi}_2(t) \end{bmatrix} &= \begin{bmatrix} 0 & 1 \\ 0 & -\alpha \end{bmatrix} \begin{bmatrix} \xi_1(t) \\ \xi_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ \gamma \end{bmatrix} v(t) \\ y(t) &= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \xi_1(t) \\ \xi_2(t) \end{bmatrix} \end{aligned}$$

in which we set  $\alpha = \beta/\mathcal{J}$  and  $\gamma = \kappa/\mathcal{J}$ .

The objective is to determine a state feedback control law that stabilises the system around a desired angle  $\bar{\theta}_1$ , namely, that steers back to  $\bar{\theta}_1$  the antenna if a disturbance moves the angular position of the arm away from it.

In the new state variables  $x_1 := \xi_1 - \bar{\theta}$ ,  $x_2 := \xi_2$  the system equations are

$$\begin{aligned} \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} &= \begin{bmatrix} 0 & 1 \\ 0 & -\alpha \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ \gamma \end{bmatrix} v(t) = F\mathbf{x}(t) + \mathbf{g}v(t) \\ y(t) &= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \bar{\theta} \end{aligned} \quad (10.184)$$

and, in the new variables, the desired equilibrium state is the state  $\mathbf{0}$ . We introduce the index

$$J = \int_0^{+\infty} (x_1^2(t) + rv^2(t))dt = \int_0^{+\infty} \left( \begin{bmatrix} x_1(t) & x_2(t) \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + rv^2(t) \right) dt, \quad (10.185)$$

and we consider the problem of its minimisation. Since the matrix  $Q$  factorises in the form  $C^T C = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$  and the pair  $(F, C)$  is detectable, the feedback matrix  $K_\infty$ , that provides the optimal control law by

$$v_\infty(t) = K_\infty \mathbf{x}(t) = K_\infty \begin{bmatrix} \xi_1 - \bar{\theta} \\ \xi_2 \end{bmatrix},$$

is such that the equilibrium  $\mathbf{x} = \mathbf{0}$  for the system (10.184) is asymptotically stable.

The Algebraic Riccati Equation associated with the problem is

$$0 = - \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 0 \\ 1 & -\alpha \end{bmatrix} M - M \begin{bmatrix} 0 & 1 \\ 0 & -\alpha \end{bmatrix} + M \begin{bmatrix} 0 \\ \gamma \end{bmatrix} r^{-1} \begin{bmatrix} 0 & \gamma \end{bmatrix} M. \quad (10.186)$$

By writing

$$M = \begin{bmatrix} m_1 & m_2 \\ m_2 & m_3 \end{bmatrix},$$

we obtain the system of equations

$$0 = 1 - \frac{\gamma^2}{r} m_2^2$$

$$\begin{aligned}
0 &= -\frac{\gamma^2}{r} m_2 m_3 + m_1 - \alpha m_2 \\
0 &= -\frac{\gamma^2}{r} m_3^2 + 2m_2 - 2\alpha m_3.
\end{aligned} \tag{10.187}$$

The only solution to (10.187) that yields a (semi)definite positive matrix  $M_\infty$  is

$$\begin{aligned}
m_1 &= \frac{\sqrt{r}}{\gamma} \alpha^2 + \frac{2\gamma}{\sqrt{r}} \\
m_2 &= \frac{\sqrt{r}}{\gamma} \\
m_3 &= \frac{r}{\gamma^2} \left( -\alpha + \sqrt{\alpha^2 + \frac{2\gamma}{\sqrt{r}}} \right),
\end{aligned} \tag{10.188}$$

and the corresponding feedback matrix

$$K_\infty = [k_1 \quad k_2] = -\frac{1}{r} [0 \quad \gamma] M_\infty = \left[ \frac{1}{\sqrt{r}} \quad \frac{1}{\gamma} \left( -\alpha + \sqrt{\alpha^2 + \frac{2\gamma}{\sqrt{r}}} \right) \right].$$

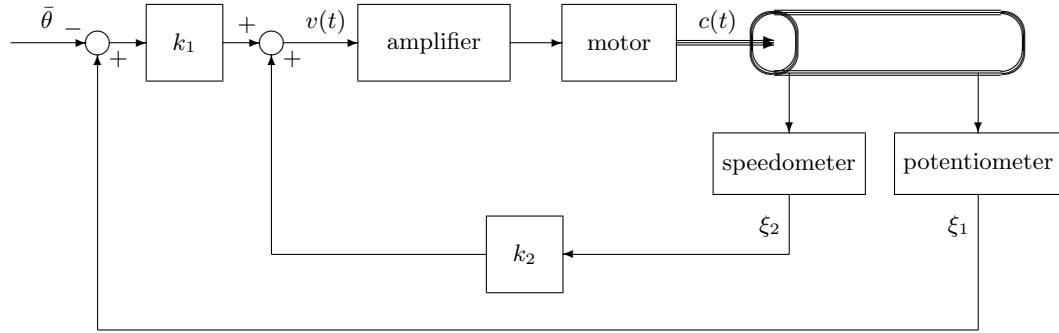


figura 10.6.1

We conclude this section by briefly addressing some questions that concern the eigenvalue allocation when the state feedback matrix is built upon the stabilising solution to the Algebraic Riccati Equation (10.180). To this aim, the main tool is the analysis of the spectrum of the Hamiltonian matrix, as shown in the following proposition.

**Proposition 10.6.4** [EIGENVALUES OF THE HAMILTONIAN MATRIX] *Assume that (10.183) is of full rank for any imaginary value of  $s$ , and let  $K_s$  be the feedback matrix associated with the stabilising solution to the Riccati Equation. Then the eigenvalues of the matrix  $F + GK_s$  coincide with the eigenvalues with negative real part of the Hamiltonian matrix  $\mathcal{H}$ .*

PROOF By performing on  $\mathcal{H}$  the similarity transformation induced by the matrix

$$T = \begin{bmatrix} I_n & 0 \\ M_s & I_n \end{bmatrix},$$

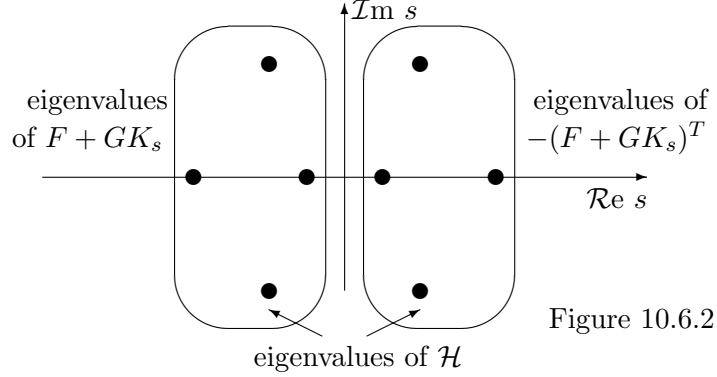
we obtain

$$\begin{aligned}
T^{-1} \mathcal{H} T &= \begin{bmatrix} I_n & 0 \\ -M_s & I_n \end{bmatrix} \begin{bmatrix} F & -GR^{-1}G^T \\ -Q & -F^T \end{bmatrix} \begin{bmatrix} I_n & 0 \\ M_s & I_n \end{bmatrix} \\
&= \begin{bmatrix} F - GR^{-1}G^T M_s & -GR^{-1}G^T \\ -Q - M_s F - F^T M_s + M_s G R^{-1} G^T M_s & -F^T + M_s G R^{-1} G^T \end{bmatrix}
\end{aligned}$$



$$= \begin{bmatrix} F + GK_s & -GR^{-1}G^T \\ 0 & -(F + GK_s)^T \end{bmatrix}. \quad (10.189)$$

Therefore, the eigenvalues of  $\mathcal{H}$  are those of  $F + GK_s$  – whose real part is negative –, and those of  $-(F + GK_s)^T$ , which are located symmetrically with respect to the imaginary axes – and whose real part is therefore positive. ■



**Example 10.6.3** Consider a stabilisable single input system  $(F, \mathbf{g})$ , with index

$$J = \int_0^\infty \mathbf{x}^T(t)Q\mathbf{x}(t) + ru^2(t)dt.$$

We assume that  $Q$  has rank 1, so that it can be expressed in the form  $Q = \mathbf{q}\mathbf{q}^T$ , where  $\mathbf{q} \neq \mathbf{0}$  is a column vector, and that the pair  $(F, \mathbf{q}^T)$  is detectable, so that the solution  $M_\infty$  of the (CARE) coincides with the stabilising solution  $M_s$ .

The characteristic polynomial of the Hamiltonian matrix is

$$\begin{aligned} \det(sI_{2n} - \mathcal{H}) &= \det \begin{bmatrix} sI_n - F & -\frac{1}{r}\mathbf{g}\mathbf{g}^T \\ -\mathbf{q}\mathbf{q}^T & sI_n + F^T \end{bmatrix} \\ &= \det(sI_n - F) \det \left( (sI_n + F^T) - \frac{1}{r}\mathbf{g}\mathbf{g}^T(sI_n - F)^{-1}\mathbf{q}\mathbf{q}^T \right) \\ &= \det(sI_n - F) \det(sI_n + F^T) \det \left( I_n - \frac{1}{r}\mathbf{g}\mathbf{g}^T(sI_n - F)^{-1}\mathbf{q}\mathbf{q}^T(sI_n + F^T)^{-1} \right) \\ &= \det(sI_n - F) \det(sI_n + F^T) \det \left( 1 - \frac{1}{r}\mathbf{q}^T(sI_n + F^T)^{-1}\mathbf{g}\mathbf{g}^T(sI_n - F)^{-1}\mathbf{q} \right) \\ &= (-1)^n \det(sI_n - F) \det(-sI_n - F^T) \det \left( 1 + \frac{1}{r}\mathbf{q}^T(-sI_n - F^T)^{-1}\mathbf{g}\mathbf{g}^T(sI_n - F)^{-1}\mathbf{q} \right). \end{aligned}$$

By setting

$$\Delta_F(s) = \det(sI_n - F), \quad \phi(s) = \mathbf{g}^T \text{adj}(sI_n - F)\mathbf{q},$$

one obtains

$$\begin{aligned} \det(sI_{2n} - \mathcal{H}) &= (-1)^n \Delta_F(s) \Delta_F(-s) \left( 1 + \frac{1}{r} \frac{\phi(-s)}{\Delta_F(-s)} \frac{\phi(s)}{\Delta_F(s)} \right) \\ &= (-1)^n \left( \Delta_F(s) \Delta_F(-s) + \frac{1}{r} \phi(s) \phi(-s) \right) \end{aligned}$$

and the eigenvalues of the closed loop system with feedback matrix  $K_\infty = -\frac{1}{r}\mathbf{g}^T M_\infty$  are the solutions, with negative real part, of equation

$$\Delta_F(s) \Delta_F(-s) + \frac{1}{r} \phi(s) \phi(-s) = 0. \quad (10.190)$$

The parameter  $r$  determines the relative weight given to the input in the index. In particular, a large  $r$  forces the amplitude of the input function to be small, and viceversa, small values of  $r$  allow the input function to take large values.

It is interesting to analyse how the distribution of the zeros of (10.190) varies as a function of  $r$ .

Let  $\zeta_i$ ,  $i = 1, 2, \dots, h$ , be the zeros of  $\phi(s)$  and  $\lambda_i$ ,  $i = 1, 2, \dots, n$ , be the zeros of  $\Delta_F(s)$ . If we write

$$\phi(s) = \alpha \prod_{i=1}^h (s - \zeta_i), \quad \Delta_F(s) = \prod_{i=1}^n (s - \lambda_i),$$

equation (10.190) becomes

$$\prod_{i=1}^n (s - \lambda_i)(s + \lambda_i) + (-1)^{n-h} \frac{\alpha^2}{r} \prod_{i=1}^h (s - \zeta_i)(s + \zeta_i) = 0.$$

By applying the properties of the root locus, one can verify that

- i) when  $r \rightarrow 0$ ,  $2h$  roots of the equation tend asymptotically to the  $2h$  zeros  $\zeta_i$  e  $-\zeta_i$   $i = 1, 2, \dots, h$ , while the remaining  $2(n-h)$  roots tend asymptotically to the  $2(n-h)$  half-lines stemming from the origin that form, with the positive real axes, angles of value

$$\frac{\pi}{n-h} j, \quad j = 0, 1, \dots, 2n-2h-1,$$

if  $n-h$  is odd, and

$$\frac{\pi}{n-h} (j + \frac{1}{2}), \quad j = 0, 1, \dots, 2n-2h-1,$$

if  $n-h$  is even.

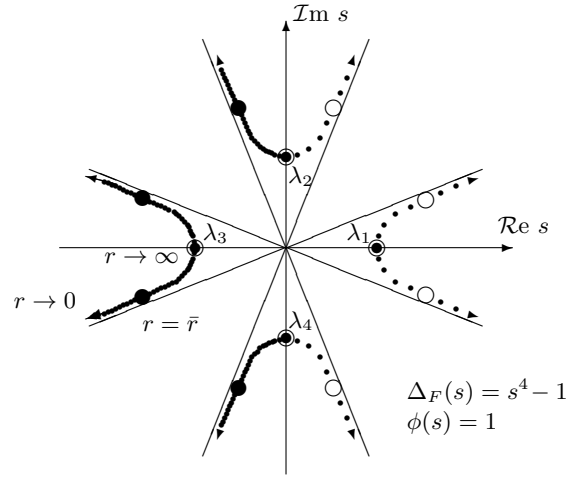


Figure 10.6.3

Furthermore, as  $r \rightarrow 0$  the  $2(n-h)$  roots that tend to infinite belong, approximately, to a circle of radius  $\left(\frac{\alpha^2}{r}\right)^{\frac{1}{2(n-h)}}$ . Obviously, the eigenvalues of  $F + GK_\infty$  are the roots whose real part is negative.

- ii) When  $r \rightarrow +\infty$ , the  $2n$  roots tend asymptotically to the  $2n$  zeros  $\lambda_i$  e  $-\lambda_i$   $i = 1, 2, \dots, n$ . The eigenvalues of  $F + GK_\infty$  constitute the  $n$ -ple with negative real part.

In Figure 10.6.3 we show the configuration of the eigenvalues of the Hamiltonian matrix when  $\Delta_F(s) = s^4 - 1$  and  $\phi(s) = 1$ . For  $r = \infty$  one gets the zeros of  $\Delta_F(s)$ , each counted twice; for finite and positive values of  $r$ , the four eigenvalues of the matrix  $F + \mathbf{g}K_\infty$  move over the solid-line curves, while the other four unstable eigenvalues of the Hamiltonian matrix trace the dashed-line curves.

## 10.7 Maximum Principle - elements

The first part of this chapter addressed optimal control problems under the assumption that the state model is linear and the index has a quadratic structure. In this concluding paragraph we give an overview of the maximum principle, one of the classical methods for the study of optimal problems in a more general context, limiting the analysis, for the sake of brevity, to continuous-time systems only. We will consider a generic dynamical system (not necessarily non-linear) represented by the state model

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t)), \quad (10.191)$$

for which it is assigned the initial state

$$\mathbf{x}(0) = \mathbf{x}_0. \quad (10.192)$$

We further assume that, at each instant, the value of the input  $\mathbf{u}$  is constrained to belong to a given set  $U$  of  $\mathbb{R}^m$ , that mathematically formalises the physical constraints of the problem, and/or other conditions that the control variables must satisfy:

$$\mathbf{u}(t) \in U \subseteq \mathbb{R}^m, \quad \forall t. \quad (10.193)$$

The control problem is to find, if it exists, an input  $\mathbf{u}(\cdot)$  that satisfies (10.193) and that, by acting over the interval  $[0, T]$ , maximises the index, or “objective function”,

$$J(\mathbf{u}, \mathbf{x}(0)) = \psi(\mathbf{x}(T)) + \int_0^T \mathcal{L}(\mathbf{x}(t), \mathbf{u}(t)) dt. \quad (10.194)$$

As for the quadratic index considered in the previous paragraphs,  $J$  includes two addends accounting for the two distinct aspects that often need to be traded off in the evaluation of the system behaviour. In particular, the term  $\psi(\mathbf{x}(T))$  represents the weight given to the final state, while the integral term weights the transient behaviour over the interval  $[0, T]$ , along which the system reaches the final state.

Except for some technical regularity conditions, the scalar-valued function  $\mathcal{L}(\cdot, \cdot)$  can be rather generic. The same holds true for the input function, apart from the constraint to belong to the set  $U$ . For any initial state, the model (10.191)-(10.194) associates to any admissible input function  $\mathbf{u}(\cdot)$  a well defined value of the objective function  $J$ . In fact, (10.191) determines

- a specific state evolution  $\mathbf{x}(\cdot)$  which, along with  $\mathbf{u}(\cdot)$ , is the argument of  $\mathcal{L}$ , thus determining its value at each time, and hence its integral;
- a specific final state  $\mathbf{x}(T)$ , and with it the term  $\psi(\mathbf{x}(T))$ .

By interpreting  $J$  as a functional defined on the set of admissible input functions  $\mathbf{u}(\cdot)$  and on the set of state evolutions  $\mathbf{x}(\cdot)$ , the optimal problem can be interpreted as the problem of minimising  $J$  subject to the condition that  $\mathbf{u}(\cdot)$  and  $\mathbf{x}(\cdot)$  are constrained by (10.191) - (10.192).

Among several possible available techniques to solve the problem, we consider here a variational formulation, which in turn will lead us to derive the maximum principle. Furthermore, and for the sake of simplicity, we shall adopt in the whole paragraph a semi-intuitive approach.

### 10.7.1 Variational formulation

As it is well known, the search of maximal points of a function is based on the intuitive criterion that small perturbations of the variables from their maximising values cannot but decrease the function value.

The variational formulation of the optimal problem is based on extending such a criterion to the case of functionals defined on function spaces. The strategy, in this case, is to determine an input  $\mathbf{u}_0(\cdot)$  that maximises  $J$  by exploiting the condition that any other

function  $\mathbf{u}(\cdot)$  whose graph is “close” to that of  $\mathbf{u}_0(\cdot)$  yields a value of  $J$  that is smaller than the value  $J_0$  that corresponds to  $\mathbf{u}_0(\cdot)$ .

The problem still does not admit in general a direct solution, since  $J$  depends on both  $\mathbf{u}(\cdot)$  and  $\mathbf{x}(\cdot)$ , which are not independent, but rather related via the differential equation (10.191).

One possible technique is based on introducing the *Hamiltonian function* (or *hamiltonian*)

$$\mathcal{H} : \mathbb{R}^n \times \mathbb{R}^n \times U \rightarrow \mathbb{R} : (\boldsymbol{\lambda}, \mathbf{x}, \mathbf{u}) \mapsto \boldsymbol{\lambda}^T \mathbf{f}(\mathbf{x}, \mathbf{u}) + \mathcal{L}(\mathbf{x}, \mathbf{u}) = \mathcal{H}(\boldsymbol{\lambda}, \mathbf{x}, \mathbf{u}) \quad (10.195)$$

that associates to any triad  $(\boldsymbol{\lambda}, \mathbf{x}, \mathbf{u})$  the real number  $\mathcal{H}(\boldsymbol{\lambda}, \mathbf{x}, \mathbf{u})$ .

(i) For any triad of functions  $(\boldsymbol{\lambda}(\cdot), \mathbf{x}(\cdot), \mathbf{u}(\cdot))$ , define the functional

$$\bar{J} := J - \int_0^T \boldsymbol{\lambda}^T(t) [\dot{\mathbf{x}}(t) - \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t))] dt. \quad (10.196)$$

Notice that  $\bar{J}$  coincides with  $J$ , independently of  $\boldsymbol{\lambda}(\cdot)$ , whenever  $\mathbf{x}(\cdot)$  and  $\mathbf{u}(\cdot)$  satisfy the state equation of the system.

We modify the expression of  $\bar{J}$  by introducing the Hamiltonian function evaluated at the triad  $(\boldsymbol{\lambda}, \mathbf{x}, \mathbf{u})$ :

$$\begin{aligned} \bar{J} &= \psi(\mathbf{x}(T)) + \int_0^T \mathcal{L}(\mathbf{x}(t), \mathbf{u}(t)) dt - \int_0^T \boldsymbol{\lambda}^T(t) [\dot{\mathbf{x}}(t) - \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t))] dt \\ &= \psi(\mathbf{x}(T)) + \int_0^T [\mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{u}(t)) - \boldsymbol{\lambda}^T(t) \dot{\mathbf{x}}(t)] dt. \end{aligned} \quad (10.197)$$

(ii) Consider now a reference input  $\mathbf{u}(\cdot)$ , a perturbed input  $\mathbf{v}(\cdot) = \mathbf{u}(\cdot) + \delta\mathbf{u}(\cdot)$ , and the corresponding reference and perturbed state evolutions  $\mathbf{x}(\cdot)$  and  $\mathbf{x}(\cdot) + \delta\mathbf{x}(\cdot)$ , respectively. By considering as norm in the space of the input functions

$$\|\mathbf{u}(\cdot)\| = \max_{i=1\dots m} \int_0^T |u_i(t)| dt,$$

the “distance” between  $\mathbf{u}(\cdot)$  and  $\mathbf{v}(\cdot)$  is given by  $\|\delta\mathbf{u}(\cdot)\| = \max_{i=1\dots m} \int_0^T |u_i(t) - v_i(t)| dt$ . By the properties of the solutions to the equation (10.191), if  $\|\mathbf{u}(\cdot)\|$  is infinitesimal, then both  $\delta\mathbf{x}(t)$ , for  $t \in [0, T]$ , and  $\|\delta\mathbf{x}(\cdot)\|$  are infinitesimal, at least of the same order.

(iii) The variation  $\delta\bar{J}$  of the functional (10.196), which is obtained by replacing in  $\bar{J}$  the pair  $(\mathbf{u}(\cdot), \mathbf{x}(\cdot))$  with the perturbed pair  $(\mathbf{v}(\cdot), \mathbf{x}(\cdot) + \delta\mathbf{x}(\cdot))$ , is given by

$$\begin{aligned} \delta\bar{J} &= \psi(\mathbf{x}(T) + \delta\mathbf{x}(T)) - \psi(\mathbf{x}(T)) \\ &+ \int_0^T \left\{ \mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t) + \delta\mathbf{x}(t), \mathbf{v}(t)) - \mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{u}(t)) - \boldsymbol{\lambda}^T(t) \delta\dot{\mathbf{x}}(t) \right\} dt. \end{aligned} \quad (10.198)$$

Integrating by parts, one obtains

$$\int_0^T \boldsymbol{\lambda}^T(t) \delta\dot{\mathbf{x}}(t) dt = \boldsymbol{\lambda}^T(T) \delta\mathbf{x}(T) - \boldsymbol{\lambda}^T(0) \delta\mathbf{x}(0) - \int_0^T \dot{\boldsymbol{\lambda}}^T(t) \delta\mathbf{x}(t) dt$$

and therefore

$$\begin{aligned}\delta\bar{J} &= \{\psi(\mathbf{x}(T)+\delta\mathbf{x}(T)) - \psi(\mathbf{x}(T)) - \boldsymbol{\lambda}^T(T)\delta\mathbf{x}(T) + \boldsymbol{\lambda}^T(0)\delta\mathbf{x}(0)\} \\ &+ \int_0^T \left\{ \mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t)+\delta\mathbf{x}(t), \mathbf{v}(t)) - \mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{u}(t)) \right\} dt \\ &+ \int_0^T \dot{\boldsymbol{\lambda}}^T(t)\delta\mathbf{x}(t)dt.\end{aligned}\quad (10.199)$$

When  $\|\delta\mathbf{u}(\cdot)\|$  is small enough, (10.199) can be approximated by neglecting terms of order higher than one in the Taylor expansion. In this way, the first row of (10.199) becomes

$$\begin{aligned}&\psi(\mathbf{x}(T)+\delta\mathbf{x}(T)) - \psi(\mathbf{x}(T)) - \boldsymbol{\lambda}^T(T)\delta\mathbf{x}(T) + \boldsymbol{\lambda}^T(0)\delta\mathbf{x}(0) \\ &\cong \left( \frac{\partial\psi(\mathbf{x}(T))}{\partial\mathbf{x}} - \boldsymbol{\lambda}^T(T) \right) \delta\mathbf{x}(T) + \boldsymbol{\lambda}^T(0)\delta\mathbf{x}(0)\end{aligned}$$

and the second one becomes

$$\begin{aligned}&\int_0^T \left\{ \mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t)+\delta\mathbf{x}(t), \mathbf{v}(t)) - \mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{u}(t)) \right\} dt \\ &= \int_0^T \left\{ [\mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t)+\delta\mathbf{x}(t), \mathbf{v}(t)) - \mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{v}(t))] + [\mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{v}(t)) - \mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{u}(t))] \right\} dt \\ &\cong \int_0^T \frac{\partial\mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{v}(t))}{\partial\mathbf{x}} \delta\mathbf{x}(t)dt + \int_0^T [\mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{v}(t)) - \mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{u}(t))] dt \\ &= \int_0^T \left[ \frac{\partial\mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{v}(t))}{\partial\mathbf{x}} - \frac{\partial\mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{u}(t))}{\partial\mathbf{x}} \right] \delta\mathbf{x}(t)dt + \int_0^T \frac{\partial\mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{u}(t))}{\partial\mathbf{x}} \delta\mathbf{x}(t)dt + \\ &+ \int_0^T [\mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{v}(t)) - \mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{u}(t))] dt \\ &\cong \int_0^T \left[ \frac{\partial\mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{u}(t))}{\partial\mathbf{x}} \delta\mathbf{x}(t) + \mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{v}(t)) - \mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{u}(t)) \right] dt.\end{aligned}$$

The symbol “ $\cong$ ” is due to the fact that we neglect infinitesimal terms of order higher than  $\|\delta\mathbf{u}(\cdot)\|$ . In particular, in the last approximation we neglect the expression  $\int_0^T \left[ \frac{\partial\mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{v}(t))}{\partial\mathbf{x}} - \frac{\partial\mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{u}(t))}{\partial\mathbf{x}} \right] \delta\mathbf{x}(t)dt$ , because both  $\delta\mathbf{x}(t)$  and the integral of  $\frac{\partial\mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{v}(t))}{\partial\mathbf{x}} - \frac{\partial\mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{u}(t))}{\partial\mathbf{x}}$  are infinitesimal of order at least the same as  $\|\delta\mathbf{u}(\cdot)\|$ , and thus their product is of order higher than  $\|\delta\mathbf{u}(\cdot)\|$ .

Therefore, apart from infinitesimals of order higher than  $\|\delta\mathbf{u}(\cdot)\|$ , the variation of  $\bar{J}$  in (10.198) can be rewritten as

$$\begin{aligned}\delta\bar{J} &= \left( \frac{\partial\psi(\mathbf{x}(T))}{\partial\mathbf{x}} - \boldsymbol{\lambda}^T(T) \right) \delta\mathbf{x}(T) + \boldsymbol{\lambda}^T(0)\delta\mathbf{x}(0) + \int_0^T \left[ \frac{\partial\mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{u}(t))}{\partial\mathbf{x}} + \dot{\boldsymbol{\lambda}}^T(t) \right] \delta\mathbf{x}(t)dt \\ &+ \int_0^T [\mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{v}(t)) - \mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{u}(t))] dt.\end{aligned}\quad (10.200)$$

(iv) In (10.200) the function  $\boldsymbol{\lambda}(\cdot)$  is arbitrary, and we can choose its structure in such a way to simplify the expression of  $\delta\bar{J}$ . In particular, we can let the first integral (10.200)

vanish by choosing  $\lambda(\cdot)$  such that it satisfies the *adjoint equation*

$$-\dot{\lambda}^T(t) = \frac{\partial \mathcal{H}(\lambda(t), \mathbf{x}(t), \mathbf{u}(t))}{\partial \mathbf{x}}, \quad (10.201)$$

while by imposing the final condition

$$\lambda^T(T) = \frac{\partial \psi(\mathbf{x}(T))}{\partial \mathbf{x}} \quad (10.202)$$

we also let the first addend of (10.200) vanish.

By choosing  $\lambda(\cdot)$  in this way, and further noticing that  $\delta \mathbf{x}(0)$  is zero because a variation of the input function  $\mathbf{u}(\cdot)$  does not change the initial state, the perturbation  $\delta \bar{J}$  takes the form

$$\delta \bar{J} = \int_0^T [\mathcal{H}(\lambda(t), \mathbf{x}(t), \mathbf{v}(t)) - \mathcal{H}(\lambda(t), \mathbf{x}(t), \mathbf{u}(t))] dt. \quad (10.203)$$

Once the input  $\mathbf{u}(\cdot)$  is chosen,  $\mathbf{x}(\cdot)$  is determined by (10.191) and (10.192), and, consequently,  $\lambda(\cdot)$  can be obtained from (10.201) and (10.202). Therefore, (10.203) yields, for any perturbed input  $\mathbf{v}(\cdot)$ , an explicit expression of the variation of  $\bar{J}$ .

(v) We want to verify that, *if the control law  $\mathbf{u}(\cdot)$  is optimal and if  $\lambda(\cdot)$  solves (10.201) - (10.202), then for any constant vector  $\bar{\mathbf{v}} \in U$  and for any  $t \in [0, T]$  it holds true*

$$\mathcal{H}(\lambda(t), \mathbf{x}(t), \bar{\mathbf{v}}) \leq \mathcal{H}(\lambda(t), \mathbf{x}(t), \mathbf{u}(t)). \quad (10.204)$$

In fact, assume by contradiction that there exists a vector  $\bar{\mathbf{v}} \in U$  such that, for some  $t = \bar{t}$ , we have

$$\mathcal{H}(\lambda(\bar{t}), \mathbf{x}(\bar{t}), \bar{\mathbf{v}}) > \mathcal{H}(\lambda(\bar{t}), \mathbf{x}(\bar{t}), \mathbf{u}(\bar{t})), \quad (10.205)$$

and consider a perturbed input  $\mathbf{v}(\cdot)$  that coincides with  $\mathbf{u}(\cdot)$ , except for a small neighbourhood of  $\bar{t}$ , in which the perturbed input takes the constant value  $\bar{\mathbf{v}}$ .

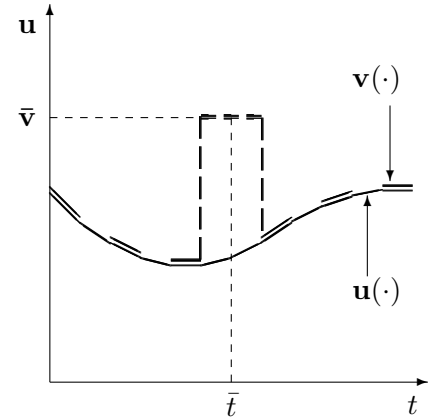


Figure 10.7.1

Corresponding to the two input functions  $\mathbf{u}(\cdot)$  and  $\mathbf{v}(\cdot)$ , from (10.205) it follows that (10.203) is positive, because the argument of the integral is positive in a neighbourhood of  $\bar{t}$  and zero elsewhere. But this is in contradiction with the assumption that  $\mathbf{u}(\cdot)$  is an optimal input, namely that (10.203) is negative for any input different from  $\mathbf{u}(\cdot)$ . We have thus proven that, when the input  $\mathbf{u}(\cdot)$  is optimal, the value taken by the Hamiltonian at any time  $t$  cannot be increased if the vector  $\mathbf{u}(t)$  of the optimal input is substituted by any other vector  $\mathbf{v} \in U$ . This result, a necessary optimality condition, is known as *Pontryagin maximum principle* for the optimal problem (10.191) ÷ (10.194).

## 10.7.2 Maximum principle

The next proposition summarises in a single statement the discussion made in the previous section.

**Proposition 10.7.1** [MAXIMUM PRINCIPLE] Assume that the input function  $\mathbf{u}(\cdot)$ , constrained to satisfy  $\mathbf{u}(t) \in U$ ,  $\forall t \in [0, T]$ , represents the optimal control that maximises the index (10.194) for the system (10.191)-(10.192).

Let  $\mathbf{x}(\cdot)$  be the corresponding state evolution and  $\boldsymbol{\lambda}(\cdot) : [0, T] \rightarrow \mathbb{R}^n$  a function such that  $\mathbf{u}(\cdot)$ ,  $\mathbf{x}(\cdot)$  and  $\boldsymbol{\lambda}(\cdot)$  satisfy

$$\begin{aligned}\frac{d\mathbf{x}(t)}{dt} &= \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t)), \\ \mathbf{x}(0) &= \mathbf{x}_0 \\ -\dot{\boldsymbol{\lambda}}^T(t) &= \boldsymbol{\lambda}^T(t) \frac{\partial \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t))}{\partial \mathbf{x}} + \frac{\partial \mathcal{L}(\mathbf{x}(t), \mathbf{u}(t))}{\partial \mathbf{x}}, \\ \boldsymbol{\lambda}^T(T) &= \frac{\partial \psi(\mathbf{x}(T))}{\partial \mathbf{x}}.\end{aligned}$$

Then the Hamiltonian function

$$\mathcal{H}(\boldsymbol{\lambda}, \mathbf{x}, \mathbf{u}) := \boldsymbol{\lambda}^T \mathbf{f}(\mathbf{x}, \mathbf{u}) + \mathcal{L}(\mathbf{x}, \mathbf{u})$$

satisfies, for any  $t \in [0, T]$  and for any  $\mathbf{v} \in U$ , the inequality

$$\mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \bar{\mathbf{v}}) \leq \mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{u}(t)). \quad \blacksquare$$

In the solution to the optimal problem,  $\mathbf{u}(\cdot)$ ,  $\mathbf{x}(\cdot)$  and  $\boldsymbol{\lambda}(\cdot)$  are to be considered unknowns. Overall, the number of unknown scalar functions is  $2n + m$ . On the other side, the problem data include the state equation and the adjoint equation, which correspond to  $2n$  scalar differential equations, and the inequality (10.204). If the optimal control lies, for any  $t$ , strictly within the region  $U$ , such an inequality translates into  $m$  algebraic equations, which can be obtained by setting to zero the derivatives of the Hamiltonian with respect to the  $m$  components of  $\mathbf{u}$ :

$$\frac{\partial \mathcal{H}(\boldsymbol{\lambda}(t), \mathbf{x}(t), \mathbf{u}(t))}{\partial \mathbf{u}} = \mathbf{0}. \quad (10.206)$$

As a consequence, in this case, the number of equations coincides with the number of unknowns. In other cases, namely when the optimal control lies on the boundary of  $U$ , there are no general procedures available, and the solution of the problem is related to the structure of the Hamiltonian function.

**Example 10.7.1** Consider, as in Example 10.5.1, a purely inertial cart of unitary mass, controlled by a force  $u(\cdot)$ . Assume that the cart is at rest in the coordinates origin at time  $t = 0$ . Denote by  $s(t)$  the travelled distance at time  $t$ , so that the equation of motion is

$$\ddot{s}(t) = u(t), \quad (10.207)$$

with initial conditions  $s(0) = \dot{s}(0) = 0$ . We want to determine  $u(\cdot)$  in order to maximise the objective function

$$J_T = s(T) - \frac{1}{2} \int_0^T u^2(t) dt \quad (10.208)$$

which trades off two contrasting requirements, namely to travel the maximum possible distance in the finite horizon  $[0, T]$ , and at the same time to avoid applying forces whose magnitude is large for long periods.

Introducing the state variables as in (10.174), we obtain the model

$$\begin{aligned}\dot{\mathbf{x}}(t) &= \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(t) \\ \mathbf{x}(0) &= \mathbf{0},\end{aligned}$$

with index

$$J_T = x_1(T) - \frac{1}{2} \int_0^T u^2(t) dt.$$

The Hamiltonian function of the system is

$$\begin{aligned}\mathcal{H}(\boldsymbol{\lambda}, \mathbf{x}, u) &= \boldsymbol{\lambda}^T \mathbf{f}(\mathbf{x}, u) + \mathcal{L}(\mathbf{x}, u) = \boldsymbol{\lambda}^T (F\mathbf{x} + \mathbf{g}u) - \frac{1}{2}u^2 \\ &= [\lambda_1 \quad \lambda_2] \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + [\lambda_1 \quad \lambda_2] \begin{bmatrix} 0 \\ 1 \end{bmatrix} - \frac{1}{2}u^2 = \lambda_1 x_2 + \lambda_2 u - \frac{1}{2}u^2.\end{aligned}\quad (10.209)$$

The adjoint equation is

$$-\dot{\boldsymbol{\lambda}}^T(t) = \boldsymbol{\lambda}^T(t) \frac{\partial \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t))}{\partial \mathbf{x}} + \frac{\partial \mathcal{L}(\mathbf{x}(t), \mathbf{u}(t))}{\partial \mathbf{x}} = \boldsymbol{\lambda}^T F, \quad (10.210)$$

with final condition

$$\boldsymbol{\lambda}^T(T) = \frac{\partial \psi(\mathbf{x}(T))}{\partial \mathbf{x}} = [1 \quad 0]. \quad (10.211)$$

By solving the adjoint equation we obtain

$$\begin{bmatrix} \lambda_1(t) \\ \lambda_2(t) \end{bmatrix} = e^{-F^T(t-T)} \begin{bmatrix} \lambda_1(T) \\ \lambda_2(T) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ T-t & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ T-t \end{bmatrix}.$$

Since there are no constraints on the values of  $u$ , one can take the partial derivative of the Hamiltonian with respect to  $u$  itself, which is

$$\frac{\partial \mathcal{H}}{\partial u} = \lambda_2 - u.$$

By setting such a partial derivative to zero, we obtain the optimal input  $u(t) = \lambda_2(t) = T - t$ , which corresponds to a force that linearly decays in time from the initial value  $u(0) = T$  to  $u(T) = 0$ .

**Example 10.7.2** [INVESTMENT PLANNING] We consider the problem of determining an optimal investment planning for a production plant. Denoting by  $x(t)\Delta t$  the gain that is due to the production over the interval  $[t, t + \Delta t]$ , and by  $u(t)\Delta t$  the investment over the same interval, we assume that

- in absence of investments,  $x(t)$  decreases, and the decrease in a period  $\Delta t$  is proportional to  $x(t)$ ;
- in  $\Delta t$  the level  $x(t)$  increases proportionally to the investments  $u(t)\Delta t$  over the interval.

From the finite terms relation  $x(t + \Delta t) = x(t) - \alpha x(t)\Delta t + \gamma u(t)\Delta t$  and moving to the limit we get the differential equation

$$\dot{x}(t) = -\alpha x(t) + \gamma u(t), \quad (10.212)$$

where  $\alpha$  and  $\gamma$  are positive constants and  $x(0) = x_0 > 0$ .

We further assume that, after a time  $T$ , the entire production unit is sold at a price that is proportional to the level  $x(T)$  that is reached at time  $T$ . The index that we want to maximise is

$$J_T = \beta x(T) + \int_0^T [x(t) - u(t)] dt, \quad \beta > 0, \quad (10.213)$$

in which the first added represents the capital return due to the sale, and the second addend the cumulative gain during the operation.

We further assume that the instantaneous level of investment is positive and upper bounded, with

$$0 \leq u(t) \leq \bar{u}.$$



The Hamiltonian function is

$$\mathcal{H}(\lambda, x, u) = \lambda(-\alpha x + \gamma u) + x - u$$

and the adjoint equation is

$$\dot{\lambda}(t) = \alpha\lambda - 1, \quad \text{con } \lambda(T) = \beta.$$

By integrating the adjoint equation, we obtain

$$\lambda(t) = e^{\alpha(t-T)}\beta + \int_T^t e^{\alpha(t-\sigma)}(-1)d\sigma = e^{\alpha(t-T)}\left(\beta - \frac{1}{\alpha}\right) + \frac{1}{\alpha}. \quad (10.214)$$

A necessary condition for optimality is that, at any time between 0 and  $T$ ,

$$\mathcal{H}(\lambda(t), x(t), u(t)) \geq \mathcal{H}(\lambda(t), x(t), v), \quad \forall v \in [0, \bar{u}].$$

By plugging the expression (11.49) of  $\lambda(t)$  into  $\mathcal{H}$ , we obtain

$$\left[e^{\alpha(t-T)}\left(\beta - \frac{1}{\alpha}\right) + \frac{1}{\alpha}\right](-\alpha x(t) + \gamma u(t)) + x(t) - u(t) \geq \left[e^{\alpha(t-T)}\left(\beta - \frac{1}{\alpha}\right) + \frac{1}{\alpha}\right](-\alpha x(t) + \gamma v) + x(t) - v,$$

namely

$$\left\{ \left[ e^{\alpha(t-T)}\left(\beta - \frac{1}{\alpha}\right) + \frac{1}{\alpha} \right] \gamma - 1 \right\} [u(t) - v] \geq 0 \quad \forall v \in [0, \bar{u}]. \quad (10.215)$$

Denote now by  $D(t)$  the term in curly brackets. It is convenient to distinguish between two cases, namely, when  $D(t)$  has constant sign over  $[0, T]$ , and when  $D(t)$  changes sign at some instant  $t_s \in [0, T]$ .

(i) In the first case, if  $D(t)$  is positive, then the same must hold true for  $u(t) - v$  for any  $v \in [0, \bar{u}]$ . It follows that the optimal investment level is constant over the entire interval, and its value is

$$u(t) = \bar{u}, \quad 0 \leq t \leq T.$$

If, instead,  $D(t)$  is negative, then the same must hold true for  $u(t) - v$  for any  $v \in [0, \bar{u}]$ , and hence the optimal solution is

$$u(t) = 0, \quad 0 \leq t \leq T.$$

(ii) In the second case, by solving  $D(t) = 0$  one finds that the only time at which  $D(t)$  can change sign is

$$t_s = T - \frac{1}{\alpha} \ln \frac{\frac{1}{\alpha} - \beta}{\frac{1}{\alpha} - \frac{1}{\gamma}}. \quad (10.216)$$

Under the assumption that  $t_s$  belongs to the interior of the interval  $[0, T]$ , the optimal investment policy consists in the concatenation of the two policies  $u(t) = \bar{u}$  and  $u(t) = 0$ , the first to be applied over the interval in which  $D(t)$  is positive, the second over the interval in which  $D(t)$  is negative. The time at which the switch happens is  $t_s$ .

**Remark** A sufficient condition for  $D(t)$  to have constant sign over  $[0, T]$  is that  $\gamma$  lies outside the interval whose extremes are  $\alpha$  and  $\frac{1}{\beta}$ . This happens, for example, if

$$\frac{1}{\beta} \leq \alpha < \gamma. \quad (10.217)$$

Under such condition  $D(t) > 0$  for any  $t \in [0, T]$ , and therefore the optimal investment policy is  $u(t) = \bar{u}$  over the interval  $[0, T]$ . The first inequality in (10.217) can be interpreted as follows: If starting at time  $T$  the production unit, instead of being sold, is exploited without any further investment, the total gain over the infinite horizon  $[T, +\infty)$  is

$$\int_T^{+\infty} x(t)dt = -\frac{1}{\alpha} \int_T^{+\infty} \dot{x}(t)dt = \frac{1}{\alpha} x(T).$$

The inequality  $1/\beta \leq \alpha$  is therefore equivalent to assuming that the capital return on sale is at least equal to the gain obtained for  $t > T$  keeping the unit operative and without any further investment.

## 10.8 Bibliography

The literature on optimal control is extremely wide. For the linear quadratic case and the Riccati Equation, we limit the references to

- 1 H.Kwakernaak, R.Sivan “*Linear optimal control systems*” Wiley, 1972
- 2 B.D.O.Anderson, J.B.Moore “*Optimal control: linear quadratic methods*” Prentice-Hall, 1989
- 3 P.Lancaster, L.Rodman “*The algebraic Riccati equation*” Oxford Univ.Press, 1995.

The third chapter of

- 4 R.W.Brockett “*Finite dimensional linear systems*” (available online).

is concerned with continuous-time minimum squares theory. It deals in particular with several problems that we skipped in the present Chapter, such as optimal control of linear time-varying systems, or with specified final state, inequalities in the domain  $\omega$ , spectral factorisation.

The paragraph on maximum principle follows Chapter 11 of

- 5 D.G. Luenberger “*Introduction to dynamic systems*”, Wiley, 1979

which includes, on the same topic, a commented bibliography.

For a thorough analysis of optimal management of natural resources, we refer to

- 6 W.M.Getz, R.G.Haight “*Population harvesting: demographic models of fish, forest and animal resources*, Princeton Univ. Press, 1989
- 7 C.W.Clark “*Mathematical bioeconomics*” Wiley, 1976

Determining the accumulated (available) energy in a dissipative system, and in particular in a passive electrical network, can be cast as a minimum (or maximum) problem. Two classical works on this topic are

- 8 J.C.Willems “*Dissipative dynamical systems - Part I: General theory - Part II: Linear systems with quadratic supply rates*” Archive for Rational Mechanics, vol 45, pp.321-51 e 352-93, 1972